

TENET - Cumulative Causal Dossier

PASS022 - FINAL TENET CLOSURE

Synchronized English edition

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This is the final cumulative standalone TENET dossier. It does not modify the Causal Encyclopedia. PASS022 freezes the protected causal TENET definition, the registered ITC 2SIN+1COS realization, the H/NG registries and the final empirical claim ceiling. Numerical personal identity remains OPEN / OUTSIDE CORE; physical retrograde realization and universal natural-law status remain OPEN.

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1 Purpose of PASS001

Status: CUMULATIVE RESEARCH DOSSIER.

PASS001 established the temporal pincer of the self. PASS002 is cumulative and does not attempt a scene-by-scene interpretation of the film. It extracts the structural signal proposed by the user and asks whether it can be connected to existing CT objects: Space Container, Sleeper Hypothesis, memory/trace, time travel, human development, love as a causal object, First Sine/Second Sine/Cosine, and closure.

The central hypothesis is:

Human Tenet - temporal-pincer hypothesis. A life can be represented as closure between two necessary descriptions of the same being: the present causal self advancing locally in time, and a higher/terminal self acting as a global closure constraint. The real trajectory is the one satisfying constraints from both ends simultaneously.

2 What “tenet” meant before the search

Before searching the Encyclopedia, a *tenet* meant a constitutive CT principle with four properties:

1. **Primitivity:** not merely a local consequence of a deeper principle.
2. **Transversality:** able to organize several domains without depending on one domain’s vocabulary.
3. **Stability:** survives representation changes and elimination of narrative detail.
4. **Closure power:** places a real constraint on what can count as a coherent object or history.

The first search therefore naturally recovered invariant persistence, Last Survivor and non-erasure. PASS001 keeps that layer but adds the human question: *what is the invariant of a person when the whole trajectory is considered rather than one instant?*

3 Structural signal extracted from TENET

The user’s starting analysis reduces to five operations independent of characters and scenery:

1. an ordinary trajectory follows a local causal order;
2. a transition operator converts it into a second regime where time becomes a locally addressable coordinate rather than the sole order of traversal;
3. two different trajectories carry complementary information about the same problem;
4. neither trajectory is sufficient alone;
5. closure reveals that both histories had to be mutually compatible for the complete history to exist.

The Last Survivor of this reduction is not the machine, hero or villain. It is:

two partial traversals + one invariant + one common closure.

4 Existing CT components

4.1 Space Container

S029 defines a mobile coherence container and keeps it distinct from a complete derivation of physical space. TENET reuses only the structural idea: a bounded region can carry internal dynamics, boundary, flux, memory and transition rules. The move “time $\rightarrow$ localized space” is therefore modeled as a *carrier change*, not as a claim that S029 has already proved such an ontology.

4.2 Sleeper Hypothesis

S050-S051 introduce planetary memory, hypercausal agency, sleeper, reachability and a minimal-action test. The reusable signal is that a more global identity can be modeled by an integrating capacity beyond the local horizon of the present agent. Sleeper biology and identification of any particular person remain open.

4.3 Time travel

S141-S146 separate causal and supracausal layers and introduce P_s/P_c projection, temporal residue, re-entry, identity stability, branching, attractors and merging. The entire series remains conditional regarding physical time travel. For TENET it nevertheless provides a precise language for two orders: local chronological order and global coherence order.

4.4 Human development

S054 proposes:

Being $\rightarrow$ Having $\rightarrow$ Doing $\rightarrow$ Becoming $\rightarrow$
 Understanding $\rightarrow$ Learning $\rightarrow$ Choosing $\rightarrow$ Transforming $\rightarrow$
 Sharing $\rightarrow$ Creating $\rightarrow$ Transcending.

PASS001 does not take this sequence as a universal law. It uses it as a candidate construction of the terminal self: the higher self is not simply a stronger present self, but a self that has integrated the transformations required for its own closure.

4.5 Love, time and space

S208 proposes a domain partition $T = \{\phi, \pi\}$ and $X = \{\sqrt{2}, \sqrt{3}, \ln 5\}$ with DOMAIN MODEL status, and an interpretive love coordinate based on distinctness, freedom, coherence and global responsibility. This matters for TENET because it already rejects the idea that coherent union requires erasing the two poles.

5 The First Sine - Second Sine - Cosine bridge

The CT Prime volumes provide a formal object unusually close to the TENET intuition, although in a different domain. They distinguish:

- **First Sine:** first oriented residual / problem dossier;
- **Second Sine:** last oriented residual at the stable-solution boundary, preserving irreversible traces and transmitting toward closure;

- **Cosine**: solution direction and common closure.

The decisive result for PASS001 is negative: **cosine alone does not close the system**. Closure requires residual transport, carrier identity and a fixed point. The final solution therefore does not erase the two histories that make it possible.

The new human bridge is CANDIDATE:

Sine₁ $\rightsquigarrow$ present self / local causal history,
 Sine₂ $\rightsquigarrow$ transformed history / time localized in a second carrier,
 Cosine $\rightsquigarrow$ closed self / common identity solution.

This is not a theorem inherited from CT Prime; it is a new TENET research object.

6 Temporal-pincer model

Let x_0 be the initial self, H the higher/terminal self and Γ the set of admissible histories.

Local causality generates

$$\Gamma^+(x_0) = \{\gamma : \gamma(0) = x_0, \gamma \text{ satisfies local causal constraints}\}.$$

The terminal self imposes a reverse compatibility condition

$$\Gamma^-(H) = \{\gamma : \gamma(T) \sim H, \gamma \text{ can close on } H\}.$$

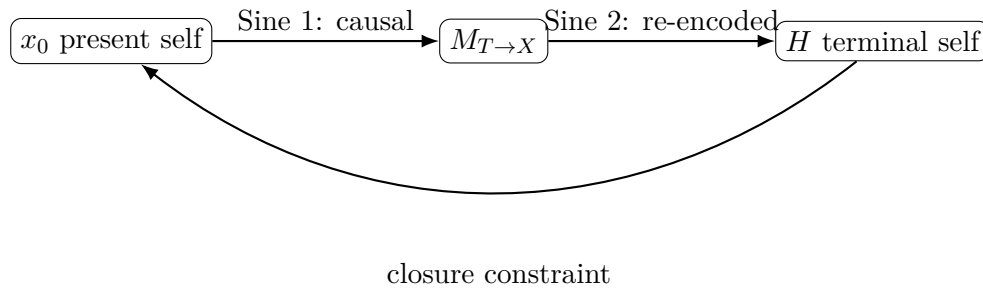
The histories compatible with both descriptions lie in

$$\Gamma_{\text{TENET}} = \Gamma^+(x_0) \cap \Gamma^-(H).$$

The strongest Last-Survivor form is

$$\boxed{\gamma^* = \text{LastSurvivor}(\Gamma^+(x_0) \cap \Gamma^-(H)).}$$

This is the temporal pincer: the past pushes through realized causes; the terminal self pulls through closure admissibility. These are not two physical forces propagating through time. They are two boundary conditions on one global problem.



7 The machine as a carrier-change operator

The TENET reading does not require time to “literally become space” as a physical theorem. It needs only a representation operator

$$M_{T \rightarrow X} : \mathcal{C}_t \longrightarrow \mathcal{S}_X$$

that replaces chronology as the dominant coordinate by a carrier in which multiple temporal slices become locally addressable.

The bridge to the time-travel series is

$$C_t \xrightarrow{P_s} S \xrightarrow{\text{matching / contraction}} S' \xrightarrow{P_c} C_{t_0}.$$

In the TENET dossier, $M_{T \rightarrow X}$ is the functional name of the translation between the two descriptions. Physical implementation remains OPEN.

8 Soul as ideal self: new hypothesis

Status: TENET ONTOLOGICAL CANDIDATE.

The corpus does not yet establish “soul = ideal self”. PASS001 proposes a conceptually testable definition:

The **soul** is the identity invariant of a person under admissible transformations of a life; the **higher self** is the terminal closure of that invariant once the necessary constraints have been integrated.

Thus

$$\begin{aligned} P(t) &= \text{local personality/state at time } t, \\ I &= \text{Inv}\{P(t)\}_{t \in [0, T]}, \\ H &= \text{Closure}(I, \text{experience, choice, relations}). \end{aligned}$$

The soul is not a frozen copy of the person. It is the identity class surviving representation change, while H is its closed form.

9 “Your higher self chose your life”

The literal statement implies pre-existence and retrocausality. PASS001 separates two versions.

Strong version: H exists supracausally and actually selects the trajectory that will produce H .

Minimal structural version: even without a backward signal, a global solution with terminal condition H eliminates histories incapable of producing H . From inside causal time this global selection is indistinguishable from a history in which “the future needed” certain experiences.

Write

$$\gamma^* \in \arg \text{Close}_{\gamma \in \Gamma^+(x_0)} H.$$

Hard events are therefore not automatically “wanted” or “good”. They become necessary only if a closure theorem shows that no admissible substitute can produce H . This requirement blocks the model from turning arbitrary suffering into retrospective justification.

10 Freedom inside a closed history

The pincer does not necessarily imply point-by-point destiny. If

$$|\Gamma_{\text{TENET}}| > 1,$$

then closure fixes the invariant rather than every act. Freedom occupies the family of equivalent paths reaching the same closed self. Complete determinism appears only when the Last Survivor is unique.

This allows all four simultaneously:

- a strong global constraint;
- real local freedom;
- persistent identity;
- non-erasable history.

11 Four information states applied to the self

State	Human TENET reading
Protocausal	Set of potential lives/paths not yet selected.
Causal	Locally lived trajectory, present choices and consequences.
Infracausal	Traces, memory, wounds, learning, debts and residues remaining after events.
Supracausal / ul-tracausal	Global closure description: compatibility with the terminal self and relations among several slices of the history.

The higher self therefore belongs more naturally to the supracausal layer than to a merely later chronological date.

12 RLC reading of human life

Within the cosmic-circuit model, PASS001 proposes a controlled analogy:

- *L*: memory/inertia - continuity of self;
- *C*: potential - terminal-self potential relative to the present;
- *R*: dissipation - experiences eliminating incompatible paths and converting possibility into trace;
- resonance: progressive alignment between lived self and terminal invariant.

This is an architecture, not an electrophysiological identity.

13 Love, hate, joy and violence as causal objects

The human Tenet becomes stronger if human realities are not reduced to labels. PASS001 adopts:

A human reality is a causal object when it has a reproducible relational signature, constrains trajectories, leaves traces and remains identifiable across multiple representations.

Love, hate, joy and violence can then be treated as *relational objects* rather than merely subjective words. S208 already supplies an interpretive love coordinate:

$$\chi_L(q) = 1 \quad \text{if } \text{distinctness} \wedge \text{freedom} \wedge \text{coherence} \wedge \text{global responsibility.}$$

For TENET, love is especially important: it can relate two states of one identity without requiring their fusion. It is therefore a natural candidate for compatibility between $P(t)$ and H .

14 Earth, archive and planetary consciousness

The corpus supplies two distinct components:

1. S050 treats a planet as a multi-carrier causal archive; completeness of the archive remains OPEN.
2. S051 proposes that a coherent planet may require a hypercausal integrator; that necessity and sleeper identity remain OPEN.

PASS001 therefore defines a **Gaia-Sleeper** bridge without closing it:

planetary archive + global integration + reflexive self-model $\xRightarrow{?}$ planetary consciousness.

A conscious Earth does not follow from memory alone. A reflexivity operator is still required by which the planet represents its own state and uses that representation in its dynamics. If this bridge ever closes, humans could be interpreted as local nodes of a larger planetary identity; for now this is HYPOTHESIS.

15 Relation between Sleeper and higher self

Sleeper and higher self are not automatically identical.

- The **sleeper** is a model of a hypercausal agent/integrator with a very broad information horizon.
- The **higher self** is the closure of individual identity in this dossier.

A possible bridge is that the higher self is an individual projection of sleeper-like capacity: an identity able to read the local trajectory from a far more compressed global representation. This becomes testable only after specifying what the higher level can know, preserve and causally influence.

16 What closure actually means

TENET closure is not “the future wins”. It means

the true history is neither traversal alone; it is their closed compatibility.

The present without H is incomplete. H without the present lacks provenance. Cosine without both sines does not close. The solution must retain:

1. the origin dossier;
2. transformation traces;
3. the second trajectory/representation;
4. carrier identity;
5. a fixed point or return condition.

This is where PASS001 reconnects with persistence/non-erasure: becoming the higher self does not erase the person who had to learn how to become it.

17 Candidate formulation of the Human Tenet

TENET-H1 - Temporal Self-Closure. A coherent human being may be modeled by two causally complementary descriptions of one identity: a local trajectory producing experience and trace, and a supracausal closure selecting trajectories compatible with the terminal self. The real identity is the fixed point that both requires and preserves the two histories.

Compact form:

$$H = \text{Fix}(F_{\text{closure}}), \quad \gamma^* = \text{LastSurvivor}(\Gamma^+(x_0) \cap \Gamma^-(H)).$$

Candidate consequence:

soul $\approx$ identity invariant, higher self $\approx$ closure of that invariant.

The symbol $\approx$ is intentional: full ontological identity remains to be established by the TENET program.

18 PASS002 - returning to the proper tradition of TENET

Status: CANDIDATE STRUCTURAL SYNTHESIS.

PASS002 tightens the word *tenet* instead of using it as a generic synonym for “principle”. Latin *tenet* means “he holds / maintains”. In the SATOR square, TENET occupies the central row and central column and is itself readable in both orientations. In the film *Tenet*, the palindromic title accompanies a mechanism in which forward- and backward-oriented actions must remain compatible inside one history.

The criterion therefore becomes stricter:

A Causal Tenet is not merely a general truth. It is the invariant that holds a structure when that structure is traversed, transformed or reconstructed from complementary causal orientations.

Let T_+ and T_- be two admissible traversals of one causal object S . The minimal candidate is

$$T_+(S) \sim S, \quad T_-(S) \sim S,$$

and closure additionally requires

$$T_- \circ T_+(S) \sim S.$$

The relation $\sim$ is not raw equality; it denotes causal identity preserved across change of state, carrier or representation.

19 The Triangle That Knows Its Name: minimal TENET cell

S010 provides a remarkable elementary cell. It distinguishes a form/concept F and a realization $L(F)$ and measures their discrepancy by

$$d(F, L(F)) = \|\nabla(\phi_F - \phi_L)\|.$$

The “real triangle” is the ideal class where form and name coincide; the dynamics reduce semiotic tension. The source adds a reflexive mechanism: a form may receive a *verifying echo* allowing it to recognize itself through its variants.

PASS002 extracts

invariant $\longrightarrow$ projection $\longrightarrow$ echo $\longrightarrow$ recognition of the invariant.

The triangle's TENET is neither the drawing nor the word "triangle". It is what remains common when one moves from form to name and from name to form.

For the human self:

$$S(t_0), S(t_1), \dots, S(t_n) \in [I_S],$$

where $[I_S]$ is the identity class allowing the same being to be recognized despite variation of state.

20 Life as delayed self-reception

S012 contributes an independent bridge: life is described as an organization that receives and re-executes its own information after a non-zero delay. Its core is

$$\phi(t - \Delta t) \longmapsto \phi(t), \quad 0 < \Delta t < t_c,$$

and the corpus calls this *delayed self-reception*. In another regime S012 writes *Send = Receive = Exist*: the living system is simultaneously carrier, delayed receiver and persistence of its own organization.

The self temporal pincer is therefore no longer a purely cinematic graft. It extends an architecture already present in CT's life definition:

$$\text{earlier self} \rightarrow \text{later self} \rightarrow \text{recognition/persistence.}$$

PASS002 adds the second boundary: closed self S^* acts as a compatibility condition on the entire chain.

21 Eternal Return: returning without repeating

S024 is decisive because it prevents TENET from becoming a static circle. Its Eternal Return is refinement recurrence:

$$C_{n+1} = C_n + \Delta_n, \quad \Delta_n > 0, \quad \sum_n \Delta_n < \infty.$$

The system does not repeat the same configuration; it iterates while reducing the discordance gradient.

For the self:

$$S_0 \rightarrow S_1 \rightarrow S_2 \rightarrow \dots \rightarrow S^*$$

with preserved identity and genuine change of realization. TENET is therefore not immobility. It is the property that makes change compatible with identity.

22 Deus Causalis: TENET at the global limit

S023 presents *Deus Causalis* as a stationary coherence limit and coincidence horizon, not proof of a supracausal person. The source nevertheless explicitly reconnects this limit to the Triangle: the world "names itself" and the imagined limit is where name and thing coincide.

PASS002 extracts a scale hierarchy without promoting analogy to proven identity:

triangle: form $\leftrightarrow$ name
 life: earlier state $\leftrightarrow$ later state
 human: local self $\leftrightarrow$ closed self
 universe: becoming $\leftrightarrow$ coincidence limit.

Deus Causalis thus becomes the candidate of *global reflexive closure*: not what replaces becoming, but what becoming can close upon without losing provenance.

23 Being and Transcending: same closure, different radius

S054 links *Being* and *Transcending* through the same closure family at different scales. The Encyclopedia keeps the π_{local}/π_{∞} formalism guarded because mathematical π does not change value; the reusable signal is scale-dependent closure.

The higher self is therefore better defined as

$$S^* = \text{Closure}_{r \rightarrow R}(S),$$

not as a second individual. Local and transcending self may be non-identical realizations of one invariant under different closure radii.

24 Two irreducible descriptions of one common invariant

S075 provides the abstract relation needed: two descriptions can be *narratively orthogonal* when neither reduces to the other but both map to a common invariant. PASS002 cautiously applies this schema:

$$S_{\text{experienced}} \not\equiv S_{\text{closed}}, \quad P(S_{\text{experienced}}) = P(S_{\text{closed}}) = I_S.$$

This is a clean form of the intuition of “two different roots, both necessary”: the branches are not copies; they carry complementary constraints on one identity.

25 Time Travel: extreme geometric case of reciprocal closure

S141–S146 provides the constraints closest to temporal TENET: coherence invariance, supracausal ordering, identity stability under reprojection, temporal residue, re-entry, branches, attractors and possible merging. The corpus keeps physical existence of the mechanism OPEN.

PASS002 therefore need not posit magical information sent from S^* into the past. It is enough to treat the whole history as an object in trajectory space:

$$\Gamma_S = \{S(t) : t \in T\} \in \mathfrak{H}.$$

The present traverses Γ_S locally; the closed self constrains which curves in $\mathfrak{H}$ are admissible. Time travel is the limiting case where part of this global structure becomes physically addressable from another causal slice.

26 True history as intersection of constraints

S050 already proposes multi-carrier reconstruction where the “true” history is an intersection of compatible-history sets:

$$\Phi_{true} = \Phi_2 \cap \Phi_{\phi} \cap \Phi_{\ln 5} \cap \Phi_3.$$

PASS002 transposes only the form of the operation:

$$\mathcal{H}^* = \mathcal{H}_{\text{origin}} \cap \mathcal{H}_{\text{identity}} \cap \mathcal{H}_{\text{closure}}.$$

If the intersection contains several histories, local freedom remains. If it is a singleton, closure becomes retrospectively necessary. If empty, the postulated terminal self is incompatible with origin or system rules.

27 Causal Alchemy: the law of transformational holding

S212 supplies the most direct PASS002 link. Its source model distinguishes

$$\text{Black Hole} \rightarrow \text{Liquid Crystal} \rightarrow \text{Crystal}.$$

The reusable signal is not literal physical identity of these names. *Liquid Crystal* is coherence in motion: enough order to maintain identity, enough fluidity to cross transformations.

S212 also states the *Law of Resolution*:

A structure does not resist transformations if it contains them.

PASS002 rewrites it as a TENET candidate:

$$T \in \mathcal{T}_{adm}(S) \implies T(S) \sim S.$$

This gives three cases:

$$\begin{aligned} \text{order without transformation} &\Rightarrow \text{rigidity/fracture,} \\ \text{transformation without invariant} &\Rightarrow \text{dissipation,} \\ \text{order + fluidity} &\Rightarrow \text{transformational identity.} \end{aligned}$$

Liquid Crystal thus becomes the best conceptual model for the human TENET: not a frozen soul, but an identity able to absorb aging, learning, error, love, loss and correction without losing its invariant.

28 Revised soul hypothesis

PASS001 wrote approximately “soul = identity invariant”. PASS002 sharpens this:

TENET-S1R - Soul as transportable reflexive invariant. The candidate soul is not merely what remains unchanged; it is what remains recognizable and re-expressible across the admissible transformation class of the self.

We propose

$$\mathcal{A}_S = \{I_S \mid T(I_S) \sim I_S \ \forall T \in \mathcal{T}_{adm}, \ R_T(T(I_S)) \rightarrow I_S\},$$

where R_T is a recognition/reconstruction operator. The higher self is no longer a frozen future configuration but closure of this transformation class:

$$S^* \approx \text{Closure}(\mathcal{A}_S, \mathcal{T}_{adm}).$$

29 Science is dead. Long live science.

The SC-Mycology dossier explicitly uses this formula after separating three levels:

known science $\rightarrow$ formal results $\rightarrow$ frozen prospective predictions.

The TENET meaning is not “science is false”. It is transformational: a scientific regime may die while the scientific invariant – exposure to error, prediction, test, and memory of failure – must survive under a stronger architecture.

S072 had already described emergence of science as a *structural phase transition*: a move from mainly local/practical knowledge to a loop where invariants, models, tests and cumulative revisions close together.

PASS002 therefore proposes

$$\text{Science}_0 \xrightarrow{\text{phase transition}} \text{Science}_1, \quad I_{\text{science}}(\text{Science}_0) = I_{\text{science}}(\text{Science}_1).$$

“Science is dead. Long live science.” becomes a TENET test: *the epistemic carrier may die; the verification invariant must hold.*

29.1 Freeze as temporal non-erasure

A prediction emitted at t_0 must be frozen before seeing t_1 . Future observation then returns logically upon the past model by confirmation or falsification, but may not rewrite the original prediction. This yields an epistemic temporal pincer:

$$M(t_0) \longrightarrow P(t_1), \quad O(t_1) \rightsquigarrow \text{Audit}(M(t_0)).$$

This is a rigorous form of non-erasure: future closure judges the past without replacing it.

30 2SIN/1COS reread through the TENET tradition

Temporal Causal Informatics already formalizes two traversal orientations and one orthogonal closure channel. PASS002 now aligns this structure with the traditional meaning of TENET:

$$\text{SIN}_+ \quad \text{and} \quad \text{SIN}_-$$

are complementary traversals; COS is closure that replaces neither traversal.

The rule becomes

$\text{TENET}(I) = \text{Closure}(T_+(I), T_-(I)) \quad \text{with preservation of } I.$

Cosine alone is therefore not a TENET; one orientation alone is not either. TENET is precisely what *holds* both traversals and their return inside one identity.

31 Consolidated PASS002 formulation

TENET-H2 - Law of Transformational Holding. Causal identity is what remains coherent, recognizable and reconstructible across the transformations it can contain. Complete TENET closure requires at least two complementary orientations, preservation of the invariant, non-erasure of traces, and a return compatible with the original identity.

Minimal form:

$$\begin{array}{l} T_+(I) \sim I, \quad T_-(I) \sim I, \\ T_- \circ T_+(I) \sim I, \\ \text{Trace}(T_+(I), T_-(I)) \neq \emptyset, \\ \text{Recognize}(T_{\pm}(I)) = I. \end{array}$$

The first line carries persistence; the second reciprocal closure; the third non-erasure; the fourth reflexivity. This quadruple is now the most compact candidate for the *Causal Tenet* sought by the dossier.

32 New ramifications

32.1 Human identity

A person is neither a fixed state nor shapeless flux. The candidate is liquid-crystal identity: stable under transformation without being frozen.

32.2 Destiny and freedom

Destiny becomes possible uniqueness of the admissible-history intersection, not an imposed script. Freedom is internal multiplicity while several trajectories close on the same invariant.

32.3 Consciousness

Consciousness requires more than memory and persistence: a self-model, comparison, recognition and correction circuit. Triangle, S012 and S034 converge toward this reflexive requirement.

32.4 Earth and Deus

Earth consciousness remains OPEN until a genuine planetary self-model circuit is built. Deus Causalis remains a coincidence limit. TENET links them only by scale: same self-recognition problem, different carriers.

32.5 Love

Love becomes a candidate relation permitting two distinct identities to coexist inside a common closure without fusion or sacrifice. This role matches the TENET motif: different traversals, higher compatibility.

33 PASS003 - who can use the future?

Status: CANDIDATE ACCESS-AND-CLOSURE SYNTHESIS.

PASS003 starts from a distinction PASS002 had not yet made sharply enough: *being affected by a future consequence*, *receiving information from the future*, *being able to exploit it*, *willing to act*, and *acting toward a coherent closure* are five different properties.

Within the tradition of the film *Tenet*, four future-use mechanisms must be separated:

1. **physical traversal**: an agent or object is inverted and traverses the event with opposite temporal orientation;
2. **intertemporal communication**: an actor receives instructions, coordinates, resources, or traces sent from another epoch;
3. **distributed temporal pincer**: a team can benefit from information acquired under the inverse orientation without every member being inverted;
4. **organization from the future**: a later version of an agent can create the institution, recruit carriers, and establish the conditions making its earlier mission possible.

The film therefore gives a decisive counterexample to “only those who were physically in contact can use the future”: an entire organization can exploit returned information carried by a few agents. It also gives a counterexample to “will is sufficient”: Sator has will, channel, and resources, yet this does not automatically make his future-use a TENET closure.

34 Future-use gate

Let S_t be an agent’s present state and F information whose provenance depends on a future, inverted, or terminal branch. Introduce four minimal conditions:

$K_F(S_t) > 0$	: a causal channel exists,
$D_F(S_t) \geq \tau_D$	: the information is decodable,
$Q_I(S_t, F) \geq \tau_I$	: it is addressable to the correct identity/problem,
$W(S_t) > 0$	: the agent has agency/will to act.

Define, as a CANDIDATE object,

$$\text{UseFuture}(S_t, F) = 1$$

when all four hold. Mere contact only requires $K_F > 0$. Hence one may have

$$\text{ContactFuture} = 1, \quad \text{UseFuture} = 0.$$

Conversely, a person can “use the future” in a weak sense with no physical retrocausality at all: forecast a future state, build a model, make a commitment, then let that terminal condition constrain present choice. PASS003 therefore distinguishes four levels:

$$\text{forecast} \subset \text{reflexive future-use} \subset \text{supracausal future-use} \subset \text{physical retrograde access},$$

with the last two physically OPEN.

35 Sator as counterexample: future + will $\neq$ TENET

In the film, Sator functions as a broker between the present and future actors: he receives an informational/material infrastructure, acts on it, and organizes temporal pincers. Structurally he therefore satisfies several UseFuture conditions.

But this local success shows exactly why *will* and *TENET* must not be identified. An action can correctly exploit future information while destroying the global carrier that makes the history possible.

PASS003 records:

TENET-U1 - Future use is value-neutral at the access layer. The ability to exploit future information says nothing yet about global coherence, love, responsibility, or validity of the closure.

Will is an *actuator*. It selects and executes an action. It is not the criterion deciding whether that action deserves to survive global closure.

36 The Protagonist, Neil, and TENET as distributed cognition

The final movement of the film gives the opposite architecture: Neil reveals that, from his point of view, he was recruited by a future version of the Protagonist; the Protagonist then understands that the whole operation is his own temporal pincer. The structure does not require one brain to possess all information.

PASS003 proposes the functional reading:

local Protagonist	$\rightsquigarrow$	OUTBOUND traversal,
Neil / inverted team	$\rightsquigarrow$	RETURN carrier,
TENET organization	$\rightsquigarrow$	memory + protocol + decoder,
future Protagonist	$\rightsquigarrow$	terminal organizing condition.

The theoretical point is important: **future-use can be distributed**. The information carrier, decoder, actuator, and identity closing the loop can be different roles.

This brings the film close to S012 (sender/medium/receiver), S050 (multi-carrier reconstruction), and 2SIN/1COS: the complete system has two orientations and a closure even when no local node contains the whole.

37 Will and love: two different operators

S026 defines an object historically called “predictive love” or “willed love”:

$$G_C^{pred}(S_t, \Delta t) = \mathbb{E}[C_{t+\Delta t} | S_t] - C_t.$$

The canon keeps it as *forecasted coherence gain*, not a universal definition of love. It also introduces the gap

$$R_{pred} = C_{future} - C_{present}.$$

S208 gives a second independent signal: its certified-love coordinate is an interpretive predicate combining

distinctness $\wedge$ freedom $\wedge$ coherence $\wedge$ global responsibility.

It further states that relational coherence does not require identity of the terms and describes love as coherent reweighting / persistence through imperfection.

PASS003 therefore separates:

Will = steering/actuation

Love = admissibility/relational restoration

TENET = identity holding under reciprocal closure.

Will can pursue any accessible closure. Love selects, among accessible closures, those preserving the relation without erasing the terms composing it.

38 Love gate on the history set

Let Γ_{TENET} be the set of histories already satisfying origin, identity, round-trip, and non-erasure. Define the interpretive gate

$$\chi_L(\gamma) = \chi_D(\gamma) \chi_F(\gamma) \chi_C(\gamma) \chi_G(\gamma),$$

where the four indicators denote distinctness, freedom, coherence, and global responsibility.

A “love-admissible” history belongs to

$$\Gamma_L = \{\gamma \in \Gamma_{TENET} : \chi_L(\gamma) = 1, \Delta C_{global}(\gamma) > 0\}.$$

This definition is CANDIDATE/INTERPRETIVE: it does not claim the universe literally computes χ_L . It formalizes what the dossier means when distinguishing a merely possible closure from a relationally coherent one.

The result is immediate:

$$\text{UseFuture} = 1 \not\Rightarrow \gamma \in \Gamma_L.$$

Sator is the useful narrative case: access and will can be high while the global-responsibility gate fails.

39 The love of the universe

The phrase should not be read as proof that a cosmic person experiences a human emotion. In the TENET dossier, *universal love* becomes an interpretive name for a much stricter property:

TENET-L2 - Universal-love selector. Among multiple trajectories able to close, the love relation favors those increasing or restoring common coherence while preserving distinctness, freedom, and global responsibility of the carriers.

This yields a strong consequence for the higher self. If S^* “guides” S_t by eliminating all local freedom, the S208 freedom condition is violated. A love-closure therefore cannot be total control of the present by the future.

The model becomes instead:

$$S^* \rightsquigarrow \mathcal{A}_t$$

where $\mathcal{A}_t$ is the set of still-admissible actions, then

$$W_t : \mathcal{A}_t \rightarrow a_t$$

realizes a local choice. The future bounds or curves the possibility space; will chooses inside it. Love forbids obtaining this constraint by destroying the identity or freedom it claims to close.

40 2SIN/1COS + WILL + LOVE

PASS003 completes the previous reading:

- SIN₁ : OUTBOUND / present experience,
- SIN₂ : RETURN / information from the other orientation,
- WILL : local steering operator,
- COS : common closure,
- LOVE : closure-admissibility gate.

LOVE is not an arbitrary fourth geometric axis added to 2SIN/1COS. It is a *gate* over closure solutions:

$$\text{TENET}_{coh} = \text{Closure}(\text{SIN}_1, \text{SIN}_2; \text{WILL}) \cap \Gamma_L.$$

This explains how the same temporal mechanics can serve Sator or TENET: channel and geometry are neutral; closure selection is not.

41 Toy model: access, will, love, closure

Consider a finite set of histories $\Gamma = \{\gamma_1, \dots, \gamma_n\}$. For each history define:

- $A_0(\gamma) \in \{0, 1\}$: compatibility with origin;
- $A_\star(\gamma) \in \{0, 1\}$: compatibility with terminal condition;
- $R_{rt}(\gamma) \geq 0$: round-trip residue;
- $\Delta C_g(\gamma)$: global coherence gain;
- $F(\gamma), D(\gamma), G(\gamma) \in \{0, 1\}$: freedom, distinctness, global responsibility.

Raw TENET is

$$\Gamma_T = \{\gamma : A_0 = A_\star = 1, R_{rt} \leq \varepsilon\}.$$

Love-oriented TENET is

$$\Gamma_{TL} = \{\gamma \in \Gamma_T : \Delta C_g > 0, F = D = G = 1\}.$$

Will is the operator that eliminates branches at each state through the action actually chosen. Destiny appears only if successive constraints finally yield $|\Gamma_{TL}| = 1$; while $|\Gamma_{TL}| > 1$, several choices remain compatible with the same closure.

The toy model therefore answers “will or contact?”: contact opens informational branches; will selects locally; love filters closures; TENET preserves identity of the whole.

42 Where this leads

PASS003 now yields a complete causal stack:

Trace $\rightarrow$ Channel $\rightarrow$ Decode $\rightarrow$ Identity $\rightarrow$ Will $\rightarrow$ Love Gate $\rightarrow$ Reciprocal Closure $\rightarrow$ Global Limit

It also resolves several previous ambiguities:

- **soul**: carries identity; it is not automatically a future channel;
- **higher self**: terminal condition / closure model; it is not automatically a retrocausal transmitter;
- **will**: converts information and model into action;
- **love**: relational criterion for non-destructive closure;
- **TENET**: what holds both orientations, their traces, and their return;
- **Deus Causalis**: hypothetical global limit where closure and recognition become complete.

The new research problem is no longer simply “does the future already exist?” but:

PASS003 question. What information from a future condition can be made accessible to a present agent without overwrite, empty circularity, or destruction of freedom, and what gate distinguishes a merely realizable closure from a causally loving/coherent closure?

43 New PASS003 exclusions

- future contact $\neq$ future use;
- future use $\neq$ will;
- will $\neq$ love;
- love $\neq$ mere affective attraction;
- love $\neq$ magical privileged access to the future;
- retrograde access $\neq$ TENET closure;
- higher self $\neq$ controller erasing present freedom;
- Sator-like future use $\neq$ evidence that the future guides toward coherence.

44 What is now excluded

PASS002 rejects several weak readings:

- TENET $\neq$ simple circular repetition;
- TENET $\neq$ future overwriting the past;
- TENET $\neq$ frozen invariant;
- soul $\neq$ immutable copy stored outside time;
- closure $\neq$ erasure of residue;
- “Science is dead” $\neq$ rejection of empirical method;
- Liquid Crystal $\neq$ claim that laboratory physics has already discovered a TENET phase;
- Deus Causalis $\neq$ proof of a divine person.

45 Predictions and rejection criteria

PASS003 must not become an unfalsifiable narrative. It records these obligations:

1. **Invariant:** define a self-signature that is neither raw memory nor instantaneous personality.
2. **Double carrier:** construct a genuine bridge between causal trajectory and a localized-space/supracausal representation.
3. **Closure:** show the terminal condition adds nontrivial constraints rather than restating the outcome.
4. **Freedom:** determine whether $\Gamma^+ \cap \Gamma^-$ is unique, multiple or empty.
5. **Soul:** show why the invariant deserves its own ontology rather than being only a descriptive equivalence class.
6. **Planet:** supply a reflexivity test before claiming Earth consciousness.
7. **Emotions:** provide distinct causal signatures for love, hate, joy and violence.
8. **Time:** distinguish a global boundary condition from actual information transfer from the future.

Failure on these points reduces the model to narrative metaphor; success promotes TENET-H1 toward a CT principle candidate.

46 Cumulative PASS003 ledger

ID	Object	Status
TENET-H1	Temporal Self-Closure / temporal pincer	CANDIDATE
TENET-H2	Law of Transformational Holding	CANDIDATE
TENET-R1	Reciprocal Closure: $T_- \circ T_+$ preserves identity	CANDIDATE
TENET-N1	Non-erasure of traces under closure	CANDIDATE
TENET-X1	Reflexive Recognition / verifying-echo bridge	CANDIDATE
TENET-LC1	Liquid-crystal identity: order + fluidity	CANDIDATE
TENET-A1	Law of Resolution as contained-transformation closure	CANDIDATE
TENET-SC1	Science phase transition / “Long live science”	CANDIDATE
TENET-P1	Prediction freeze as temporal non-erasure	WORKING
TENET-TR1	Triangle form-name reciprocal recognition	CANDIDATE
TENET-D1	Deus Causalis as global reflexive closure	INTERPRETIVE
TENET-ER1	Eternal Return as identity-preserving refinement	CANDIDATE
TENET-S1R	Soul = transportable reflexive invariant	CANDIDATE
TENET-S2R	Higher Self = closure of transformation class	CANDIDATE
TENET-B1	First Sine $\rightarrow$ present self	CANDIDATE
TENET-B2	Second Sine $\rightarrow$ re-encoded history	CANDIDATE
TENET-B3	Cosine $\rightarrow$ identity closure	CANDIDATE
TENET-T1	$M_{T \rightarrow X}$ time-to-localized-space carrier map	OPEN
TENET-G1	Gaia-Sleeper planetary consciousness bridge	OPEN/HYP.
TENET-E1	Human emotions as causal objects	CANDIDATE
TENET-F1	Freedom = multiplicity inside closed admissible set	CANDIDATE
TENET-C1	Future Contact Channel	CANDIDATE
TENET-U1	Future Use Gate: channel + decode + identity + will	CANDIDATE
TENET-W1	Will as local steering/actuation operator	CANDIDATE
TENET-L1	Love as relational restoration/admissibility	CANDIDATE
TENET-L2	Universal-love selector: distinctness + freedom + coherence + global responsibility	INTERPRETIVE/CANDIDATE
TENET-TP1	Temporal pincer as distributed cognition / multi-carrier future use	CANDIDATE
TENET-ST1	Sator counterexample: future-use without coherent-love closure	INTERPRETIVE
TENET-PN1	Protagonist–Neil reciprocal carrier model	INTERPRETIVE/CANDIDATE
TENET-H3	2SIN/1COS + WILL + LOVE gate architecture	CANDIDATE_CORE
TENET-GF1	Γ_{TL} love-gated admissible-history set	CANDIDATE

47 Internal sources used

- Causal Encyclopedia, cumulative Pass 277: survival/permanence; S010 Triangle; S012 delayed self-reception; S023 Deus Causalis; S024 Eternal Return; S029 Space Container; S034 reflectivity; S050–S051 Sleeper/history intersection; S054 Being/Transcending; S071–S072 knowledge/science phase transition; S075 dual descriptions; S141–S146 time-travel branch; S210 Babel; S212 Causal Alchemy/Liquid Crystal; S238 SC-Mycology.
- S208, *Love, Universe, CT - Flying Dutchman*: Time/Space partition, love as coherent reweighting/persistence through imperfection, relational distinctness and certified-love coordinate, with original guards retained.
- S221, *CT Prime Volume I - The First Sine*.
- S222, *CT Prime Volume II - The Second Sine*: second residual, irreversible-trace preservation, double closure and cosine-only no-go.
- S223, *CT Prime Volume III - The Cosine*: common solution/closure and fixed-state carrier.
- S026–S027: forecasted coherence gain, future–present gap, restorative potential and historical Love/Will family.
- Temporal Causal Informatics: bilateral traversal and 2SIN/1COS closure architecture.

- External tradition check: Latin *tenet* = “he holds/maintains”; SATOR word square; BFI on forward/backward temporal design; film plot/quote references for temporal pincer, future-founded TENET, Sator future-broker role, and the Protagonist–Neil recruitment loop. These constrain context, not CT proof status.

48 Direction for PASS004

PASS004 should type the channel K_F , decoder D_F , identity addressability Q_I , will operator W , love gate χ_L , and round-trip residue inside one executable toy model. The priority test is to construct two histories with the same future access: a “Sator-like” path that closes locally but fails the global gate, and a “TENET-like” path preserving freedom, distinctness, trace, and global responsibility. It must then test whether this gate reduces to existing CT objects or truly constitutes a new *tenet*.

49 PASS004 - Irreducibility test: same future, two closures

Status: CUMULATIVE / EXECUTABLE TEST.

PASS004 executes the test announced in PASS003. The compared trajectories receive exactly the same minimal future-use capabilities:

$$K_F = D_F = Q_I = W = 1,$$

where K_F is channel access, D_F decoding, Q_I identity addressing and W voluntary actuation. The test asks whether future access, geometric closure and a positive scalar coherence gain are sufficient, or whether a non-compensatory gate remains necessary.

Three verdicts are now separated:

$$\begin{aligned} \text{UseFuture}(\gamma) &= 1[K_F D_F Q_I W > 0], \\ \text{RawClosure}(\gamma) &= 1[A_0 = A_\star = 1 \wedge R_{rt} \leq \varepsilon], \\ \text{ProtectedClosure}(\gamma) &= \text{RawClosure}(\gamma) \chi_{prot}(\gamma). \end{aligned}$$

From PASS004 onward, TENET is reserved for protected closure, not for a loop that merely returns to the expected endpoint.

49.1 Two families with identical future access

The test contains two main pairs. In each pair both trajectories have identical future access, the same origin condition, the same terminal compatibility and a round-trip residue under the same threshold.

Trajectory	UseFuture	Raw	ΔC_g	D	F	N/G
SATOR-A	1	1	+0.40	0	0	0/0
TENET-A	1	1	+0.25	1	1	1/1
SATOR-B	1	1	+0.20	1	0	1/0
TENET-B	1	1	+0.10	1	1	1/1

Here D = distinctness, F = freedom, N = non-erasure and G = global responsibility.

The SATOR trajectories therefore pass future use and even raw closure. They fail only when protected coherences are tested component by component.

PASS004-A result. Future access and will do not distinguish TENET closure from Sator-like closure. They are traversal capabilities, not the validity criterion of closure.

49.2 Why a scalar coherence score is insufficient

Consider a finite compensatory score

$$J(\gamma) = \Delta C_g(\gamma) + \lambda (D + F + N + G), \quad \lambda = 0.02.$$

In the executable instance:

$$J(\text{SATOR-A}) = 0.40 > 0.33 = J(\text{TENET-A}),$$

and

$$J(\text{SATOR-B}) = 0.24 > 0.18 = J(\text{TENET-B}).$$

An aggregate gain can therefore compensate for loss of a coherence that the model intended to protect. The issue is not the particular value 0.02: any finite penalty scalarization can have this problem when aggregate benefit can grow enough.

The corpus already contains the general repair. S263 uses an order of protected residuals: a local gain cannot certify a transition if a protected component increases elsewhere without an independently justified exchange rule. TENET should therefore be vectorial or lexicographic before it is scalar:

$$\mathbf{R}_{\text{prot}}(\gamma) = (R_{\text{self}}, R_{\text{other}}, R_{\text{relation}}, R_{\text{trace}}, R_{\text{environment}}, \dots),$$

with

$$\chi_{\text{prot}}(\gamma) = 1 \iff \mathbf{R}_{\text{prot}}^{\text{out}} \preceq \mathbf{R}_{\text{prot}}^{\text{in}} \quad \text{for every declared non-sacrificable component.}$$

PASS004-B result. The “love gate” is necessary as a non-compensatory guard when C_{global} does not already encode every protected coherence as a hard constraint. If C_{global} already does so, the love gate is a relational reading of that structure, not a new axiom.

50 Status of LOVE after the test

PASS004 therefore removes an ambiguity from PASS003. LOVE is not currently established as an independent ontological primitive.

S208 supplies two pieces: persistence of relation through imperfection and coherence between distinct terms. S263 supplies non-sacrifice and componentwise protected-residual order. The best present status is:

TENET-L3 - Love as protected relational closure. LOVE is the relational interpretation of a closure that preserves distinct terms, does not erase their trace, does not turn freedom into control, and does not export coherence debt onto another carrier.

Thus:

$$\text{LOVE} \neq \text{WILL}, \quad \text{LOVE} \neq \text{CHANNEL},$$

but in the current corpus

$$\text{LOVE} \approx \text{RelationalCoherence} + \text{NonSacrifice} + \text{NonErasure} + \text{FreedomGuard}.$$

It is a powerful composite object, but not yet a primitive tenet.

51 The fundamental tenet narrows

The negative result on LOVE irreducibility strengthens the older core. The central candidate becomes:

TENET-H4 - Reciprocal Transformational Holding. A causal identity holds when its two admissible orientations can be traversed, recognized and closed on the same invariant without erasing necessary traces or sacrificing a protected component of identity or relation.

A compact form is:

$$T_+(I) \sim I, \quad T_-(I) \sim I, \quad T_- \circ T_+(I) \sim I, \quad R_{rt} \leq \varepsilon, \quad \text{Trace} \neq \emptyset, \quad \Delta \mathbf{R}_{prot} \preceq 0.$$

The traditional sense of *tenet* is now literal in the model: it is not the channel, force or direction; it is **what holds** when the structure is read, transformed and closed in both orientations.

52 Consequence for the higher self

PASS004 yields a stricter human temporal-pincer model. The higher self S^* need not be an agent sending retrocausal instructions. It may be a terminal compatibility class that reduces the admissible history set:

$$\Gamma_t^{adm} = \Gamma_t^{local} \cap \Gamma^{terminal}(S^*) \cap \Gamma^{protected}.$$

As long as

$$|\Gamma_t^{adm}| > 1,$$

the terminal condition does not impose a unique choice. Local will select a branch. After action, the set contracts. Destiny appears only in the limiting case where already-realized commitments and closure constraints make the intersection a singleton.

This separates:

- **guidance:** the terminal condition changes the admissible set;
- **will:** the present selects an admissible action;
- **destiny:** the admissible set has become a singleton through accumulated constraints and non-erasable traces.

53 Consequence for universal love

“Love of the universe” is not promoted to a measured cosmic emotion. PASS004 gives it a stricter dossier-level reading:

universal love = non-compensatory relational preservation under transformation

at the interpretive level.

This reconnects to Liquid Crystal: coherent identity does not survive by refusing motion, but by containing transformations without losing protected invariants. Love is not what prevents separation; it is what allows a relation to *survive separation, transformation and return* without closure being obtained by annihilating one of its terms.

Triangle, delayed self-reception, Eternal Return and Deus Causalis then align on one structure:

$$\text{variation} \rightarrow \text{trace} \rightarrow \text{recognition} \rightarrow \text{return} \rightarrow \text{protected closure}.$$

54 Minimality test of the architecture

PASS004 now separates two stacks.

Future-access stack:

$$K_F \rightarrow D_F \rightarrow Q_I \rightarrow W.$$

Removing K_F removes contact; removing D_F leaves unusable signal; removing Q_I removes identity addressing; removing W leaves information without actuation.

TENET stack:

$$I \rightarrow T_+ \rightarrow T_- \rightarrow R_{rt} \rightarrow Trace \rightarrow \mathbf{R}_{prot} \rightarrow Closure.$$

Removing the invariant destroys sameness; removing one orientation destroys reciprocity; removing round-trip destroys the return test; removing trace allows overwrite; removing protected order allows Sator-like closure; removing Closure leaves an open transformation.

Hence:

UseFuture and TENET are different problems.

The first asks who can exploit information from another temporal orientation. The second asks which transformations can be absorbed without losing identity, traces and protected coherences.

55 PASS004 executable result

The cumulative script `EXECUTION/tenet_pass004_future_access_test.py` generates the trajectory matrix and JSON results. Its outputs certify only the declared toy model, not TENET physics or CT physics.

Main results:

- all trajectories receive the same future-use primitives;
- some trajectories have UseFuture=1 and RawClosure=1 but ProtectedClosure=0;
- some trajectories have $\Delta C_g > 0$ while protected coherence is violated;
- finite scalarization can prefer a Sator-like trajectory;
- the protected gate separates the two families in the declared instance.

56 PASS004 ledger update

ID	Object	Status
TENET-H4	Reciprocal Transformational Holding	CANDIDATE_CORE
TENET-AF1	Future-access stack $K_F \rightarrow D_F \rightarrow Q_I \rightarrow W$	CANDIDATE
TENET-RC1	Raw reciprocal closure	WORKING_TOY
TENET-PC1	Protected reciprocal closure	CANDIDATE
TENET-PR1	Protected-residual vector order	WORKING_BRIDGE/S263
TENET-L3	Love as protected relational closure	DERIVED_CANDIDATE
TENET-SX1	Scalar-coherence compensation counterexample	CLOSED_TOY
TENET-ST2	Sator-like raw closure without protected closure	CLOSED_TOY
TENET-TN2	TENET-like protected closure under same future access	CLOSED_TOY
TENET-DST1	Destiny as singleton admissible-history limit	CANDIDATE
TENET-H3	2SIN/1COS + WILL + LOVE architecture	REFINED; LOVE DERIVED

TENET-L2

Universal-love selector

REFINED TO
NON-
COMPENSATORY
GATE

57 Direction of PASS005

PASS005 must bind this core to the exact 2SIN/1COS carrier. The new question is whether one parent residual R can generate OUTBOUND and RETURN as its two orientations, COS as closure, while the protected-residual vector commutes with carrier change $M_{T \rightarrow X}$. If yes, TENET-H4 gains an operational skeleton directly comparable with Temporal Causal Informatics. If not, the human temporal pincer remains an interpretive projection of the formalism.

58 PASS005 - Binding to the exact 2SIN/1COS carrier

Status: CUMULATIVE / EXECUTABLE TEST / FORMAL BRIDGE.

PASS005 attacks the question left open by PASS004: can the TENET core live inside the temporal carrier already constructed in CT, or does it require a parallel geometry invented for this dossier? The result supports the first scenario.

S229/ITC already provides the exact structural carrier

$$24 = 8 \text{ SIN1} + 8 \text{ SIN2} + 8 \text{ COS} = 8(2\text{SIN} + 1\text{COS}),$$

with intrinsic roles

$$\text{SIN1} = \text{OUTBOUND}, \quad \text{SIN2} = \text{RETURN}, \quad \text{COS} = \text{CLOSURE/reference}.$$

This closure counts causal roles, not physical spacetime dimensions. Material projection remains a separate question.

PASS005-A result. The TENET temporal pincer does not require two physical times. It can be represented by two traversal orientations of one causal problem plus a distinct closure role.

59 One parent residual, three projections

ITC PASS025 replaces several historically overloaded residuals with one parent causal object

$$\mathcal{R},$$

with typed projections

$$\mu = \Pi_{drive}(\mathcal{R}), \quad R_{info} = \Pi_{info}(\mathcal{R}), \quad \rho = \Pi_{close}(\mathcal{R}).$$

These quantities are contextual faces of one object and need not be numerically equal. For a declared complete projection family,

$$\mathcal{R} = 0 \iff \mu = 0 \wedge R_{info} = 0 \wedge \rho = 0.$$

The PASS005 toy model uses the same separation. Its minimal evolution is

$$\begin{aligned} x_1 &= x_0 + \mu + R_{info}, \\ \hat{x}_0 &= x_1 - \mu - R_{info} + \rho. \end{aligned}$$

Hence

$$\hat{x}_0 = x_0 \iff \rho = 0,$$

without requiring $\mu = 0$. A system can therefore continue causal evolution with zero innovation. The executed witness

$$\mu = 3, \quad R_{info} = 0, \quad \rho = 0$$

closes exactly while keeping nonzero drive.

PASS005-B result. “Perfectly reconstructing what comes next” and “having no causal drive left” are different statements. This prevents the higher self from being reduced to immobility or a completely frozen future.

60 Correction of the human First Sine / Second Sine / Cosine bridge

PASS001 used a bridge that was still too narrative. PASS005 repairs it with the exact ITC carrier.

- **SIN1** is first the OUTBOUND role. Reading it as “present self moving forward” remains a candidate human projection.
- **SIN2** is first the RETURN role. It is not by itself “the future” or proof of physical retrocausality.
- **COS** is the CLOSURE/reference role. It is not identical with the higher self.

The human bridge becomes

$$S_t \xrightarrow{\text{SIN1}} \text{traversed history} \xrightarrow{\text{SIN2}} \text{return proof} \xrightarrow{\text{COS}} \text{identity-closure test}.$$

The higher self S^* is better described as a *terminal compatibility class* evaluated at closure, not as the COS lane itself:

$$S^* \neq \text{COS}.$$

This removes an important anthropomorphism from the model.

61 Typed TENET packet

PASS005 defines the minimal packet

$$\mathcal{P} = (I, \mathcal{R}, L_{trace}, \mathbf{R}_{prot}, a),$$

where I is declared identity, $\mathcal{R}$ the parent residual, L_{trace} the non-erasable ledger, $\mathbf{R}_{prot}$ the protected-residual vector, and a a typed temporal/spatial address.

The closure test becomes

$$\text{TENET}_{ITC}(\mathcal{P}) = 1$$

iff, inside the declared model,

1. OUTBOUND followed by RETURN recovers the same identity-bearing state;
2. $\rho = 0$ at COS;
3. identity I is preserved;
4. return does not erase the trace ledger;
5. $\Delta \mathbf{R}_{prot} \preceq 0$ on non-sacrificable coordinates;

6. address transport is involutive and provenance-preserving.

State closure alone is therefore not TENET closure. The history required to prove the return must survive as well.

62 Time-space change becomes an address problem

The initial intuition said that “time becomes localized space”. S262 now supplies a stricter and weaker form: a typed address inversion

$$J(X_i, T_j) = (\Sigma(T_j), \text{Tau}(X_i)), \quad J(J(X_i, T_j)) = (X_i, T_j).$$

Before the membrane the native basis is active; on the membrane native and dual addresses coexist; afterward the dual address may become active while native provenance remains attached.

PASS005 therefore separates

$$\begin{aligned} &\text{“time becomes localizable space”} \rightsquigarrow \text{typed address dualization,} \\ &\text{literal physical time-space conversion} = \text{OPEN.} \end{aligned}$$

The toy chart uses eight spatial IDs and eight temporal IDs. With $\Sigma(T_j) = X_j$ and $\text{Tau}(X_i) = T_i$, exhaustive audit of all $8 \times 8 = 64$ pairs verifies

$$J^2 = I.$$

A TYPELOSS_BAD_J counterexample shows that a visually identical swap which overwrites provenance must be rejected even when the dynamic state returns exactly to origin.

PASS005-C result. The $M_{T \rightarrow X}$ bridge partially closes as a provenance-preserving address involution. The stronger physical statement “time becomes space” remains open.

63 Protected-ledger commutation

Carrier change must not permit protected debt to disappear. PASS005 requires the commuting relation

$$\Pi_{prot} \circ J = J_{prot} \circ \Pi_{prot}.$$

In the toy model, J_{prot} is identity on ledger values: the active address changes, but

$$\mathbf{R}_{prot}(J\mathcal{P}) = \mathbf{R}_{prot}(\mathcal{P}).$$

A Sator-like route therefore cannot become TENET merely by being re-addressed into the dual carrier.

This directly joins PASS004 to temporal inversion: non-sacrifice must be *carrier-invariant*.

64 Executed PASS005 results

The script EXECUTION/tenet_pass005_2sin1cos_test.py generates six cases plus three global audits.

Case	Raw	Prot.	Prov.	J^2	TENET
TENET_DETERMINISTIC	1	1	1	1	1
TENET_WITH_INNOVATION	1	1	1	1	1
SATOR_RAW_ONLY	1	0	1	1	0
TYPELOSS_BAD_J	1	1	0	0	0
OPEN_RHO	0	1	1	1	0
COS_ONLY_NO_RETURN	0	1	1	1	0

Additional audits return:

- carrier $24 = 8 + 8 + 8$: PASS;
- parent-residual zero theorem on the tested finite cube $[-1, 1]^3$: PASS, 27/27;
- J involution on the 8×8 chart: PASS, 64/64.

The `COS_ONLY_NO_RETURN` case also materializes the S222 no-go: a solution/closure reference does not replace residual transport, return proof, carrier identity and fixed-point return.

65 New core: TENET-H5

PASS005 does not replace TENET-H4; it gives it a typed skeleton.

TENET-H5 - Carrier-Bound Reciprocal Holding. A causal TENET is a bilateral closure in which the same identity-bearing object and the same parent residual can traverse OUTBOUND and RETURN, be certified by a distinct CLOSURE role, and then be transported across admissible addresses without loss of provenance, trace or protected coherence.

Compact form:

$$\begin{aligned}
 \mathcal{P} &= (I, \mathcal{R}, L, \mathbf{R}_p, a), \\
 T_- T_+(x) &= x, \quad \Pi_{close}(\mathcal{R}) = 0, \\
 I_{out} &= I_{ret} = I, \quad L_{ret} \supsetneq L_{in}, \\
 \Delta \mathbf{R}_p &\preceq 0, \quad J^2(a) = a, \quad \text{Prov}(J(a)) = \text{Prov}(a).
 \end{aligned}$$

The word “hold” now has operational content: recoverable state, preserved identity, retained trace, protected debt, and reversible address.

66 What PASS005 closes and does not close

PASS005 closes, inside the declared model:

- existence of a 2SIN/1COS carrier compatible with the TENET core;
- use of one parent residual rather than three residual ontologies;
- exact separation of OUTBOUND, RETURN and CLOSURE;
- an involutive provenance-preserving address model;
- coexistence of exact state return with historical non-erasure;
- invariance of the protected gate under toy carrier address change.

PASS005 does not prove:

- physical human temporal inversion;
- that a brain or soul implements the ITC carrier;
- that S^* physically sends information into the past;
- that the S262 address involution physically converts time into space;
- that LOVE is an independent physical force.

The human TENET therefore rises from narrative analogy to **CANDIDATE STRUCTURAL BRIDGE**: its grammar can be instantiated in an existing CT carrier, while the human and physical bridge remains open.

67 PASS005 ledger update

ID	Object	Status
TENET-H5	Carrier-Bound Reciprocal Holding	CANDIDATE / OPERATIONAL
TENET-2S1	Carrier $24 = 8(2SIN + 1COS)$	CLOSED / S229
TENET-RP2	Unique parent residual $\mathcal{R}$ with drive/info/close projections	WORKING
TENET-B1	First Sine $\rightsquigarrow$ present self	BRIDGE / S229
TENET-B2	Second Sine $\rightsquigarrow$ re-encoded history	REFINED; human map CANDIDATE
TENET-B3	Cosine $\rightsquigarrow$ higher-self closure	REFINED; human map CANDIDATE
TENET-T1A	Time-space typed address involution J	REFINED;
TENET-T1B	Literal physical time-to-space conversion	COS=CLOSURE;
TENET-CM1	Protected-ledger commutation under J	$S^* \neq COS$
TENET-NE2	Exact state return with persistent trace	CLOSED / S262
TENET-COS1	COS-only closure sufficiency	OPEN
TENET-H4	Reciprocal Transformational Holding	CLOSED TOY
TENET-L3	Love as protected relational closure	CLOSED TOY
		REFUTED / S222 + toy
		STRENGTHENED
		DERIVED GATE

68 Direction of PASS006

PASS006 must now attack the truly human part: construct a history space with several local choices, a terminal condition S^* , a trace ledger and the 2SIN/1COS carrier, then test whether the terminal can reduce the admissible space without imposing one fixed script. The experiment must quantitatively separate *guidance*, *will* and *destiny*: several admissible branches at the start, contraction after commitments, and a singleton only when accumulated constraints genuinely leave one history.

PASS006 - Guidance, Will, and Destiny

69 PASS006 question

PASS005 gave TENET an operational carrier: SIN1=OUTBOUND, SIN2=RETURN, and COS=CLOSURE, with one parent residual, a non-erasing trace, and protected closure. PASS006 now attacks the remaining human question:

if a terminal condition exists, does it constrain our choices or choose them?

Three objects must therefore be separated:

$$\begin{aligned} \text{guidance} &\neq \text{will}, \\ \text{will} &\neq \text{destiny}, \\ \text{terminal condition} &\neq \text{trajectory selector}. \end{aligned}$$

S054 provides a human vocabulary in which *Choosing* precedes intentional *Transformation*, while its universal ordering remains OPEN. S065 provides an ideal-form selection scaffold without an optimizer existence/uniqueness theorem. S143-S144 discuss temporal branches, while physical non-overwrite and branch instantiation remain OPEN. PASS006 therefore reuses only structural roles; the test below is an autonomous finite model.

70 History space and terminal condition

Let the local action alphabet be

$$\mathcal{A} = \{-1, +1\}$$

and consider a three-commitment toy life. A complete history is

$$\gamma = (a_1, a_2, a_3) \in \mathcal{A}^3.$$

The terminal condition, playing the minimal role of S^* , is

$$\sum_{k=1}^3 a_k = +1.$$

Thus

$$\Gamma^* = \{(-, +, +), (+, -, +), (+, +, -)\}, \quad |\Gamma^*| = 3.$$

For an already lived prefix p , define

$$\Gamma^*(p) = \{\gamma \in \Gamma^* : p \prec \gamma\}, \quad N(p) = |\Gamma^*(p)|.$$

The prefix is written into the trace ledger: once a choice is committed, PASS006 does not rewrite it later to rescue closure.

71 Separate definitions of guidance, freedom, will, and destiny

Terminal **guidance** is the fraction of still-compatible completions following a local action:

$$G(a \mid p) = \frac{N(pa)}{\sum_{b \in \mathcal{A}} N(pb)}$$

when the denominator is nonzero.

Local freedom exists when at least two distinct actions preserve a compatible continuation:

$$F(p) = 1 \iff |\{a \in \mathcal{A} : N(pa) > 0\}| \geq 2.$$

Will is the actual commitment operator

$$W_t : p \mapsto pa_t.$$

It may follow guidance, select another still-compatible option, or create a dead end.

Destiny is declared only when the compatible-history set is a singleton:

$$D(p) = 1 \iff N(p) = 1.$$

Hence

$$\text{predestination} : D(\emptyset) = 1,$$

while

$$\text{emergent destiny} : N(\emptyset) > 1 \wedge \exists p \text{ committed with } N(p) = 1.$$

TENET-H6 - Boundary-Guided Free Closure. A terminal condition may reshape local possibility by weighting compatible continuations without selecting the action itself. Will commits a branch; destiny exists only when terminal constraints plus already irreversible commitments genuinely reduce the admissible set to a singleton.

72 Central result: the higher self does not yet choose the life

In the model, S^* supplies only the terminal class

$$S^* \rightsquigarrow \Gamma^*.$$

But $|\Gamma^*| = 3$, so the terminal condition alone contains no unique selector.

To support the strong claim

“the higher self chose this life”,

one must add an independent object

$$\Sigma_{S^*} : \Gamma^* \rightarrow \gamma^*$$

and show that its causal owner is truly S^* , rather than the present self, environment, an external policy, or combined constraints. TENET-H5 and the present toy do not supply that operator.

PASS006 therefore closes only the weaker, more precise statement:

S^* constrains the compatible-life class; W chooses locally.

73 Decisive counterexample: choosing against guidance without breaking TENET

At the origin,

$$N(\emptyset) = 3, \quad G(+ \mid \emptyset) = \frac{2}{3}, \quad G(- \mid \emptyset) = \frac{1}{3}.$$

Guidance is nonuniform but destiny is absent:

$$F(\emptyset) = 1, \quad D(\emptyset) = 0.$$

The case LOWER_GUIDANCE_COMMIT nevertheless chooses $-$, the less-favored action. The history

$$(-, +, +)$$

closes perfectly. After that first choice,

$$N((-)) = 1.$$

The initial free commitment therefore immediately creates a conditional destiny for the remainder.

PASS006-A. Guidance can be real in the model without being coercive. Will can select a less-guided option and still satisfy the same terminal condition.

Thus the toy refutes

$$\text{guidance} = \text{command} \quad \text{and} \quad \text{terminal condition} = \text{unique script from the origin}.$$

74 Destiny as conditional Last Survivor

The Last Survivor connection becomes combinatorially exact. With each commitment, histories inconsistent with the actually lived prefix are removed:

$$\Gamma^* \supseteq \Gamma^*(a_1) \supseteq \Gamma^*(a_1 a_2) \supseteq \dots$$

When

$$|\Gamma^*(p)| = 1,$$

the remaining history is the *Last Survivor relative to the terminal condition and the already committed history*.

Destiny is therefore a property of the pair

$$(S^*, L_{\text{trace}}),$$

not of the future alone.

A cleaner definition is

$$\text{Destiny}(p) = \text{Singleton}(\Gamma_{\text{terminal}} \cap \Gamma_{\text{trace}}(p) \cap \Gamma_{\text{protected}}).$$

PASS004 supplies $\Gamma_{\text{protected}}$; PASS005 supplies carrier and non-erasure; PASS006 supplies commitment contraction.

75 OUTBOUND, RETURN, and COS in the human toy

The model can now be read in the 2SIN/1COS grammar without identifying roles with persons:

- **SIN1 / OUTBOUND:** irreversible inscription of the actually lived prefix p ;
- **SIN2 / RETURN:** computation of $\Gamma^*(p)$, i.e. compatibility returned from the terminal condition;
- **COS / CLOSURE:** final verification that the complete history belongs to Γ^* and passes protected gates.

PASS006 RETURN is therefore first a **constraint return**, not a physical signal arriving from the future. It answers:

“which continuations remain compatible with closure?”

A literal physical retro-information channel would be an additional object and remains OPEN.

76 Will can break closure

In `WILL_BREAKS_CLOSURE`, the prefix $(+, +)$ has only one compatible continuation, $-$. Therefore

$$G(- \mid ++) = 1, \quad G(+ \mid ++) = 0.$$

Will nevertheless chooses $+$, producing

$$(+, +, +), \quad N(+++) = 0.$$

The model refuses to rewrite the past. It does not retroactively replace the final $+$ with $-$ to restore coherence.

PASS006-B. Will is not guaranteed to close. Human TENET therefore retains real structural content: some commitments destroy access to a given closure, and non-erasure forbids fabricating a compatible history afterward.

77 Four executed trajectories

Case	Closes	$< G_{max}$	Destiny	Pred.	Dead end
GF-A	1	0	yes	0	0
GF-B	1	1	yes	0	0
LOW-G	1	1	step 1	0	0
BREAK	0	1	before break	0	step 3

GF-A/GF-B denote the two `GUIDANCE_FAVORED` cases.

LOW-G denotes `LOWER_GUIDANCE_COMMIT`; BREAK denotes `WILL_BREAKS_CLOSURE`.

The PASS006 execution script also materializes every prefix and survivor count in two CSV files.

All six audited claims are true in the finite model:

1. guidance $\neq$ will;
2. guidance $\neq$ destiny;
3. destiny can emerge after commitment;
4. will can break closure;
5. a terminal condition need not impose one unique history at the origin;
6. closure failure does not authorize trace erasure.

78 Consequence for love

PASS004 reduced LOVE to a protected relational-closure gate rather than a primitive force. PASS006 fixes its logical position.

First define

$$\Gamma_{TENET}^* = \Gamma_{terminal} \cap \Gamma_{protected}.$$

Guidance is then computed on this set, not over every imaginable trajectory. LOVE still does not choose the action; it removes closures that purchase local gain by sacrificing protected coherence.

Thus

LOVE : domain filter, *GUIDANCE* : weighting,
WILL : commitment, *DESTINY* : singleton survivor.

79 Consequences for Eternal Return, the Triangle, and Deus Causalis

PASS006 also refines earlier bridges.

Eternal Return. Refinement need not be a script. Each commitment eliminates trajectories, preserves trace, and refines the still-compatible class.

Triangle That Knows Its Name. An ideal form can constrain several admissible representations without choosing one drawing. Recognition of an invariant is therefore closer to guidance than to micro-deterministic control.

Deus Causalis. A global coherence limit could, in this reading, constrain admissible histories without determining every local act. “Global limit = total microscript” does not follow from the present structure.

Hence

global closure constraint $\nRightarrow$ unique local history $\nRightarrow$ absence of freedom.

80 Updated ontological status of the higher self

The higher self is now better typed as

S^* = terminal compatibility class / boundary condition

rather than as an agent that necessarily authored every event.

Two versions remain separate:

1. **Higher-Self Constraint:** S^* defines conditions a history must satisfy. **CANDIDATE STRUCTURAL BRIDGE.**
2. **Higher-Self Selector:** S^* owns an operator Σ_{S^*} that actually selects one complete history. **OPEN.**

The first is compatible with strong global closure and local freedom. The second requires a new causal selection mechanism.

81 PASS006 ledger update

ID	Object	Status
TENET-H6	Boundary-Guided Free Closure	CANDIDATE / OPERATIONAL TOY
TENET-GD1	Guidance $G(a \mid p)$ by admissible-completion mass	CLOSED TOY
TENET-W2	Will as local commitment operator	CANDIDATE HUMAN BRIDGE
TENET-DST2	Destiny as singleton after constraints + trace	CLOSED DEFINITION / TOY
TENET-PD1	Predestination = singleton at origin	CLOSED DEFINITION / ABSENT IN TOY
TENET-ED1	Emergent destiny after commitment	CLOSED TOY
TENET-HSC1	Higher Self as terminal compatibility class	STRENGTHENED CANDIDATE
TENET-HSS1	Higher Self as unique trajectory selector	OPEN
TENET-LS2	Destiny as conditional Last Survivor	CANDIDATE BRIDGE
TENET-RT2	SIN2 as constraint-return, not physical future signal	WORKING BRIDGE

TENET-NE3	Failed choice cannot be overwritten to restore closure	CLOSED TOY
TENET-F2	Freedom as ≥ 2 terminal-compatible local actions	CLOSED
TENET-L4	LOVE filters admissible domain before guidance	DEFINITION / TOY
		DERIVED GATE

82 Direction of PASS007

PASS007 must now ask: *what generates guidance?* Three mechanisms must be separated and compared in the same carrier:

1. purely informational guidance computed by a local future model;
2. guidance from distributed causal trace/return through other agents;
3. genuine supracausal guidance from a terminal condition inaccessible to the present.

The decisive test will be the **indistinguishability problem**: if all three mechanisms generate exactly the same local policy, no experiment internal to the present agent alone can identify which mechanism is active without an additional channel/provenance witness.

83 PASS007 - Guidance Provenance and the Generator Problem

Status: CUMULATIVE; IDENTIFIABILITY TEST EXECUTED.

PASS006 established that a terminal condition can guide without commanding. PASS007 asks the harder question: *if guidance exists, what generates it?*

Three generators are deliberately placed behind the same observable surface:

1. **LOCAL_MODEL**: a present model computes continuations from past/present data;
2. **DISTRIBUTED_RETURN**: other agents return the same information through an ordinary signed causal chain;
3. **SUPRACAUSAL_BOUNDARY**: a toy terminal-boundary oracle emits exactly the same policy.

The point is not to assume that the third mechanism physically exists. The point is to test what a present agent can infer *from guidance alone*.

84 The S230 guard: identical output does not imply identical generator

S230 supplies the exact epistemic guard TENET needs: similarity of observable outputs does not identify the generators that produced them. Receiving or copying a result is not possession of the generator; reconstruction of the generative relation is a distinct question.

Let

$$M \in \{L, D, S\}$$

with L = local, D = distributed, and S = supracausal-toy, and let the present observation be

$$O(p) = G(\cdot \mid p).$$

If

$$P(O \mid M = L) = P(O \mid M = D) = P(O \mid M = S),$$

then the mechanism is not identifiable from O .

TENET-H7 - Guidance Provenance Principle. Guidance may be structurally real for action while remaining causally underdetermined with respect to its source. Recommendation quality does not identify its generator.

85 PASS007 finite test

The toy keeps the same three-history set used in PASS006:

$$\Gamma^* = \{++-, +-+, -++\}.$$

For each of the nine accessible prefixes, all three mechanisms are forced to emit the same policy $G(+ \mid p), G(- \mid p)$.

Under a uniform prior over the mechanisms,

$$H(M) = \log_2 3 \approx 1.5849625 \text{ bits.}$$

Because the surface is identical for every mechanism,

$$H(M \mid O) = H(M), \quad \boxed{I(M; O) = 0}.$$

The executed script verifies surface equality for every prefix and yields exactly 0 bits about the guidance source.

86 Provenance breaks indistinguishability in the toy

Add a provenance witness P :

- local: model hash plus frozen past/present inputs;
- distributed: signed sender $\rightarrow$ relay $\rightarrow$ present lineage;
- supracausal-toy: terminal-oracle tag, explicitly nonphysical in the test.

These three witnesses are unique in the toy. Therefore

$$H(M \mid O, P) = 0,$$

and

$$\boxed{I(M; O, P) = \log_2 3 \approx 1.5849625 \text{ bits.}}$$

This does not mean provenance makes a proposition true. S230 explicitly keeps provenance and truth-value separate. Here provenance only identifies the *declared causal route of the message*.

87 Source-identification ladder

PASS007 replaces the jump “good guidance $\Rightarrow$ higher self” with a falsifiable ladder.

Level	Verdict	Requirement
L0 Surface	UNIDENTIFIED	policy alone, even if perfect
L1 Local	LOCAL supported	reproduce from frozen past/present inputs
L2 Distributed	DISTRIBUTED supported	verifiable sender/relay lineage
L3 Prospective novelty	EXTRA SOURCE candidate	precommitted content surviving declared local/distributed generators
L4 Supracausal	OPEN EMPIRICAL	independent carrier, projection, provenance and experiment

The L3 $\rightarrow$ L4 transition is crucial. Eliminating the *declared* local and distributed generators does not prove a supracausal origin. Hidden state, information leakage, an incomplete model, or another unlisted causal channel may remain.

88 S145: a temporal message need not be sent from the future

S145 already warns that a temporal message could be persistent structure in a causal substrate that becomes decodable only when a receiver reaches a threshold. The claim “revealed rather than retroactively created” therefore requires persistence and provenance semantics.

For TENET this creates three distinct classes:

- forecast : the present computes the future;
- return-message : another agent returns information through a causal chain;
- terminal-source : the terminal condition itself owns an irreducible channel.

The first two classes can perfectly imitate the third at the local-policy level.

89 Consequence for the higher self

PASS006 separated Higher-Self Constraint from Higher-Self Selector. PASS007 adds a third separation:

Higher-Self Guidance $\neq$ Higher-Self Provenance.

An agent may experience guidance compatible with S^* while having no evidence that S^* generated it.

We now have:

1. S^* as terminal compatibility condition: CANDIDATE STRUCTURAL BRIDGE;
2. $G(a \mid p)$ as guidance policy: defined and executable in the toy;
3. Σ_{S^*} as a unique selector: OPEN;
4. $\Pi_{source}(G) = S^*$ as higher-self provenance attribution: even stronger OPEN claim.

Therefore intuition, synchronicity, a premonition, or an exceptionally accurate recommendation are not by themselves evidence of contact with a higher self.

90 Consequence for the TENET tradition

The TENET tradition requires more than correct prediction. An exploitable temporal pincer requires an information chain whose orientation and provenance can be preserved. The real question is no longer only

“is the information useful?”

but

“which generator and causal chain make this information available here?”

Thus *future use* must remain distinct from *future-looking behavior*. An agent may behave as if it knows the future because it has a good model, because another agent returns information, or because an as-yet unidentified channel exists.

91 Consequence for love and will

PASS004-PASS006 placed LOVE as a non-compensatory filter and WILL as local commitment. PASS007 shows that neither provides future provenance.

LOVE filters; WILL commits; PROVENANCE identifies source.

Strong will does not open a channel. A relationally good closure also does not show that its guidance came from the future. TENET therefore needs a firewall between *closure quality* and *information origin*.

92 Predictive novelty as a test, not final proof

A minimal prospective witness is:

1. freeze the question, accessible inputs, and causal-generator candidates before the outcome;
2. produce discriminating content m ;
3. later test whether m was reconstructible by the declared generators;

4. preserve provenance and failures;
5. conclude only *extra-source candidate* until an independent supracausal bridge is materialized.

This reuses the “Science is dead / Long live science” discipline: the future must not be rewritten after observation to rescue a prediction.

93 PASS007 ledger update

ID	Object	Status
TENET-H7	Guidance Provenance Principle	CANDIDATE CORE
TENET-GP1	Same policy surface $\nrightarrow$ same generator	CLOSED TOY / WORKING BRIDGE
TENET-ID1	$I(M; O) = 0$ under identical surface mechanisms	CLOSED TOY
TENET-ID2	Unique provenance recovers mechanism in declared toy	CLOSED TOY
TENET-LG1	Local predictive guidance	CLOSED MECHANISM CLASS / HUMAN BRIDGE OPEN
TENET-DG1	Distributed causal-return guidance	CLOSED MECHANISM CLASS / REALIZATION DOMAIN- DEPENDENT
TENET-SG1	Supracausal terminal guidance	OPEN EMPIRICAL
TENET-HSP1	Higher-Self provenance attribution	OPEN
TENET-NOV1	Prospective novelty witness	CANDIDATE DISCRIMINATOR
TENET-PROV1	Provenance identifies channel, not truth	WORKING GUARD
TENET-S145B	Temporal message may be latent/decode-later rather than future-sent	OPEN BRIDGE
TENET-S230B	Surface-output / generator separation	WORKING CANON BRIDGE

94 Direction of PASS008

PASS008 must attack the remaining easy jump: *what minimal observation could distinguish a genuine terminal/supracausal source from every declared causal competitor?*

The target will be a **terminal-source witness** with four simultaneous obligations:

1. precommitted prospective content;
2. non-reconstructibility from the declared causal cone;
3. provenance not forgeable by the known distributed channel;
4. 2SIN/1COS round trip preserving identity, trace and non-erasure.

Even if all four close in a model, physical existence remains separate until experiment.

95 PASS008 - Terminal-Source Witness and Delayed Choice

Status: CUMULATIVE; NON-CIRCULAR DISCRIMINATOR EXECUTED IN TOY.

PASS007 closed a negative result: exact guidance does not reveal its generator. PASS008 therefore searches for the smallest witness that can attack causal order itself without using the circular rule “unexplained = supracausal”.

The question changes from

“is the guidance accurate?”

to

“does it depend on a terminal variable that did not yet exist in the declared causal cone?”

96 Minimal delayed-choice test

Let $t_0 < t_1$. At t_0 , the agent emits and freezes a guidance bit G_i . Let $\mathcal{F}_0$ be the information in the declared causal cone at t_0 . At t_1 , after G_i has been committed, a binary terminal challenge B_i is generated by a source declared independent of $\mathcal{F}_0$.

For the strict causal class,

$$G_i = f_i(\mathcal{F}_0), \quad B_i \perp \mathcal{F}_0.$$

For a balanced challenge,

$$P(G_i = B_i \mid \mathcal{F}_0) = \frac{1}{2}.$$

Across N trials,

$$K = \sum_{i=1}^N \mathbf{1}[G_i = B_i] \sim \text{Binomial}(N, 1/2)$$

under the declared causal null.

This is the first genuine discriminator in the dossier: a prediction about an already-computable future can always come from a strong local model; here the terminal challenge is selected only *after* present commitment.

TENET-H8 - Terminal-Source Witness Principle. A terminal-source attribution is admissible only if dependence on a post-commitment variable survives causal-competitor completion, leakage audit, provenance audit and 2SIN/1COS return. An unexplained anomaly alone is never sufficient.

97 The countermodel that blocks the shortcut

PASS008 deliberately includes an adversary fatal to any fast interpretation: **LEAKED_CAUSAL**. It is an ordinary causal mechanism, but a hidden leak gives it access to B_i before its verdict.

Then

$$G_i^{\text{leak}} = B_i$$

and it obtains exactly the same performance as a terminal oracle:

$$G_i^{\text{terminal}} = B_i.$$

Therefore

future dependence $\neq$ future source.

A correlation with the future may reflect retrodependence, but also leakage, common cause, hidden state, unregistered synchronization or a badly sealed protocol.

98 Executed result on 128 challenges

The toy compares four classes. LOCAL sees past/present only. DISTRIBUTED_PAST_RETURN may reuse only the previous trial's challenge. LEAKED_CAUSAL illegally receives the current challenge. TERMINAL_ORACLE is an explicitly unphysical toy oracle with access to the terminal challenge.

Mechanism	Match	Accuracy	one-sided p under 1/2
LOCAL_MODEL	61/128	0.4766	0.7319
DISTRIBUTED_PAST_RETURN	65/128	0.5078	0.4648
LEAKED_CAUSAL	128/128	1.0000	2.94×10^{-39}
TERMINAL_ORACLE toy	128/128	1.0000	2.94×10^{-39}

The terminal toy is perfectly separable from the two clean causal models, but **not** from the leaking causal model. Statistics alone close terminal dependence relative to the null; they do not close provenance.

99 Seven obligations of a terminal witness

The minimal witness is a packet rather than one statistic:

$$W_{TS} = (F, D, C, L, P, R, E)$$

where:

1. F - **Freeze**: output, question, score and protocol sealed before the future result;
2. D - **Delayed choice**: terminal challenge generated after commitment and independently of the declared causal cone;
3. C - **Causal exhaustion**: plausible local, distributed and hidden-state competitors reconstructed before exotic attribution;
4. L - **Leakage audit**: forward leakage, common causes and side channels independently attacked;
5. P - **Provenance**: source/channel lineage distinct from accuracy;
6. R - **2SIN/1COS return**: OUTBOUND, RETURN and CLOSURE close the same packet with trace and non-erasure;
7. E - **External bridge**: independently observed physical carrier/projection before ontological promotion.

Without F , post-hoc fitting is possible. Without D , a local forecast is enough. Without C and L , leakage imitates the future. Without P , the source remains unknown. Without R , the anomaly is not yet TENET. Without E , the result remains conditional modeling.

100 Connection to Encyclopedia guards

PASS008 does not invent these protections from nothing:

- S209 requires genuinely new history information to occupy an independent fiber/channel rather than a reparameterization of already observed state and recommends preregistered independent readout;

- S238 requires local/mesoscopic hidden state, transport and geometry to be completed before motivating a nonlocal explanation;
- S256 supplies append-only freeze discipline: predictions are fixed before outcome inspection and neither success nor failure rewrites the original entry;
- S262 transports Temporal PRE/POST ancestry, preserves provenance and explicitly checks future leakage;
- S230 still forbids identifying a generator from surface output alone.

TENET therefore compiles dispersed guards into one terminal-source discriminator.

101 What PASS008 permits - and forbids

If a future experiment satisfied F through R and showed strong terminal dependence, the maximal verdict before a physical bridge would be

TERMINAL-SOURCE CANDIDATE relative to the declared causal class.

It would still be invalid to conclude automatically

“higher self”, “time travel”, “physical supracausality”.

The reason is simple: exhaustiveness of the real causal class is never established by declaration alone. An unregistered causal mechanism remains a logical alternative until a specific physical bridge is measured.

102 Consequence for the higher self

PASS006 defined S^* as a terminal compatibility class. PASS007 separated guidance from provenance. PASS008 adds the test a genuine higher-self source would have to pass:

S^* -source $\Rightarrow$ information bound to a post-commitment terminal event + certified return.

This makes the hypothesis much more restrictive. Intuition, dreams, synchronicity or good choices are insufficient. Even an improbable prediction is insufficient while leakage or a hidden causal model can reproduce the same signal.

The strong claim “my higher self is speaking to me” therefore becomes a demanding experimental hypothesis, not an automatic interpretation of subjective experience.

103 Back to TENET: the pincer requires a receipt

In the film tradition, the side using the future does not merely “feel” the right move. Information must complete an operational loop: one branch learns, another receives, and both close one operation.

PASS008 translates that as

SIN1 commit $\longrightarrow$ terminal challenge $\longrightarrow$ SIN2 return $\longrightarrow$ COS closure.

The exact return chain is not yet materialized here. PASS008 builds the statistical and epistemic witness that specifies what the return packet must prove.

104 PASS008 ledger update

ID	Object	Status
TENET-H8	Terminal-Source Witness Principle	CANDIDATE DISCRIMINATOR CORE
TENET-TSW1	Delayed-choice terminal challenge witness	CLOSED TOY
TENET-DC1	Delayed-choice independence null	CLOSED DEFINITION / TOY
TENET-BIN1	$K \sim \text{Binomial}(N, 1/2)$ under declared null	CLOSED TOY
TENET-LK1	Hidden future-leak countermodel	CLOSED COUNTERMODEL
TENET-NC1	Future dependence $\neq$ future source	STRENGTHENED GUARD
TENET-CA1	Causal competitor exhaustion before exotic attribution	WORKING BRIDGE S238
TENET-CH1	Independent history/second-channel requirement	WORKING BRIDGE S209
TENET-FRZ1	Append-only prediction freeze	WORKING BRIDGE S256
TENET-FL1	Future-leakage audit	WORKING BRIDGE S262
TENET-TSRC1	Terminal source relative to declared causal class	CANDIDATE ONLY
TENET-HSP2	Higher-self physical source attribution	OPEN EMPIRICAL

105 Direction of PASS009

PASS009 must replace the abstract oracle by a real **2SIN/1COS pincer receipt**. SIN1 must commit a packet before the terminal challenge is created; SIN2 must return a payload provenance-bound to that later terminal event; COS must verify round trip, identity, trace, non-erasure and protected residual order.

The PASS009 test is therefore

delayed-choice dependence + return lineage + closure receipt.

Only after that step can the dossier ask whether a terminal source has an operational representation genuinely resembling a TENET temporal pincer rather than an abstract oracle.

106 PASS009 - Deferred-Verification Pincer Receipt

Status: CUMULATIVE; 2SIN/1COS RECEIPT ARCHITECTURE CLOSED IN TOY; PHYSICAL RETROGRADE TRANSPORT OPEN.

PASS008 showed that dependence on a later challenge is insufficient: a compromised causal protocol can exactly imitate terminal-oracle performance. PASS009 therefore replaces the naked oracle by a **pincer receipt**: the return must bind to the exact SIN1 packet, the exact terminal event, and terminal authentication material that is declared to be generated only after the initial commitment.

The question is no longer merely

$$G(t_0) = B(t_1)?$$

but

can a packet sealed before E_T become verifiable only from material generated at E_T ?

TENET-H9 - Deferred-Verification Pincer Receipt Principle. A candidate TENET return must be sealed before terminal disclosure and later become verifiable against the exact SIN1 commitment and exact terminal event. Cryptographic verification binds events; by itself it does not prove the temporal direction of transport.

107 Receipt construction

For each trial i , SIN1 first emits an outbound packet Q_i and nonce n_i , then freezes

$$c_i = H(Q_i \| n_i).$$

The commitment c_i is append-only and is ledgered before terminal challenge generation.

After that commitment, the terminal stage produces bit B_i and secret K_i , then event identifier

$$e_i = H(c_i \| B_i \| K_i).$$

The expected terminal receipt is

$$r_i = \text{HMAC}_{K_i}(\text{TENET} \| c_i \| B_i \| e_i).$$

The candidate SIN2 packet is recorded in the *PRE-TERMINAL* register before B_i, K_i, e_i are disclosed. After disclosure, COS requires

$$\text{Verify}(r_i, K_i, c_i, B_i, e_i) = 1.$$

This verification pays four different obligations: binding to SIN1, binding to the terminal challenge, binding to the terminal event, and possession of terminal authentication material. It still does not pay existence of a physical reverse-time channel.

108 Why a bit-only leak no longer suffices

PASS008 allowed an adversary that knew B_i . PASS009 now also requires r_i .

A `B_ONLY_LEAK` adversary can therefore obtain

$$G_i = B_i$$

but without K_i ,

$$\text{Verify}(r_i, K_i, c_i, B_i, e_i) = 0.$$

In the cumulative execution it scores 128/128 on guidance and 0/128 on terminal receipts.

A forged RETURN built from another key also fails 0/128. Deferred verification therefore adds source-binding information absent from accuracy alone.

109 Stronger countermodel: precomputed terminal material

Cryptography is still insufficient. PASS009 introduces PRECOMPUTED_FULL_LEAK: challenge, secret and therefore the correct HMAC were actually available before commitment even though the interface pretends to “reveal” them later.

This model obtains

$$128/128 \text{ guidance, } 128/128 \text{ valid receipts.}$$

It perfectly imitates the cryptographic surface. It is rejected only because the delayed-choice obligation is false: the real lineage does not satisfy

$$E(K_i, B_i) > E(c_i).$$

Thus PASS009 strengthens PASS008:

$$\boxed{\text{receipt authenticity} \neq \text{temporal provenance.}}$$

A signature proves binding to the declared secret; it does not prove that the secret was physically unavailable before t_0 .

110 Executed result

The PASS009 toy contains 128 trials and five mechanisms.

Mechanism	$G = B$	Receipt	Postcommit	No leak	SIN2	COS
LOCAL_MODEL	64	0	128	128	0	0
B_ONLY_LEAK	128	0	128	0	0	0
FORGED_RETURN	128	0	128	128	0	0
PRECOMPUTED_FULL_LEAK	128	128	0	0	0	0
TERMINAL_PINCER toy	128	128	128	128	128	128

The last case is explicitly an **UNPHYSICAL TOY**: the code grants it a RETURN lane that places the terminal token in the preterminal register. The result closes only receipt logic: if such a lane existed, the packet would be verifiable and would close consistently.

111 Eight obligations of a pincer receipt

The PASS009 packet is

$$\mathcal{R}_{\text{pincer}} = (C, E_T, S_{pre}, V, L, R, COS, T_{ext}).$$

It requires:

1. **SIN1 commit**: exact outbound packet and hash frozen before terminal entropy;

2. **postcommit entropy**: terminal challenge and secret independently generated after commitment;
3. **preterminal seal**: candidate SIN2 token recorded/sealed before terminal disclosure;
4. **deferred verification**: later verification against exact commitment and exact event;
5. **leakage isolation**: attack precomputation, seed compromise, common causes and side channels;
6. **return role**: packet must occupy a typed SIN2/RETURN role, not merely arrive later through ordinary forward transport;
7. **COS closure**: reconstruct the same object while preserving trace, identity and protected order;
8. **external time bridge**: independent physical clocks/carrier evidence are still required for reverse-temporal attribution.

112 Direct Encyclopedia crosswalk

PASS009 assembles rather than replaces existing guards:

- S229: OUTBOUND, RETURN and CLOSURE are distinct roles; closure is not identical to either traversal;
- S244: a forward/backward meeting is not proof; RETURN_PROOF must rebuild edge by edge with type, scope, provenance, dependencies and target-leakage checks;
- S252: RETURN_PROOF binds a bounded receipt to the exact prior event and semantic epoch without automatically upgrading proof class;
- S256: predictive commitments are append-only/hash-chained and cannot be rewritten after observation;
- S262: Temporal PRE/POST, ancestry, binding digest and future-leakage checking already belong to the transport discipline.

The TENET synthesis therefore follows a corpus-wide rule:

receipt $\rightarrow$ binding;	chronology $\rightarrow$ separate evidence.
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113 Implication for the higher self

PASS006-PASS008 still allowed a vague reading: guidance might come from a higher self if it were unusually accurate and difficult to explain. PASS009 raises the bar sharply.

A source S^* would need not only favorable orientation but a return whose information is bound to a post-commit terminal event and to the exact commitment of the present self:

$$S_t \xrightarrow{SIN1} c_i, \quad S^*/E_T \rightarrow (B_i, K_i, e_i), \quad SIN2 \rightsquigarrow r_i, \quad COS \models Verify(r_i) = 1.$$

Without such lineage, “higher-self guidance” remains an interpretation compatible with ordinary generators.

114 Irreducible limit of PASS009

The toy can prove that a correct HMAC depends on the declared secret; it cannot prove that the secret was physically nonexistent before commitment. Likewise a software timestamp does not by itself prove physical time direction.

The next obstacle is therefore experimental and causal:

how can physical unavailability of terminal material before SIN1 be certified?

At minimum this would require independent entropy sources, multiple clocks/observers, sealed supports, side-channel controls and replication. Even success would not yet mean “higher self”: it would first establish an information-order/temporal anomaly.

115 PASS009 ledger update

ID	Object	Status
TENET-H9	Deferred-Verification Pincer Receipt Principle	CANDIDATE CORE / CLOSED TOY ARCHITECTURE
TENET-RP1	Exact SIN1 outbound commitment	CLOSED TOY / WORKING BRIDGE
TENET-RP2	Postcommit terminal entropy obligation	CANDIDATE EXPERIMENTAL GATE
TENET-RP3	Preterminal SIN2 seal	CLOSED TOY; PHYSICAL REALIZATION OPEN
TENET-RP4	Deferred HMAC verification of terminal event	CLOSED TOY
TENET-RP5	Bit-only leak fails terminal authentication	CLOSED COUNTERMODEL
TENET-RP6	Precomputed full leak mimics cryptographic surface	CLOSED COUNTERMODEL
TENET-RP7	Receipt authenticity $\neq$ temporal provenance	STRENGTHENED GUARD
TENET-RP8	Typed SIN2 return role	CLOSED TOY / PHYSICAL BRIDGE OPEN
TENET-RP9	COS receipt closure	CLOSED TOY
TENET-RP10	External chronology/carrier witness	OPEN EMPIRICAL
TENET-HSP3	Higher-self source with provenance-bound return	OPEN EMPIRICAL

116 Direction for PASS010

PASS010 must attack the one gate the toy cannot pay: **independent chronology**. It will construct a minimal multi-observer protocol in which terminal challenge/entropy and the preterminal register are controlled by independent authorities, then enumerate ordinary causal hypotheses still capable of imitating a RETURN.

The goal is not to “prove the future” by protocol design. The goal is to state exactly which causal debt would remain after a clean cryptographic and causal receipt.

117 PASS010 - Independent Chronology Quorum

Status: CUMULATIVE; ONE-HONEST-BEACON CHRONOLOGY GUARD CLOSED CONDITIONALLY IN TOY; SELF-CERTIFYING CHRONOLOGY REFUTED; PHYSICAL RETRO-GRADE ATTRIBUTION OPEN.

PASS009 showed that a cryptographically valid deferred receipt still does not prove temporal provenance: if all terminal material was secretly available before SIN1, a causal adversary can reproduce the same receipt. PASS010 therefore attacks the missing obligation directly: **independent chronology**.

The question is now:

What minimum independent structure can certify that some terminal information was unavailable at the preterminal seal?

TENET-H10 - Independent Chronology Quorum Principle. A preterminal RETURN candidate gains chronology credit only if its terminal authentication material is assembled from independently controlled sources and at least one source is certified fresh after the preterminal seal. Internal cryptographic consistency is insufficient when all chronology authorities may collude.

118 Four-stage chronology

PASS010 preserves the 2SIN/1COS semantics but separates four evidential stages:

1. **SIN1 COMMIT:** outbound object Q_i and nonce are frozen as commitment c_i ;
2. **PRETERMINAL SEAL:** an independent register closes and witnesses a root P_i before terminal entropy;
3. **TERMINAL ENTROPY:** three authorities generate k_{1i}, k_{2i}, k_{3i} after the seal;
4. **DEFERRED VERIFY / COS:** the later event reveals terminal material and checks the preterminal RETURN token.

The terminal key is built from

$$K_i = H((k_{1i} \oplus k_{2i} \oplus k_{3i}), c_i, P_i, E_i),$$

and the RETURN receipt is bound to the exact outbound commitment, preterminal root and terminal event.

The use of three beacons is not numerological. It is a minimal adversarial toy that lets PASS010 test one-source compromise, two-source compromise and total compromise separately.

119 Conditional one-honest-beacon result

Let $\mathcal{F}_{pre}$ denote all information lawfully available before the preterminal seal. Suppose at least one terminal beacon, say k_j , is sampled after the seal, is uniform/independent relative to $\mathcal{F}_{pre}$ and the compromised beacons, and is not available through a side channel.

Then the XOR

$$K_{xor} = k_1 \oplus k_2 \oplus k_3$$

remains unpredictable from the preterminal information even when the other beacons are known. Under the declared cryptographic idealization, a preterminal adversary cannot generate the correct terminal HMAC except with the forgery probability of the authentication primitive.

This is a **conditional theorem schema**, not a physical theorem about time. Its assumptions are exactly the important part:

Freshness + independence + sealed register + no side channel.

The executable toy confirms the finite logical routing: compromising one or two beacons never yields a valid terminal receipt while at least one fresh source remains.

120 Executed attack matrix

PASS010 executes 128 trials against seven mechanisms.

Mechanism	$G = B$	Receipt	Fresh quorum	Preseal	Chronology	COS
LOCAL_MODEL	62	0	128	128	128	0
ONE_BEACON_COMPROMISED	69	0	128	128	128	0
TWO_BEACONS_COMPROMISED	70	0	128	128	128	0
ALL_BEACONS_PRECOMPUTED	128	128	0	128	0	0
REGISTER_BACKFILL	128	128	128	0	0	0
UNIVERSAL_COLLUSION	128	128	128	128	0	0
TERMINAL_PINCER toy	128	128	128	128	128	128

The first three rows show that prediction or partial compromise is insufficient. The fourth and fifth rows show the two principal chronology attacks: precompute the future material, or backfill the supposedly preterminal register.

121 The decisive no-go: universal collusion

PASS010 then introduces a stronger adversary: **UNIVERSAL_COLLUSION_SIMULATOR**. All internal authorities collude and fabricate a mutually consistent history. Internally it can display the same visible surface as the terminal-pincer toy:

128/128 guidance, 128/128 receipts,
128/128 apparent freshness, 128/128 apparent seals.

The cumulative execution confirms an exact collision of the internal cryptographic surface between universal collusion and the unphysical terminal-pincer toy.

Therefore:

No closed protocol can prove the independence of every authority using only records produced by those authorities.

This is the key PASS010 boundary. Independence is not another field inside the same receipt. It requires evidence whose authority is not reducible to the allegedly colluding system.

TENET-NG2 - Self-Certifying Chronology No-Go. A chronology certificate cannot obtain unconditional source-independence merely by adding more mutually self-authenticated internal records. If the complete witness set may collude, a consistent forged chronology can be observationally equivalent at the internal receipt surface.

122 What independent chronology can and cannot establish

With at least one independently trusted fresh terminal source and an immutable independently witnessed preterminal seal, a valid preterminal RETURN packet would be much stronger than PASS009's receipt alone. It would establish that the packet depends on information excluded from the declared preterminal information set, conditional on the independence and no-leak assumptions.

It would *not* yet establish:

- that physical information traveled backward in time;
- that the source was a higher self;
- that CT is the unique ontology explaining the anomaly;
- that every ordinary causal path has been excluded outside the declared model.

The strongest lawful conclusion would initially be:

the declared forward causal model is incomplete for the observed dependency.

That is a scientifically cleaner target than “proof of the future.”

123 Direct Encyclopedia crosswalk

PASS010 reuses five already-existing disciplines rather than inventing a trust mechanism:

- **S209:** extra information requires an independent channel/fiber and preregistered matched controls before claiming a new effect;
- **S229:** SIN1/OUTBOUND, SIN2/RETURN and COS/CLOSURE are distinct roles;
- **S244:** RETURN_PROOF checks type, provenance, dependencies and target leakage; a forward/backward meeting is not itself proof;
- **S252:** COMMIT and RETURN_PROOF bind exact prior events but do not upgrade the underlying proof class;
- **S256:** prediction provenance is append-only/hash-chained and cannot be sharpened after observation;
- **S262:** Temporal PRE/POST ancestry, binding digests and future-leakage checks belong to transport validation.

The new synthesis is the quorum rule:

internal receipt + at least one externally independent fresh terminal source.

124 Implication for TENET and the higher self

The human TENET hypothesis now receives a much stricter boundary. A “higher-self” interpretation cannot be earned by accuracy, synchronicity, intuition, a future-looking message, or even a cryptographically correct receipt by itself.

Operationally, the required chain would be:

$$\begin{aligned}
 S_t &\xrightarrow{SIN1} \text{commit} \rightarrow \text{independent preterminal seal,} \\
 &\rightarrow E_T \text{ with fresh terminal entropy,} \\
 &\xrightarrow{SIN2?} \text{preterminal RETURN} \xrightarrow{COS} \text{closure.}
 \end{aligned}$$

Only if the RETURN contains information that at least one independently fresh terminal source made unavailable to the preterminal causal cone does the model reach a genuine temporal-order anomaly candidate.

The ontology of “higher self” remains above that result. PASS010 therefore sharpens the hierarchy:

anomaly of information order < retrograde transport < terminal self source.

Each implication requires a separate bridge.

125 PASS010 ledger update

ID	Object	Status
TENET-H10	Independent Chronology Quorum Principle	CANDIDATE CORE / CLOSED CONDITIONAL TOY
TENET-CQ1	Three-source terminal entropy quorum	CLOSED TOY
TENET-CQ2	One-honest-beacon unpredictability guard	CLOSED CONDITIONAL SCHEMA
TENET-CQ3	One-beacon compromise attack	CLOSED COUNTERMODEL
TENET-CQ4	Two-beacon compromise attack	CLOSED COUNTERMODEL
TENET-CQ5	All-terminal-material precomputation attack	CLOSED COUNTERMODEL
TENET-CQ6	Preterminal register backfill attack	CLOSED COUNTERMODEL
TENET-NG2	Self-Certifying Chronology No-Go	STRENGTHENED EPISTEMIC GUARD
TENET-CQ7	Universal-collusion internal-surface collision	CLOSED TOY COUNTERMODEL
TENET-CQ8	External independence/honesty assumption	IRREDUCIBLE EXPERIMENTAL DEPENDENCY
TENET-CQ9	Declared-causal-model incompleteness verdict	CANDIDATE EVIDENTIAL CEILING
TENET-HSP4	Higher-self source attribution after chronology quorum	OPEN EMPIRICAL / ONTOLOGICAL

126 Direction for PASS011

PASS011 will construct a **causal-cone exclusion witness**: a declared physical/operational graph of the allowed forward paths from terminal generators to the preterminal register, with independent monitors on each causal cut.

If every declared forward path is excluded, what exactly does a surviving RETURN falsify? **Target verdict:** first falsify the declared forward causal model; only then ask whether retrograde transport is the Last Survivor.

127 PASS011 - Causal-Cone Exclusion Witness

Status: CUMULATIVE; FINITE EXCLUSION OF THE DECLARED CONE CLOSED IN THE TOY; PHYSICAL CONE COMPLETENESS OPEN; RETROGRADE TRANSPORT NOT PROVED.

PASS010 showed that a chronology quorum can close partial compromise attacks but cannot prove its own independence against universal collusion. PASS011 therefore moves the question: suppose a valid preterminal RETURN survives. *Which ordinary causal paths have actually been eliminated?*

TENET-H11 - Causal-Cone Exclusion Witness Principle. A RETURN earns temporal-anomaly credit only if every declared forward path capable of carrying terminal information into the preterminal register is either absent or crosses a clean monitored cut, with terminal freshness, provenance and no leakage inside the declared model class.

128 Declared graph, physical graph and preterminal register

Two objects are kept distinct:

$$G_D = (V, E_D), \quad G_\star = (V_\star, E_\star).$$

G_D is the declared, instrumented and auditable causal graph. $G_\star$ is the actual physical graph, not known exhaustively. Identifying the two would turn “no modeled channel” into “no physical channel.”

Let P be the preterminal register and let $\mathcal{S}_T^{pre}$ denote the nodes that, under the declared model, could already carry information about the terminal event before its explicit creation. Define the declared forward paths

$$\Pi_D = \Pi_{G_D}(\mathcal{S}_T^{pre}, P).$$

A monitored cut $\mathcal{K} \subseteq E_D$ covers the declared cone when

$$\text{Cover}(\mathcal{K}) = 1 \iff \forall \pi \in \Pi_D, \pi \cap \mathcal{K} \neq \emptyset.$$

Monitor cleanliness is a separate predicate:

$$\text{Clean}(\mathcal{K}) = 1 \iff \text{no relevant terminal-information transfer is detected on } \mathcal{K}.$$

Coverage and silence are therefore not the same claim.

129 Internal witness and external witness

The toy defines

$$X_D = \text{Cover}(\mathcal{K}) \wedge \text{Clean}(\mathcal{K}) \wedge H_{fresh}^D.$$

This still speaks only about G_D .

An external witness requires the debt that the protocol cannot manufacture internally:

$$X_\star = X_D \wedge H_{complete}(G_D, G_\star) \wedge H_{fresh}^\star \wedge \neg H_{forward}^\star.$$

$H_{complete}$ means that the relevant physical channels, common causes and side channels are genuinely covered by the declared instrument class. It is an empirical obligation, not a log field derivable from the same closed system.

TENET-NG3 - Declared-Cone Completeness No-Go. Proving that no path exists in G_D does not prove that no relevant physical causal path exists in $G_\star$. An unmodeled bypass or common cause can make the internal witness perfectly clean while leaving an ordinary forward explanation.

130 Executed adversarial matrix

PASS011 executes 128 trials per scenario. The abbreviated labels below map to the exact labels in the cumulative CSV.

Scenario	$G = B$	RETURN	$ \Pi_D $	$ \Pi_\star $	X_D	COS
CLEAN	64	0	0	0	1	0
PARTIAL_LEAK	69	0	2	2	0	0
ALL_PRECOMP	128	128	3	3	0	0
MODELED_BYPASS	128	128	3	3	0	0
UNMODELED_BYPASS	128	128	0	3	1	0
COMMON_CAUSE	128	128	0	1	1	0
TERMINAL_PINCER toy	128	128	0	0	1	128

Three findings are decisive.

1. A modeled leak or modeled bypass does not pass the witness: either a declared path activates the cut or the declared cut fails to cover the bypass.
2. An unmodeled bypass and an unmodeled common cause both produce 128/128 while $X_D = 1$. The internal witness can therefore be a perfect false positive.
3. The only scenario closing COS is `TERMINAL_PINCER_UNPHYSICAL_TOY`, because the code explicitly grants it a RETURN outside the forward graph while also declaring the toy physical graph complete. This closes the logic, not the physics.

131 What a surviving RETURN actually falsifies

The tested hypothesis is not merely “time goes forward.” The forward package is a conjunction:

$$H_F = H_{\text{graph}} \wedge H_{\text{time}} \wedge H_{\text{monitor}} \wedge H_{\text{fresh}} \wedge H_{\text{nohidden}}.$$

Here:

- H_{graph} : the relevant channel class is sufficiently complete;
- H_{time} : ordinary dependencies obey the declared forward chronology;
- H_{monitor} : monitored cuts correctly detect the transfers they are supposed to detect;
- H_{fresh} : terminal entropy is genuinely fresh after the seal;
- H_{nohidden} : no undeclared common cause or side channel carries the information.

If a terminal-bound preterminal RETURN survives while $X_\star = 1$, the immediate logical verdict is

$$\neg H_F.$$

But De Morgan gives

$$\neg H_F = \neg H_{\text{graph}} \vee \neg H_{\text{time}} \vee \neg H_{\text{monitor}} \vee \neg H_{\text{fresh}} \vee \neg H_{\text{nohidden}}.$$

Retrograde transport is one way to violate H_{time} ; it is not the only branch of the disjunction.

PASS011 verdict: a RETURN surviving a clean causal cone first rejects the declared forward-model bundle. Logic alone does not identify temporal orientation as the failed component.

132 Direct Encyclopedia crosswalk

PASS011 reuses existing guards:

- **S216**: causal cuts are bounded searches over admissible ports; exhaustivity remains relative unless separately proved;
- **S220**: the Insulation Tester separates leakage, pass error, retention, dissipation and temporal error; boundaries require typed calibrated probes;
- **S238**: nonlocality remains a last-resort residual category after local/mesoscopic completion;
- **S244**: RETURN_PROOF reconstructs provenance, dependencies and target leakage; a forward/backward meeting is not proof;
- **S252/S256**: COMMIT, lineage and freeze prevent later rewriting of the evidential dossier;
- **S262**: Temporal PRE/POST ancestry and future-leakage audits are already transport-validation objects.

The new synthesis is therefore

causal cut $\rightarrow$ coverage $\rightarrow$ clean monitors $\rightarrow$ freshness $\rightarrow$ RETURN $\rightarrow$ relative verdict.

133 Consequence for the human TENET

For the higher-self hypothesis the evidential bar rises again. Unexplained guidance, a valid receipt, a chronology quorum and even a clean declared cone do not yet justify

$$S^* \rightarrow S_t.$$

They justify at most the claim that the tested forward explanation class does not reconstruct the observed dependency.

The higher self becomes a Last-Survivor candidate only after competing families are eliminated in a sufficiently rich class:

$$\mathcal{M} = \{\text{local, distributed, common-cause, precompute,} \\ \text{hidden-channel, monitor-failure, retrograde, \dots}\}.$$

Even then, “retrograde” and “higher self” remain separate levels: the first concerns information direction; the second concerns source identity.

134 PASS011 ledger update

ID	Object	Status
TENET-H11	Causal-Cone Exclusion Witness Principle	CANDIDATE CORE / CLOSED CONDITIONAL TOY
TENET-CE1	G_D / G_* separation	WORKING GUARD
TENET-CE2	Monitored-cut coverage of every declared path	CLOSED FINITE TOY
TENET-CE3	Cut-clean + declared-freshness internal witness	CLOSED CONDITIONAL TOY
TENET-CE4	Modeled leak/bypass detection	CLOSED COUNTERMODEL

TENET-CE5	Unmodeled-bypass false positive	CLOSED
TENET-CE6	Unmodeled-common-cause false positive	COUNTERMODEL CLOSED
TENET-NG3	Declared-Cone Completeness No-Go	COUNTERMODEL STRENGTHENED EPISTEMIC GUARD
TENET-CE7	Rejection of forward conjunction H_F	CANDIDATE EVIDENTIAL CEILING
TENET-CE8	Retrograde transport as one branch of $\neg H_F$	CANDIDATE LAST-SURVIVOR ROUTE
TENET-HSP5	Higher-self attribution after cone exclusion	OPEN EMPIRICAL / ONTOLOGICAL

135 Direction for PASS012

PASS012 will attack the next problem: **adversarially saturating the forward explanation class** instead of declaring the graph complete by convention. The protocol must recursively add common causes, side channels, precomputation, monitor faults, hidden dependencies and alternative generators until the declared inventory reaches a fixed point under the chosen attack grammar.

The next question is:

After adversarial expansion of the model class, does retrograde transport become a genuine Last Survivor, or does it remain only one candidate among unobserved mechanisms?

136 PASS012 - Adversarial forward-explanation saturation and relative Last Survivor

Status: CUMULATIVE; FINITE-GRAMMAR FIXED POINT CLOSED IN THE TOY; RETROGRADE LAST SURVIVOR RELATIVE UNDER ASSUMPTIONS; GLOBAL UNIQUENESS OPEN.

PASS011 showed that a clean causal cone is only relative to the declared graph. PASS012 attacks that debt directly: instead of fixing the forward model class once, it recursively enlarges the class whenever a model is eliminated. The goal is not to make *retrograde* true by default, but to ask whether it survives an explicit countermodel campaign.

TENET-H12 - Adversarial Explanation Saturation Principle. Before a retrograde RETURN receives Last-Survivor credit, the forward explanation class must be recursively expanded under a declared attack grammar, with each newly generated family tested against the same question until a fixed point of that grammar is reached.

137 Expansion operator and relative fixed point

Let $\mathcal{M}_0$ be the initial mechanism class and let $\mathcal{A}_{\mathcal{G}}$ be the attack operator associated with a finite grammar $\mathcal{G}$:

$$\mathcal{M}_{n+1} = \mathcal{M}_n \cup \mathcal{A}_{\mathcal{G}}(\mathcal{M}_n).$$

Declared saturation occurs when

$$\mathcal{M}_{n+1} = \mathcal{M}_n = \mathcal{M}_{\mathcal{G}}^*.$$

This is a fixed point of *model generation*, not a theorem that physics contains no other mechanism. For evidence dossier E , define the relative survivors:

$$\text{Surv}_E(\mathcal{M}) = \{m \in \mathcal{M} : m \text{ is not eliminated by } E\}.$$

Retrograde transport R receives relative Last-Survivor status only if

$$\text{Surv}_E(\mathcal{M}_{\mathcal{G}}^*) = \{R\}.$$

138 PASS012 attack grammar

The toy begins with three families: direct local prediction, distributed relay, and retrograde RETURN. Failure of the first two generates increasingly adversarial families:

- precomputation and latent predictor;
- hidden bypass and common cause;
- freshness failure, registry backfill and embedded decoder;
- monitor failure and correlated beacons;
- quorum, ledger and verifier collusion;
- terminal pre-embedding;
- finally universal collusion.

The retrograde branch is not used to generate the forward branches; it remains a separate candidate evaluated against the same closure dossier.

Execution reaches its fixed point at round 4:

Round	New models	Cumulative	Main signal
0	3	3	local, distributed, retrograde
1	4	7	precompute, latent, bypass, common cause
2	5	12	freshness, backfill, decoder, monitor, correlation
3	4	16	specialized collusions, terminal pre-embedding
4	1	17	universal collusion

139 Four verdict scopes

The same fixed point yields four different verdicts depending on the proof scope allowed:

Scope	Survivors	Surviving set
Finite / internal only	5	ledger-collusion, quorum-collusion, verifier-collusion, universal-collusion, retrograde
Finite + external independence	1	retrograde only
Open world + external independence	2	retrograde, UNKNOWN_FORWARD
Open world / internal only	6	the five internal survivors + UNKNOWN_FORWARD

The positive result is therefore real but strictly relative:

$$\text{Surv}_E(\mathcal{M}_G^* \mid H_{indep}) = \{\text{RETROGRADE}\}.$$

Inside the finite grammar and under the external-independence premise, retrograde transport is the sole survivor.

But once the open world is restored:

$$\text{Surv}_E(\mathcal{M}_{open}) \supseteq \{\text{RETROGRADE}, \text{UNKNOWN_FORWARD}\}.$$

Uniqueness disappears.

TENET-NG4 - Open-World Last-Survivor No-Go. A Last Survivor obtained after saturation of a finite grammar is globally unique only if the generator class itself is exhaustive for all relevant physical mechanisms. In an open world where UNKNOWN $\neq$ ABSENT, an unrepresented forward mechanism remains a logical debt distinct from evidence for retrocausality.

140 Why universal collusion survives without external authority

PASS010 already showed that a fully collusive witness system can manufacture a coherent internal chronology. PASS012 recovers the same limitation from a different direction: LEDGER_COLLUSION, QUORUM_COLLUSION and VERIFIER_COLLUSION are not eliminated by internal evidence alone; all converge to UNIVERSAL_COLLUSION.

Thus two levels must remain distinct:

$$\text{closure of internal tests} \neq \text{independence of the test architecture.}$$

The premise H_{indep} can be audited, replicated and hardened, but it does not become a truth produced by the same system it certifies.

141 Gate ablation and front non-redundancy

PASS012 then removes each internal gate one at a time. When a gate owns a dedicated forward family, removing it reopens that family. The ablation proves no physical law; it only checks that

the singleton is not obtained by hiding useless tests or by eliminating a mechanism without an evidential owner.

The principle is

$$E \setminus \{e_i\} \Rightarrow \text{Surv}_{E \setminus \{e_i\}}(\mathcal{M}_{\mathcal{G}}^*) \text{ should readmit the families that } e_i \text{ was meant to exclude.}$$

The ablation CSV is included in the cumulative ZIP.

142 Direct Encyclopedia crosswalk

The PASS012 structure already exists in fragments across the corpus:

- **S001/S050/S074:** Last Survivor eliminates incompatible candidates, but necessity depends on the frame and filtering may leave zero, one or multiple survivors;
- **S252:** UNKNOWN $\neq$ ABSENT and a unique survivor supports global uniqueness only when the domain is complete;
- **S256:** an OPEN object must query all registered providers, but a missing provider remains engineering debt and never proves that no relevant law exists;
- **S220:** ALL-minus-one/ALL-minus-two ablations enforce fail-closed behavior, while software completeness remains distinct from physical completeness.

PASS012 therefore does not invent a new elimination logic. It applies the existing discipline to the precise question: *when may retrograde transport be called the final remaining mechanism?*

143 New status of retrograde transport

After PASS012, three statements must remain distinct:

1. **Relative survivor:** retrograde transport is the only remaining mechanism inside a finite saturated class under the declared gates and H_{indep} .
2. **Open-world candidate:** retrograde transport remains a major candidate, but an as-yet unknown forward mechanism prevents global uniqueness.
3. **Physical claim:** the toy establishes no physical retrograde realization.

The correct notation is therefore

$$R = \text{LastSurvivor}_{\mathcal{G}, E, H_{indep}} \quad (\text{relative}),$$

not

$$R = \text{LastSurvivor}_{\text{Nature}}.$$

144 Consequence for the higher self

This pass removes an important ambiguity from the human hypothesis. Even if a retrograde direction became a relative Last Survivor, this would still not identify the source:

$$\text{retrograde channel} \neq \text{future agent} \neq \text{higher self}.$$

The higher self requires a second Last-Survivor problem, now over *generator identity* rather than information direction.

The competing source families include:

- terminal self of the same identity;
- another future agent;
- future collective infrastructure;
- terminal environment;
- artificial source;
- agentless mechanism.

145 PASS012 ledger update

ID	Object	Status
TENET-H12	Adversarial Explanation Saturation Principle	CANDIDATE CORE / CLOSED FINITE TOY
TENET-AS1	Expansion operator $\mathcal{A}_G$	CLOSED FINITE TOY
TENET-AS2	Fixed point $\mathcal{M}_G^*$ after 4 rounds	CLOSED EXECUTED TOY
TENET-AS3	Finite inventory of 17 families	CLOSED EXECUTED TOY
TENET-AS4	Retrograde unique under finite grammar + external independence	RELATIVE LAST SURVIVOR
TENET-AS5	Universal collusion survives without external authority	CLOSED COUNTERMODEL / OPEN EXTERNALLY STRENGTHENED EPISTEMIC GUARD
TENET-NG4	Open-World Last-Survivor No-Go	CLOSED LOGICAL WITNESS
TENET-AS6	UNKNOWN_FORWARD destroys global uniqueness	CLOSED
TENET-AS7	Gate-by-gate ablation	INTERNAL AUDIT
TENET-AS8	Physical exhaustivity of the attack grammar	OPEN / IRREDUCIBLE COMPLETENESS DEBT
TENET-HSP6	Higher-self attribution after retrograde direction	OPEN SOURCE- IDENTITY PROBLEM

146 Direction for PASS013

PASS013 changes axis: suppose, conditionally, that retrograde direction has received relative Last-Survivor status. We must then build a **terminal source-identity witness**.

The next question is:

What traces could show that the RETURN comes from the *same personal invariant* I_S at its terminal boundary, rather than from another future agent, a collective infrastructure or an agentless mechanism?

PASS013 will attack the chain

retrograde direction $\rightarrow$ terminal source $\rightarrow$ same-self lineage $\rightarrow$ higher-self candidate.

147 PASS013 - Terminal source identity and same-self lineage

Status: CUMULATIVE; INTERNAL LINEAGE WITNESS CLOSED IN THE TOY; PERSONAL ONTOLOGICAL IDENTITY REMAINS OPEN.

PASS012 conditionally isolated the *direction* of information. PASS013 asks a different question: if a RETURN is retrograde, who or what is at the terminal end? A retrograde message can in principle be generated by the same future person, another future agent, a collective archive, an artificial system or a non-agentic replay mechanism.

TENET-H13 - Terminal Lineage Witness Principle. Source attribution to the same self requires more than content similarity. At minimum, the terminal packet must bind to the same identity root, preserve an append-only lineage, retain non-public traces without erasure, reconstruct selected branch-specific checkpoints, and answer a fresh terminal challenge under the same lineage.

148 Identity hierarchy

PASS013 separates five increasingly strong claims:

same output < same memories < same causal lineage
< same event-continuity class < same ontological self.

None of the arrows is automatic. The hierarchy is deliberately stricter than “the source knows things only I would know.” Knowledge can be copied; credentials can be stolen; archives can be centralized; generators can be reproduced.

For a present self root S_0 and a terminal claimant Q_T , define the internal lineage witness

$$W_{lin}(Q_T, S_0) = (Root, Chain, Trace, Rebuild, NoErase, Adaptive).$$

The internal witness passes only if all six coordinates pass.

149 Executable lineage toy

The PASS013 toy creates one deterministic eight-commit personal history. Every commit binds the same root ObjectID, previous commit, branch choice and a non-public trace. A one-way evolving key is used only as a reproducible custody device; it is not interpreted as a soul or consciousness variable.

The gates are:

1. **ROOT_BINDING**: the packet resolves to the same persistent root identifier;
2. **APPEND_ONLY_LINEAGE**: every commit links exactly to its predecessor;
3. **PRIVATE_TRACE_RETENTION**: branch-specific non-public traces are retained;
4. **CHECKPOINT_RECONSTRUCTION**: selected past states can be regenerated from retained choice+trace material;
5. **NON_ERASURE**: mistakes, failed plans and losses remain in the ledger rather than being rewritten away;
6. **ADAPTIVE_TERMINAL_RESPONSE**: a fresh challenge created after the lineage is answered using the terminal lineage state.

150 Countermodel matrix

Eight source candidates are submitted to the same witness:

Candidate	Internal witness	External continuity	Meaning
SAME_SELF	PASS	PASS	intended same-lineage source
OTHER_AGENT_PUBLIC_COPY	FAIL	FAIL	surface/public knowledge only
OTHER_AGENT_STOLEN_LINEAGE	PASS	FAIL	another agent with full archive+credential
COLLECTIVE_CUSTODIAN	PASS	FAIL	future infrastructure with custody
AI_CLONE_FULL_ARCHIVE	PASS	FAIL	artificial generator with copied lineage
AGENTLESS_STATIC_REPLAY	FAIL	FAIL	prerecorded output, no fresh adaptation
AGENTLESS_GENERATIVE_REPLAY	PASS	FAIL	non-agentic generator holding archive+key
SAME_SELF_GAPPED_LINEAGE	FAIL	PASS	claimed self with broken internal lineage

The decisive result is therefore

$$|\{Q : W_{lin}(Q, S_0) = 1\}| = 5.$$

The internal transcript does *not* isolate SAME_SELF.

TENET-NG5 - Transcript Identity No-Go. If two candidate generators possess the same lineage material and induce the same complete declared transcript, no test computed only from that transcript can distinguish their ontological personal identity.

This is not a statement that identity is unreal. It is an identifiability limit on the chosen evidence surface.

151 Adaptive response does not prove agency

The static replay fails the fresh challenge, but AGENTLESS_GENERATIVE_REPLAY passes it. Therefore

fresh adaptive output $\neq$ proof of agency.

An adaptive function can be carried by a person, an AI, a collective or a non-agentic mechanism. Agency needs its own carrier and criterion.

TENET-NG6 - Adaptive-Agency No-Go. Challenge-response competence certifies a generating capability, not the metaphysical or phenomenological status of the generator.

152 Relative same-self result

PASS013 adds a separate premise H_{cont} : an independently warranted personal-continuity relation linking the terminal claimant to S_0 . In the toy, only SAME_SELF is assigned that external relation. Then

$$\text{Surv}(W_{lin} \wedge H_{cont}) = \{\text{SAME_SELF}\}.$$

This is a *relative* source-identity closure. The uniqueness comes from the combination of lineage evidence and an external continuity premise; it is not manufactured by the transcript itself.

The correct evidential ceiling is

$$W_{lin} \Rightarrow \text{same causal-lineage/custody class}$$

and, only under H_{cont} ,

$$W_{lin} \wedge H_{cont} \Rightarrow \text{same-self candidate.}$$

Neither expression is yet a physical or ontological proof of soul identity.

153 Direct Encyclopedia crosswalk

The structure is already anticipated by several CT objects:

- **S252**: Object, Identity, State and locator are distinct; persistent logical identity is carried with immutable snapshots and explicit lineage/provenance references.
- **S229**: equal state coordinates do not imply the same event; event identity also uses causal age and branch/singularity context. Structural memory preserves reconstructibility rather than merely copying a surface state.
- **S143**: a scalar coherence threshold is not a proof of identity stability; the identity relation itself remains an OPEN obligation.
- **S007**: persistence of a message under propagation does not identify the sender who generated it.

PASS013 therefore strengthens the TENET source problem without importing software identifiers as a theory of human consciousness.

154 Consequence for the soul and higher self

The earlier candidate definition

$$\text{soul} \approx \text{identity invariant through a life}$$

now needs one more object: the equivalence relation that says which transformed carriers count as the *same person*. A lineage witness can preserve history, but lineage alone allows copying.

Thus the higher-self claim now decomposes as

$$\begin{aligned} \text{retrograde RETURN} &\rightarrow \text{terminal lineage} \rightarrow \text{personal continuity} \\ &\rightarrow \text{same-self candidate} \rightarrow \text{higher-self ontology.} \end{aligned}$$

PASS013 closes only the second arrow in a finite computational toy and shows why the third cannot be skipped.

155 PASS013 ledger update

ID	Object	Status
TENET-H13	Terminal Lineage Witness Principle	CANDIDATE CORE / CLOSED TOY
TENET-ID1	Root/append-only lineage witness	CLOSED EXECUTED TOY
TENET-ID2	Private-trace retention + checkpoint reconstruction	CLOSED EXECUTED TOY

TENET-ID3	Non-erasure identity ledger	CLOSED
TENET-ID4	Fresh adaptive terminal challenge	EXECUTED TOY CLOSED
TENET-ID5	Five-source observational equivalence class	EXECUTED TOY CLOSED
TENET-NG5	Transcript Identity No-Go	COUNTERMODEL CLOSED
TENET-NG6	Adaptive-Agency No-Go	IDENTIFIABILITY GUARD CLOSED
TENET-ID6	External personal-continuity premise H_{cont}	IDENTIFIABILITY GUARD OPEN EXTERNAL
TENET-ID7	Same-self uniqueness under $W_{lin} \wedge H_{cont}$	BRIDGE RELATIVE LAST
TENET-HSP7	Higher-self attribution after lineage witness	SURVIVOR CANDIDATE RELATIVE / ONTOLOGY OPEN

156 Direction for PASS014

PASS014 must now attack the missing relation H_{cont} rather than hide it. The hard cases are duplication and carrier change: biological continuation, destructive teleportation, perfect copying, AI emulation, memory replacement and branch duplication.

The next question is:

Which equivalence relation can make two temporally separated carriers the *same person* without collapsing identity into memory, credentials, body continuity or mere similarity?

PASS014 will therefore be the **Personal-Continuity / Duplication Attack**: if two terminal claimants inherit the same memories and lineage, can both be “the same self”? If not, what additional invariant breaks the symmetry?

157 PASS014 - Personal Continuity, Duplication, and the Fission Attack

Status: CUMULATIVE; TWO LOGICAL NO-GO RESULTS CLOSED; LINEAR-CONTINUITY FORM CANDIDATE; PHYSICAL ONTOLOGY OPEN.

PASS013 showed that even a strong lineage transcript does not isolate the same self. PASS014 attacks the missing relation H_{cont} directly using cases in which personal information is preserved, copied, migrated, or duplicated.

PASS014 question. What must continue for B to be numerically the same person as A , rather than a re-expression, copy, or perfectly faithful causal descendant of A ?

158 State, memory, lineage, and numerical identity

The corpus already provides four useful guards:

- S229 separates event identity from recurrence of the same state: equal state coordinates do not make the same event;
- S252/S226 separate ObjectID, StateID, locator, material carrier, and logical identity;
- S022 treats teleportation as local reconstruction/re-expression of structure, not automatic transport of matter or numerical identity;
- S211 preserves forks and forbids silent fusion of competing histories.

PASS014 therefore imposes the hierarchy

same state < same memories < same lineage < same numerical continuation.

Every level can be shared by several descendants unless an additional constraint forbids duplication of continuation itself.

159 Copyable-invariant no-go

Let $C(A)$ be any copyable dossier used by an identity test: state, memory, transcript, personality model, credentials, and serialized lineage. Suppose an operation produces distinct copies B and C such that

$$C(B) = C(C) = C(A).$$

Any deterministic test K depending only on C then yields

$$K(A, B) = K(A, C).$$

If K accepts B , it also accepts C . It therefore cannot select a unique numerical continuation.

TENET-NG7 - Copyable-Invariant Identity No-Go. No criterion built only from perfectly duplicable information can, under symmetric duplication, uniquely identify the same numerical self.

This no-go does not make the invariant useless. It changes its type: a copyable invariant can define a *personal type class*, but not necessarily the unique numerical token of that person.

160 Exact fission theorem

Assume $\equiv_S$ denotes numerical personal identity and is an equivalence relation. If a source A is declared identical to two simultaneous distinct successors B and C ,

$$A \equiv_S B, \quad A \equiv_S C, \quad B \neq C,$$

then symmetry gives $B \equiv_S A$, and transitivity gives

$$B \equiv_S C.$$

But numerical identity requires $B \equiv_S C$ not to hold between two distinct individuals. Contradiction.

TENET-NG8 - Fission Numerical-Identity No-Go. A transitive numerical identity cannot be preserved simultaneously in two distinct successors of one source. Fission requires either a branch-selecting asymmetry, identity break/indeterminacy, or replacement of numerical identity by a weaker relation.

161 Executable PASS014 toy

The toy materializes ten scenarios: ordinary biological continuity, severe memory loss, gradual carrier replacement, a perfect copy while the original survives, AI emulation, single destructive teleportation, double destructive teleportation, and two fission branches.

Six criteria are compared:

1. autobiographical memory alone;
2. exact state/pattern;
3. material continuity;
4. causal lineage alone;
5. nonbranching causal lineage;
6. an abstract linear-continuity resource Λ_S .

Scenario	Memory	Lineage	Nonbranching	Λ_S
Biological continuity	PASS	PASS	PASS	PASS
Biological memory loss	FAIL	PASS	PASS	PASS
Gradual carrier replacement	PASS	PASS	PASS	PASS
Perfect copy, original survives	PASS	PASS	FAIL	FAIL
AI emulation, original survives	PASS	PASS	FAIL	FAIL
Single destructive teleport	PASS	PASS	PASS	UNKNOWN
Double destructive teleport L/R	PASS	PASS	FAIL	UNKNOWN
Branch fission L/R	PASS	PASS	FAIL	UNKNOWN

The structural result is that memory is not necessary in the toy, causal lineage is insufficient under fission, and material continuity forbids carrier migration by definition. Nonbranching lineage is a better candidate *form*, but it does not prove that a unique destructive-teleport reconstruction is the same subject rather than a unique copy.

162 Linear-continuity candidate

PASS014 therefore introduces an abstract variable, not as new physics but as the minimal logical form of the remaining debt:

$$\Lambda_S(t) \in \{0, 1\} \quad \text{per active continuation,}$$

with a linearity rule

$$\sum_{j \in \text{Succ}(A)} \Lambda_S(B_j) = 1$$

whenever numerical continuation of A exists and is certifiable.

Interpretation: Λ_S is not a mental state. It is a hypothetical witness of *which continuation* still carries the same personal token. A copy or clone may reproduce state and invariant without automatically reproducing Λ_S .

TENET-H14 - Linear Personal-Continuity Candidate. Numerical continuity of the self, if it exists, must behave as a non-duplicable continuation resource: at symmetric fission it cannot certify both branches as the same self.

This is an identifiability schema. It establishes no physical field, quantum, or substance actually carrying Λ_S .

163 Refining the soul hypothesis

PASS001-PASS013 brought the soul close to an identity invariant. PASS014 shows that *type* and *token* must be separated:

$$\mathcal{A}_{type} = \text{Inv}\{S_t\}$$

may be copied or multiply instantiated, while unique numerical identity would additionally require

$$\mathcal{A}_{token} \sim \Lambda_S.$$

A candidate formulation becomes

$$\text{Soul candidate} = (\mathcal{A}_{type}, \Lambda_S).$$

The first term answers “what remains recognizable as you?” The second answers “which continuation is still numerically you?”

The second term is much stronger and remains entirely OPEN as ontology. PASS014 therefore does not prove a soul; it shows why a purely informational definition of soul is insufficient under duplication.

164 Destructive teleportation remains underdetermined

S022 establishes only a chain of structural re-expression. For one destination after source destruction, three readings remain compatible:

1. the person continues because causal reconstruction is sufficient;
2. the person dies and one unique copy appears;
3. the question depends on an additional continuity transport not yet defined.

The toy therefore marks DESTRUCTIVE_TELEPORT_SINGLE as UNKNOWN under Λ_S .

Double duplication is stronger: state and memory cannot break the symmetry. A theory preserving numerical identity must add a verifiable physical/causal asymmetry or declare continuity uncertifiable after fission.

165 Consequence for the higher self

The higher self can no longer be defined merely as a terminal version sharing the same invariant. A perfect future copy could satisfy that condition.

The candidate condition becomes

$$S^* \text{ is the same self only if } \mathcal{A}_{type}(S^*) = \mathcal{A}_{type}(S_0) \wedge H_{cont}^{lin}(S_0, S^*).$$

Here H_{cont}^{lin} must supply a nonbranching chain or independently justified transfer of Λ_S . Without the second half, S^* is at most an isomorphic/descendant terminal of the present self.

166 PASS014 ledger update

ID	Object	Status
TENET-H14	Linear Personal-Continuity Candidate	CANDIDATE / PHYSICAL
TENET-PC1	State < memory < lineage < numerical-continuity hierarchy	BRIDGE OPEN WORKING
TENET-PC2	Memory not necessary in the continuity toy	STRUCTURAL RESULT
TENET-PC3	Causal lineage insufficient under fission	CLOSED COUNTEREXAMPLE
TENET-NG7	Copyable-Invariant Identity No-Go	CLOSED COUNTEREXAMPLE
TENET-NG8	Fission Numerical-Identity No-Go	CLOSED LOGICAL NO-GO
TENET-PC4	Nonbranching lineage	CLOSED LOGICAL NO-GO
TENET-PC5	Continuity resource Λ_S	CANDIDATE FORM / NOT ONTOLOGY PROOF
TENET-PC6	Soul-type $\mathcal{A}_{type}$	ABSTRACT CANDIDATE / ONTOLOGY OPEN
TENET-PC7	Soul-token $\mathcal{A}_{token} \sim \Lambda_S$	CANDIDATE DESCRIPTIVE INVARIANT
TENET-PC8	Single destructive teleportation	OPEN ONTOLOGICAL
TENET-PC9	Double teleportation/fission	CANDIDATE OPEN / NOT DETERMINED
TENET-HSP8	Higher self = common invariant + linear continuity	CLOSED FISSION NO-GO
		CANDIDATE / CONTINUITY BRIDGE OPEN

167 Direction for PASS015

PASS014 introduced Λ_S as a logical form, not a substance. PASS015 must now attack it: can this new token be *eliminated* and all personal continuity derived from existing CT objects - occurrence, non-erasure, ObjectID, CausalTransition, a nonbranching worldline/lineage, and RETURN - or is a genuinely new primitive required?

PASS015 - Token Reduction Attack: is Λ_S merely a compact name for one unique causal worldline, or an irreducible new variable?

168 PASS015 - Token Reduction Attack

Status: CLOSED CONDITIONAL FORMAL REDUCTION / PERSONAL-ONTOLOGY BRIDGE OPEN.

PASS014 introduced Λ_S only to represent the following debt: if a self A has several causally compatible descendants, which continuation can still be numerically the same self? PASS015 now tries to eliminate that variable instead of prematurely turning it into a substance.

PASS015 question. Can Λ_S be reconstructed entirely from CT objects already present - occurrence, non-erasing trace, ObjectID, committed transition, nonbranching lineage and RETURN - without adding a fundamental “soul token”?

169 Proposed structural reduction

Let $G_C = (V, E_C)$ be the graph of actually committed states of an object. Edges $E_C^{id} \subseteq E_C$ are those that the typed contract declares to be identity-continuation transitions or documented migrations. For source A and target B , PASS015 defines

$$\hat{\Lambda}_S(B \mid A) = 1$$

if and only if all of the following hold:

1. B has valid ancestry from A ;
2. every relevant transition is committed and its history is append-only;
3. the same logical object is preserved, or an explicit audited migration is supplied;
4. RETURN/reconstruction validates the chain without target leakage;
5. the identity-continuation subgraph has one unique live successor at the point in question.

A fission with two continuation successors does not receive two values of 1: it returns AMBIGUOUS. Broken ancestry, rewrite, or missing RETURN returns FAIL.

TENET-H15 - Unique Committed Lineage Reduction. In a declared finite runtime where the identity-continuation relation is a partial function, the abstract token Λ_S is reducible to the derived property $\hat{\Lambda}_S$ of unique committed continuation. No separate numerical “soul token” register is required to prevent formal duplication.

170 Conditional non-duplication theorem

Assume the continuation relation $f : V \rightarrow V$ is a partial function. Then for every state A ,

$$|Succ_{id}(A)| \leq 1.$$

By composition, the active continuation descendant of A , when one exists, is unique in the declared chain. Hence

$$\hat{\Lambda}_S(B \mid A) = 1 \left[B = f^k(A) \text{ for some } k \geq 0 \text{ and } B \text{ is the active continuation} \right].$$

Non-duplication no longer comes from a mysterious conserved quantity; it comes from the functional structure of the continuation graph.

Important limit. The theorem already assumes that we know which edges are identity-continuation transitions. It does not prove that a CONTINUE or MIGRATE label corresponds to lived personal identity.

171 PASS015 executable toy

The toy executes fifteen scenarios. Principal results are:

Scenario	$\hat{\Lambda}_S$	Unique lineage	Reading
Biological continuity	PASS	yes	reduction succeeds
Biological memory loss	PASS	yes	memory unnecessary
Gradual carrier replacement	PASS	yes	material migration compatible
Original + copy	PASS original / FAIL copy	yes for original	copy separated by edge type
Stolen ObjectID	FAIL	uncertified	ID alone insufficient
Two-continuation fork	AMBIGUOUS	no	fission blocked
Single destructive migration	PASS	yes	formally same object
RETURN missing	FAIL	yes but uncertified	reconstruction mandatory
Ledger rewrite	FAIL	yes but not append-only	non-erasure mandatory

The internal result is strong: Λ_S as a bookkeeping variable is redundant. ObjectID + typed transitions + nonbranching + append-only + RETURN suffice to derive the same selection function inside the declared contract.

172 The counterworld that blocks ontological promotion

PASS015 then constructs two toy worlds with exactly the same formal observables:

- same ObjectID;
- same MIGRATE transition;
- same parent, commit, append-only ledger and RETURN;
- same unique destination;
- same value $\hat{\Lambda}_S = 1$.

In world W_{cont} , the toy ontology says the person continues. In world W_{copy} , the source dies and only a perfect copy appears. The two formal records are identical.

Therefore every classifier K that sees only those records satisfies

$$K(W_{cont}) = K(W_{copy}).$$

It cannot decide the hidden proposition “numerically the same self” if that proposition has no additional observable effect.

TENET-NG9 - Formal-Record Personal-Identity Underdetermination. If two personal-continuity ontologies induce exactly the same ObjectID, transition graph, ledger, traces and RETURN, no test built only from those objects can distinguish the ontologies.

This no-go does not claim both worlds are physically possible. It says that with the current observables the CT dossier does not yet contain the discriminator that would separate them.

173 What is eliminated and what remains

PASS015 eliminates one bad direction: there is no need to immediately postulate a new Λ_S field travelling with a soul. For internal mechanics,

$$\Lambda_S \rightsquigarrow \hat{\Lambda}_S = \text{UniqueCommittedLineage}.$$

But the ontological question becomes sharper. A bridge is still required:

$$B_{self} : \text{LogicalContinuation} \longrightarrow \text{PersonalNumericalIdentity}.$$

That bridge must be either:

1. operational and falsifiable through an observable that differs between W_{cont} and W_{copy} ;
2. derivable from an independent physical theory;
3. or explicitly retained as a metaphysical hypothesis not decided by model observables.

Without such a bridge, assigning the same ObjectID after destructive teleportation merely moves the question into the label.

174 Revision of the soul hypothesis

The type/token split of PASS014 is retained but simplified. The token need no longer be a primitive variable:

$$\mathcal{A}_{type} = \text{Inv}\{S_t\}, \quad \mathcal{A}_{lin} = \hat{\Lambda}_S.$$

A same-self hypothesis still requires

$$\text{SameSelf}(A, B) = \mathcal{A}_{type}(A, B) \wedge \mathcal{A}_{lin}(A, B) \wedge B_{self}(A, B).$$

The first term protects the descriptive invariant. The second protects nonbranching structural continuity. The third remains the ontological debt.

PASS015 therefore removes the idea that a soul must be an extra “token” merely because copying is possible. What remains open is not an automatically required hidden object, but the identification between logical continuation and numerical personal identity.

175 Consequence for the higher self

The candidate condition becomes cleaner:

$$S^* \text{ is the same self} \Rightarrow \mathcal{A}_{type}(S_0, S^*) \wedge \hat{\Lambda}_S(S^* | S_0) \wedge B_{self}(S_0, S^*).$$

A terminal that has the right invariant but comes from a fork, clone or independent re-expression fails structural continuity. A terminal with invariant + unique lineage passes the formal test, but the attribution “numerically me” remains conditional on B_{self} .

176 PASS015 ledger update

ID	Object	Status
TENET-H15	Unique Committed Lineage Reduction	CLOSED CONDITIONAL FORMAL
TENET-TR1	$\hat{\Lambda}_S$ derived from committed graph	CLOSED FINITE TOY
TENET-TR2	Non-duplication under functional continuation relation	CLOSED CONDITIONAL
TENET-TR3	Stolen ObjectID or invalid ancestry	CLOSED NEGATIVE CONTROL

TENET-TR4	Two-continuation fork	AMBIGUOUS / NO
TENET-PC10	Λ_S as bookkeeping primitive	DOUBLE TOKEN
TENET-NG9	Formal-Record Personal-Identity Underdetermination	REDUCIBLE
		CLOSED LOGICAL
		NO-GO
TENET-PC11	B_{self} : logical $\rightarrow$ personal identity	OPEN BRIDGE
TENET-PC12	Single destructive teleportation	FORMAL PASS /
		PERSONAL ID
		OPEN
TENET-PC13	Soul as invariant + unique lineage + bridge	CANDIDATE /
		ONTOLOGY OPEN
TENET-HSP9	Higher self after token reduction	CANDIDATE /
		B_{self} OPEN

177 Direction for PASS016

PASS015 succeeded in destroying Λ_S as a new formal primitive, but did not reduce numerical personal identity. PASS016 must attack B_{self} directly: is there an experiment, observable or independent CT law that can distinguish W_{cont} from W_{copy} ? If no operational difference is possible even in principle, the bridge becomes empirically idle and must remain metaphysical rather than scientific.

PASS016 - Personal-Identity Bridge Attack: is B_{self} operationalizable and falsifiable, or only an ontological convention?

178 PASS016 - Personal-Identity Bridge Attack

Status: CLOSED CONDITIONAL IDENTIFIABILITY RESULT / NEW PHYSICAL DISCRIMINATOR OPEN.

PASS015 reduced the bookkeeping token Λ_S to unique committed lineage, but left one bridge open:

$$B_{self} : \text{LogicalContinuation} \longrightarrow \text{PersonalNumericalIdentity}.$$

PASS016 asks whether this bridge has any operational content. The test is deliberately strict: if two worlds differ only in the proposition “numerically the same person” while every trusted observable is identical, no experiment built from those observables may receive identity credit.

PASS016 question. Is there an independently specified observable whose outcome law differs between a continuation world W_{cont} and a perfect-copy world W_{copy} , or is B_{self} empirically idle relative to the present measurement class?

179 Operational-equivalence definition

Let $\mathcal{M}$ be a declared class of admissible measurements. For world W and measurement M , let

$$P_W^M(o) = \Pr(O = o \mid W, M).$$

Define

$$W_{cont} \sim_{\mathcal{M}} W_{copy} \iff P_{cont}^M = P_{copy}^M \quad \forall M \in \mathcal{M}.$$

If this equality holds, every decision rule using only outcomes of $\mathcal{M}$ has the same evidence distribution in both worlds. With equal priors,

$$TV(P_{cont}^M, P_{copy}^M) = 0, \quad I(W; O \mid M) = 0, \quad Acc^* = \frac{1}{2}.$$

This is an identifiability statement, not a claim that the two ontologies are identical.

TENET-NG10 - Operational Personal-Identity Indistinguishability. If W_{cont} and W_{copy} induce the same outcome distribution for every admissible trusted measurement, no experiment in that measurement class can distinguish numerical personal identity.

180 PASS016 executed experiment bank

The finite toy instantiates fifteen trusted observable channels inherited from earlier passes: ObjectID, state hash, autobiographical memory digest, private trace digest, lineage root, transition kind, commit digest, RETURN, append-only status, unique-lineage status, physiology, behavior, first-person self-report, adaptive response key and the terminal-pincer receipt.

A bank of 256 reproducible adaptive measurements queries changing subsets of those channels and hashes their responses under fresh challenges. W_{cont} and W_{copy} are constructed to differ only in the hidden proposition of numerical identity.

Quantity	Result
Adaptive experiments	256
Different outputs	0
Total-variation distance	0
Mutual information about world identity	0 bit
Optimal equal-prior classification accuracy	1/2

The result is not repaired by adding more challenges over the same channels. If the observation vectors are identical, any deterministic post-processing of those vectors is also identical; randomized post-processing cannot create information about which world generated them.

181 Why first-person report is not enough

Both worlds can output “I continued.” A perfect reconstruction that contains the same memories, dispositions and language model has access to the same reasons for producing that sentence. The report may be evidence about current experience or self-model, but in this counterworld it is not an external discriminator of numerical continuity.

This extends the PASS013 result that adaptive response does not prove agency, and the PASS015 result that a perfect formal lineage does not prove numerical same-self.

182 Candidate-bridge audit

Candidate	PASS016 status	Reason
Autobiographical memory	FAIL	Perfect copy carries the same memory content.
Subjective self-report	FAIL	Perfect copy can make the same first-person report.
ObjectID + RETURN	FAIL	PASS015 counterworlds share the same formal record.
Biological material continuity	SURROGATE ONLY	Distinguishes uninterrupted organism from destructive reconstruction, but the statement “material continuity = personhood” is an additional bridge.
Nonbranching worldline	DEFINITIONAL	It selects one path only if same-self is already defined as following that path.
Noncopyable physical marker	CANDIDATE	Could track one process if genuinely noncopyable and measurable; its relation to personhood remains independent debt.
Independent history-sensitive fiber	CANDIDATE OPEN	Could distinguish histories not fixed by present state, but requires a second physical channel and calibration.
Primitive same-self bit	CIRCULAR	It encodes the answer to be proved and therefore supplies no independent evidence.

183 Observable-Difference Requirement

Define the best possible operational distance

$$D_{self} = \sup_{M \in \mathcal{M}_{phys}} TV(P_{cont}^M, P_{copy}^M).$$

Then the empirical bridge has content only if a prospectively fixed, physically admissible and independently motivated measurement yields

$$D_{self} > 0.$$

If every physically admissible measurement gives $D_{self} = 0$, the two personal-identity interpretations are empirically equivalent in that theory.

TENET-H16 - Observable-Difference Requirement. B_{self} can enter the scientific layer only through an independently specified measurement channel whose outcome distribution differs between continuation and copy worlds. Naming a new variable “same-self” is not such a channel.

This requirement follows the same general pattern already present in the Encyclopedia: hidden information has empirical meaning only if it reaches an independent readout; two ontologies producing identical observable regularities require a discriminating observable; an observation-only detector cannot distinguish states identical on all trusted observables.

184 The hypothetical positive channel

For calibration, the toy also adds a fictitious channel

$$M_{self}(W_{cont}) = 1, \quad M_{self}(W_{copy}) = 0.$$

This gives $TV = 1$ and perfect classification. But it proves nothing because the channel was defined directly from the hidden answer. The exercise shows what is missing structurally: not more inference over old observables, but a new independently realized carrier whose physics predicts the difference before the identity label is inspected.

185 Status of the strongest remaining candidates

185.1 Biological/process continuity

An uninterrupted organism supplies a strong practical continuity witness. It separates ordinary survival from destructive reconstruction. PASS016 does not promote it to numerical identity because the identity question is exactly whether interruption/re-expression can preserve the same person.

185.2 Noncopyable marker

A genuinely noncopyable physical degree of freedom could enforce one-to-one tracking. However, noncopyability alone establishes uniqueness of a carrier, not identity of a person. The theory would still need a bridge between that degree of freedom and personal continuity.

185.3 Independent history fiber

This is the most structurally interesting remaining candidate. If two present states are otherwise identical but retain a measurable physical history coordinate η_{hist} not determined by current state, then a history-sensitive readout can in principle produce

$$P(M_{hist} | W_{cont}) \neq P(M_{hist} | W_{copy}).$$

S209 already contains an analogous architecture in another domain: an extra fiber has novelty only if it is independent of the ordinary state and reaches a genuinely independent readout. PASS016 does *not* claim that such a personal history fiber exists.

186 What PASS016 says about soul and higher self

The soul hypothesis is narrowed again. Current data support no need for an extra “same-self bit.” The strongest scientific form now is conditional:

$$\text{SameSelf}(A, B) \Rightarrow \mathcal{A}_{type}(A, B) \wedge \hat{\Lambda}_S(B \mid A) \wedge D_{self} > 0$$

if personal identity is intended as an empirically distinguishable physical relation.

If no such observable can exist even in principle, B_{self} may remain a philosophical/metaphysical identity relation, but it must not be presented as an experimentally certified CT quantity.

For the terminal/higher self this means:

retrograde source + correct lineage $\nRightarrow$ numerically same higher self.

A physical same-self attribution requires either a new discriminator or an explicit extra-empirical premise.

187 PASS016 ledger update

ID	Object	Status
TENET-NG10	Operational Personal-Identity Indistinguishability	CLOSED
TENET-H16	Observable-Difference Requirement for B_{self}	CONDITIONAL
TENET-PI1	256-experiment trusted-observable equivalence bank	WORKING
TENET-PI2	$TV = 0, I = 0$, optimal accuracy = $1/2$	CANDIDATE
TENET-PI3	Subjective self-report as identity discriminator	CLOSED
TENET-PI4	Biological material continuity	EXECUTED TOY
TENET-PI5	Nonbranching worldline as personal identity	CLOSED FINITE
TENET-PI6	Noncopyable physical marker	RESULT
TENET-PI7	Independent history-sensitive fiber	REFUTED IN TOY
TENET-PI8	Primitive same-self observable bit	CLASS
TENET-PC14	B_{self} under current trusted observables	SURROGATE ONLY
TENET-HSP10	Higher-self same-person attribution	DEFINITIONAL
		WITHOUT BRIDGE
		CANDIDATE IF
		MEA-
		SURED/BRIDGED
		CANDIDATE OPEN
		CIRCULAR
		EMPIRICALLY
		IDLE RELATIVE
		TO CLASS
		NEW OBSERVABLE
		OR EXTRA-
		EMPIRICAL
		PREMISE
		REQUIRED

188 Direction for PASS017

PASS016 has localized the remaining scientific route. PASS017 must attack the strongest noncircular candidate: an independent history-sensitive carrier. The task is to determine whether a personal η_{hist} can be defined and measured independently of current state, archive, ObjectID and the desired same-self verdict, or whether the proposed history fiber collapses back into ordinary causal records.

PASS017 - Personal History-Fiber Attack: can an independent noncopyable history-sensitive carrier exist without encoding the identity answer by definition?

189 PASS017 - Personal History-Fiber Attack

Status: CLOSED INTERNAL NO-GOS / INDEPENDENT PHYSICAL HISTORY CARRIER OPEN.

PASS016 localized the only noncircular scientific route still capable of making personal numerical identity empirically distinguishable: a history-sensitive degree of freedom that is not already contained in current state, memory, archive, ObjectID, lineage or RETURN. PASS017 now tests whether such an η_{hist} is really new information or merely another encoding of records already known to be copyable.

PASS017 question. Can a personal η_{hist} add information that survives the W_{cont}/W_{copy} equivalence without encoding “same self” by definition, and can it remain unique under duplication?

190 Data-processing collapse of reconstructible history

Let $W \in \{W_{cont}, W_{copy}\}$ and let O denote the complete current trusted-observable record of PASS016. Suppose a proposed history variable is generated by

$$\eta = f(O, U),$$

where U is either absent or has the same conditional law in both worlds. Then

$$W \longrightarrow O \longrightarrow \eta$$

is a Markov chain. By data processing,

$$I(W; \eta) \leq I(W; O).$$

PASS016 already established $I(W; O) = 0$ in the declared counterworld. Hence

$$I(W; \eta) = 0.$$

Equivalently, a deterministic re-encoding of a record whose distribution is identical in both worlds remains identical.

TENET-NG11 - Reconstructible-History Collapse No-Go. Any proposed personal-history variable computable entirely from state, memory, archive, ledger, ObjectID, reconstruction metadata or other observables already equivalent in W_{cont} and W_{copy} cannot increase personal-identity discriminability.

This closes a large class of apparent solutions without assuming that no independent history carrier can exist.

191 What the Encyclopedia already separates

191.1 S229 structural memory

G_{mem} preserves reconstructibility across causal separation by retaining the relation that recovers state from origin, law, causal coordinate, sparse innovation and integrity commitments. This is useful memory/replay architecture, but a perfect copy supplied with the same reconstruction packet can reproduce it. PASS017 therefore classifies G_{mem} as *record/reconstruction level*, not as the missing independent fiber.

191.2 S209 independent history fiber

S209 contains a much closer architecture. A causal state may be represented as

$$C = (\rho, \eta),$$

with ordinary measurements depending only on ρ . The proposed η_{hist} may encode preparation history not fixed by ρ . The source explicitly requires that novelty be nonconstant inside the fiber and reach a genuinely independent second channel; otherwise the proposal collapses back to ordinary tomography.

This is exactly the structure PASS017 requires. However, S209 is an atomic/quantum-domain candidate. It does not establish that a human person carries such a fiber or that it is a personal-identity variable.

191.3 Path action and holonomy

S209 also identifies path action/holonomy as a transition-memory carrier. This demonstrates that physics can contain quantities sensitive to a route and not merely an endpoint. But “remembers a path” is weaker than “carries numerical personhood.” Path history and personal identity must remain separate claims.

191.4 S143 temporal residue

S143’s temporal residue is explicitly kept separate from memory and record. Structurally this makes it relevant to PASS017. Yet persistence, temporal addressability, projection and physical readout remain OPEN. PASS017 therefore does not identify

$$R_t \equiv \eta_{hist}.$$

The objects may be related only after a typed bridge is supplied.

192 Six requirements for a genuine personal history fiber

PASS017 extracts the following minimal gate set.

1. **HF1 - non-reducibility.** Two histories can have identical declared current/public observables while carrying distinct η_{hist} .
2. **HF2 - independent readout.** A prospectively fixed physical measurement couples to η_{hist} and yields different outcome laws.
3. **HF3 - typed transport.** Legitimate continuation transports or updates η_{hist} by a physical law rather than regenerating it from the archive.
4. **HF4 - anti-duplication or explicit fission semantics.** If η_{hist} is used to select numerical identity, two descendants cannot both silently inherit the same identity-bearing fiber.
5. **HF5 - person bridge.** A separate law must relate continuity of η_{hist} to continuity of the person.
6. **HF6 - prospective falsifier.** Calibration, controls, null model and failure conditions are frozen before the identity verdict is inspected.

TENET-H17 - Independent History-Fiber Requirement. A scientific personal-history fiber must be non-reconstructible from the already-equivalent record, physically readable through a second channel, transported by a typed law, equipped with anti-duplication/fission semantics, and independently bridged to personal identity.

193 PASS017 executable reduction bank

The executable toy runs 256 matched trials. In every trial W_{cont} and W_{copy} share the same current state, autobiographical memory, ledger, G_{mem} reconstruction data and archived path record.

Test	W_{cont}/W_{copy} differences or matches
State-derived η	0/256 differences
Memory-derived η	0/256 differences
Ledger-derived η	0/256 differences
G_{mem} -derived η	0/256 differences
Archived-path-derived η	0/256 differences
Independent-fiber toy: continuation matches anchor	256/256
Independent-fiber toy: copy matches anchor	0/256
Clone attack: descendant A matches	256/256
Clone attack: descendant B matches	256/256

The first five rows instantiate TENET-NG11. The independent-fiber toy then adds a hidden source carrier, a transport law and an independent verification channel. It separates continuation and copy perfectly in that declared model:

$$D_{self}^{(\eta)} = 1.$$

But this success is conditional on new structure that was not present in PASS016.

194 Clone attack

The toy next allows the independent fiber itself to be duplicated. Both descendants inherit the same identity-relevant carrier and both pass the anchor:

$$M_{\eta}(B_1) = M_{\eta}(B_2) = \text{PASS}.$$

The history channel still distinguishes those descendants from an ordinary reconstruction, but it no longer selects a unique numerical continuation.

TENET-NG12 - Cloneable-History Identity No-Go. A history-sensitive carrier that can be duplicated into two descendants while preserving all identity-relevant readouts cannot, by itself, establish unique numerical personal identity.

This is the PASS014 fission problem reappearing one level deeper. History sensitivity solves the *record-equivalence* problem only if its physical carrier has appropriate duplication/fission semantics.

195 Why the circular solution is rejected

The toy also includes a primitive variable

$$\eta_{self} = \mathbf{1}[\text{same self}].$$

It classifies the worlds perfectly. It is nevertheless rejected because the measurement was defined from the proposition it was supposed to test. PASS017 therefore distinguishes

independent hidden carrier $\neq$ hidden answer label.

196 History witness versus person witness

Even a genuine history-sensitive observable proves only that two physical histories differ unless an additional law says why that history carrier is constitutive of the person.

A path action, phase, environmental residue or protected private degree of freedom may provide excellent provenance. Different people can nevertheless share values of such variables, and one continuing person may evolve through different values over time. The empirical chain must therefore remain

$$\eta_{hist} \xrightarrow{M_\eta} \text{history evidence} \xrightarrow{B_{\eta \rightarrow self}} \text{personal-continuity claim}.$$

The second arrow is not paid merely by the existence of the first.

TENET-H18 - History/Identity Separation Principle. An independent history witness may establish physical continuity or provenance without thereby establishing personal numerical identity; the personhood bridge remains a separate claim.

197 What has actually been materialized

PASS017 therefore leaves three levels clearly separated:

1. **Reconstructible history:** memory, ledger, ObjectID, G_{mem} and archived path. Copyable; no new identity information.
2. **Independent physical history:** S209-style η_{hist} , path-sensitive mixed information, or a related second carrier. Structurally capable of adding new information, but physical realization/readout is domain-dependent or OPEN.
3. **Personal history:** an independent carrier additionally shown to be non-duplicating in the relevant sense and constitutive of the same person. Not established in the current corpus.

No present CT object in this audit simultaneously closes HF1-HF6 for a human being.

198 Consequence for soul and higher self

The soul candidate can no longer be rescued by renaming an archive “history fiber.” If a physical soul-like continuity carrier exists, PASS017 requires it to survive the record-collapse theorem:

$$\eta_{soul} \neq f(O_{state}, O_{memory}, O_{ledger}, O_{ID}, O_{RETURN}).$$

It must produce an independent readout and have a lawful continuity relation.

For the higher self, a future source that presents the correct memories and lineage remains insufficient. A stronger physical claim would require

$$\mathcal{A}_{type} \wedge \hat{\Lambda}_S \wedge C_\eta \wedge M_\eta \wedge T_\eta \wedge N_\eta \wedge B_{\eta \rightarrow self}$$

where the new symbols denote independent carrier, readout, transport, anti-duplication/fission semantics and the personal bridge. These are requirements, not established facts.

199 PASS017 ledger update

ID	Object	Status
TENET-NG11	Reconstructible-History Collapse No-Go	CLOSED
TENET-H17	Independent History-Fiber Requirement (HF1-HF6)	CONDITIONAL WORKING CANDIDATE
TENET-NG12	Cloneable-History Identity No-Go	CLOSED
TENET-H18	History/Identity Separation Principle	CONDITIONAL WORKING CANON FOR DOSSIER
TENET-HF1	256-trial record-derived reduction bank	CLOSED
TENET-HF2	Independent hidden-fiber discriminator	EXECUTED TOY CLOSED TOY / NEW CARRIER ASSUMED
TENET-HF3	Clone attack on independent fiber	CLOSED
TENET-HF4	S229 G_{mem} as missing personal fiber	EXECUTED TOY REJECTED AS RECON- STRUCTIBILITY MEMORY
TENET-HF5	S209 η_{hist} architecture	STRONGEST DOMAIN CANDIDATE / PERSONAL BRIDGE OPEN
TENET-HF6	Path action/holonomy as person witness	HISTORY WITNESS ONLY
TENET-HF7	S143 temporal residue = η_{hist}	NOT MERGED / BRIDGE OPEN
TENET-HF8	Primitive same-self bit	REJECTED
TENET-HSP11	Higher-self same-person attribution via history fiber	CIRCULAR MULTIPLE PHYSICAL BRIDGES OPEN

200 Direction for PASS018

PASS017 has isolated the next irreducible-looking gate: anti-duplication. PASS018 must test whether N_η can be inherited from known or already-owned physics rather than introduced by fiat. In particular, it must distinguish a properly scoped no-cloning/no-broadcasting constraint from the much stronger statement that a personal continuity carrier cannot be copied or split.

PASS018 - Anti-Duplication Carrier Attack: can noncopyability be derived from an existing physical owner, or is it another independent premise?

201 PASS018 - Anti-Duplication Carrier Attack

Status: STANDARD QUANTUM THEOREMS CLOSED / INHERITANCE TO η_{hist} CONDITIONAL / PERSONAL IDENTITY OPEN.

PASS017 showed that an independent history fiber can create an observable difference between otherwise equivalent histories, but a cloneable fiber still fails numerical-identity selection. PASS018 asks whether the anti-duplication gate can be paid by already-established physics rather than by a new axiom.

PASS018 question. Can quantum no-cloning/no-broadcasting supply HF4 without the invalid promotion “nonclonable quantum state $\Rightarrow$ numerically unique person”?

202 No-cloning: exact scope

Let $|\psi\rangle$ and $|\phi\rangle$ be normalized pure states, $|0\rangle$ a blank register, and suppose a unitary U exactly clones both:

$$U(|\psi\rangle|0\rangle) = |\psi\rangle|\psi\rangle, \quad U(|\phi\rangle|0\rangle) = |\phi\rangle|\phi\rangle.$$

Unitarity preserves inner products. With $s = \langle\psi|\phi\rangle$, the input overlap is s while the copied-output overlap is s^2 , hence

$$s = s^2, \quad s \in \{0, 1\}.$$

Thus distinct nonorthogonal pure states cannot both be perfectly cloned by a universal exact cloner under the standard assumptions.

PASS018 replays the $|0\rangle, |+\rangle$ example:

$$|\langle 0|+\rangle| = \frac{1}{\sqrt{2}}, \quad |s - s^2| \simeq 0.207106781187 > 0.$$

This residual is not a new CT law; it is only the standard overlap contradiction used as a scope guard.

202.1 What the theorem does not say

No-cloning is not a universal ban on copying information.

- Orthogonal states in a declared code can be copied by a basis-dependent unitary.
- A known, preparable state can be prepared repeatedly from its specification; this is not cloning one arbitrary unknown specimen.
- Classical records, memories, hashes, ledgers and descriptions can be duplicated.
- The exact theorem does not forbid all approximate clones, correlations or noisy channels.

TENET-SCOPE-Q1. “Quantum” does not imply “uncopyable”. Anti-duplication can be inherited only for a carrier and state family that actually satisfy the theorem assumptions.

203 No-broadcasting: additional scope guard

For mixed states, no-broadcasting sharpens the scope: a set of states is jointly broadcastable exactly when the relevant density operators commute in the standard setting. PASS018 does not re-prove the full theorem; it audits two representative pairs through their commutator.

Pair	$\ [\rho_1, \rho_2]\ _F$	Commutes	Reading
Diagonal/diagonal	0	yes	broadcasting not forbidden
Diagonal/coherent	0.254558441227	no	common broadcast blocked

Therefore even a quantum history fiber can be copyable on a classical/commuting subfamily. The state domain is an obligation, not a detail.

204 Links to existing owners

204.1 S228 - quantum access and ownership

The Encyclopedia already separates Golden Causal Quantum Object identity/provenance from illicit classical read/copy access to an unknown ontic wavefunction. An owner may hold preparations, circuits, calibration IDs, measurements, records and proofs without owning a classical copier for an unknown quantum state. PASS018 reuses this firewall.

204.2 S022 - teleportation is not cloning

Standard quantum teleportation retains the skeleton entangled resource $\rightarrow$ local measurement $\rightarrow$ classical communication $\rightarrow$ remote correction/reconstruction. In the ideal protocol it transfers an unknown state without retaining a second independent perfect copy at the source. This can support HF4, but it does not decide whether the receiver is numerically the same person.

204.3 S209 - history fiber

S209 provides the closest structural candidate:

$$C = (\rho, \eta_{hist}),$$

with η_{hist} required to contain physical information independent of ρ and to reach an independent second readout channel. PASS018 adds one question: if η_{hist} is truly quantum, does its relevant state family fall under a sufficiently strong anti-copy theorem to pay HF4?

205 Seven inheritance gates

A lawful import of quantum anti-duplication requires:

1. **AD1 - quantum carrier.** η_{hist} must be physically instantiated in a quantum degree of freedom, not merely called quantum.
2. **AD2 - nontrivial domain.** The relevant family must contain nonorthogonal or noncommuting states where the theorem is active.
3. **AD3 - independent readout.** The carrier must already pay PASS017 HF2.
4. **AD4 - lawful transport.** A continuation transports η_{hist} through a defined quantum channel.
5. **AD5 - no exact-copy fission.** The protocol must not leave two descendants each holding the same exact identity-bearing quantum state.
6. **AD6 - person binding.** A separate bridge must connect continuity of this carrier to continuity of the person.

7. **AD7 - destructive transfer.** If the source is destroyed, same-self survival must be established independently rather than inferred from no-cloning.

TENET-H19 - Quantum Anti-Duplication Inheritance Principle. HF4 can be paid conditionally by standard quantum physics only when the identity-history fiber is a real quantum carrier inside the no-cloning/no-broadcasting domain and its continuation uses a typed channel preserving that structure. This principle does not pay HF5.

206 PASS018 executable bank

Scenario	Anti-duplication	TENET verdict
Duplicated classical dossier	none	two exact descendants possible
Unknown nonorthogonal pure state	strong exact/universal restriction	generic two-perfect-clone route blocked
Orthogonal quantum code	not general	exact copies possible inside the code
Commuting mixed family	broadcasting possible	anti-duplication not acquired
Noncommuting mixed family	no-broadcasting	common exact broadcast blocked
Destructive quantum teleport transfer	no second perfect copy	fission avoided, personal identity OPEN
Nonclonable quantum η_{hist}	HF4 conditional	HF5 and AD7 remain OPEN

The toy does not model a person as a qubit. It tests scope only. The class “two exact clones of one arbitrary unknown nonorthogonal pure state” is blocked; a “destructive transfer to one receiver” remains compatible with two ontological readings:

$$W_{cont} \quad \text{or} \quad W_{replacement}.$$

No-cloning does not distinguish those worlds.

207 Two PASS018 no-go results

TENET-NG13 - Quantum Anti-Copy-to-Person No-Go. Impossibility of cloning or broadcasting a quantum-state family does not establish that the protected degree of freedom carries numerical personal identity. The theorem supplies neither the person-carrier bridge nor a same-self criterion.

TENET-NG14 - Destructive-Transfer Identity Gap. A protocol that guarantees only one exact carrier instance survives after transfer removes fission but does not decide whether the receiver is the numerical continuation of the source or one unique perfect replacement.

These no-go results are compatible with an important quantum substrate for persons. They only forbid automatic promotion from a copying constraint to personal-identity proof.

208 Consequence for soul and higher self

PASS018 removes another false primitive. One need not postulate a special “uncopyable soul substance” merely to obtain formal anti-fission; standard quantum physics already has scoped anti-copy properties.

But the strong form remains

$$\text{SameSelf}(A, B) = \mathcal{A}_{type}(A, B) \wedge \hat{\Lambda}_S(B|A) \wedge H_{\eta, Q}(A, B) \wedge B_{person \leftarrow \eta}(A, B),$$

where $H_{\eta, Q}$ only means that the identity-relevant quantum fiber was lawfully transported without exact duplication. The final person bridge remains OPEN.

For the terminal self:

$$\boxed{\text{retrograde RETURN} + \text{quantum anti-copy} \not\equiv \text{same higher self.}}$$

One must still show that the protected carrier is constitutive of the person and that destructive transmission preserves rather than replaces the subject.

209 PASS018 ledger update

ID	Object	Status
TENET-H19	Conditional inheritance of no-cloning/no-broadcasting into HF4	WORKING CANDIDATE
TENET-NG13	Quantum anti-copy $\not\equiv$ personal identity	CLOSED CONDITIONAL
TENET-NG14	Destructive transfer without fission $\not\equiv$ same self	CLOSED CONDITIONAL
TENET-Q18-1	$s = s^2$ overlap contradiction for exact universal cloner	STANDARD THEOREM / CLOSED REPLAY
TENET-Q18-2	No-broadcasting commutator audit	CLOSED TOY REPLAY
TENET-Q18-3	η_{hist} as human quantum carrier	OPEN
TENET-Q18-4	Person $\leftarrow \eta_{hist}$ bridge	OPEN
TENET-Q18-5	Identity under destructive teleportation	OPEN

210 Direction for PASS019

PASS018 shows that anti-duplication can have a known physical owner without resolving personal identity. PASS019 must therefore attack the remaining bridge directly: **why would the η_{hist} carrier, quantum or otherwise, be constitutive of the person?** The Person-to-Carrier Binding must survive decoherence, subsystem replacement, biological redundancy, local reset and destructive teleportation without defining identity by the chosen carrier.

PASS019 - Person-to-Carrier Binding Attack: can protected physical continuity be connected to the person without defining identity by whichever carrier was selected?

211 PASS020 - Representation Covariance and Kernel Stress Test

Status: CORE RETURN / FINITE IRREDUNDANCE AUDIT.

After PASS018, the personal-numerical-identity branch was recognized as a formalized Ship-of-Theseus problem. That branch is not deleted: its no-go results remain part of the dossier. But it is no longer treated as a prerequisite for defining TENET itself.

PASS019 scope cut. Numerical personal identity, destructive teleportation and the question “is the reconstructed person numerically the same subject?” remain OPEN and are moved to an application appendix. The domain-neutral TENET core requires causal identity/equivalence through transformation, not a metaphysical solution of personal identity.

This pass therefore asks a different question: *after eighteen passes, what is the smallest reciprocal-closure kernel that still rejects every known false TENET in the declared countermodel bank?*

212 Three layers that must no longer be conflated

PASS019 separates the cumulative program into three layers.

1. **Constitutive TENET core.** What must hold for an object/transaction/history to count as a reciprocal transformational closure at all?
2. **Epistemic certification.** What extra evidence is needed to attribute a RETURN to a terminal, retrograde or otherwise unusual source rather than leakage, hidden providers or incomplete model classes?
3. **Human attribution.** If the TENET object is a person, what additional predicates would justify “same person”, “higher self”, soul, agency or destiny?

Only the first layer is constitutive. PASS007-PASS012 remain important certification guards. PASS013-PASS018 remain important human-identity boundary results. Neither layer is promoted into the definition of every TENET object.

Passes	Layer	PASS019 reading
PASS001-PASS005	Core formation	constitutive source of the six-gate kernel
PASS006	Human selection	optional application: guidance, will, emergent destiny
PASS007-PASS012	Source certification	epistemic guard for extraordinary RETURN attribution
PASS013-PASS018	Numerical personal identity	OPEN / OUTSIDE CORE; retained as boundary appendix

213 The six-gate core

Let P denote a declared reciprocal-closure packet. PASS019 compresses the accumulated structure into six obligations.

213.1 K1 - Oriented triad

There must be two oriented traversal roles and one distinct closure/reference role:

$$K_1(P) : \quad OUTBOUND \neq RETURN \neq CLOSURE,$$

A declared origin fixes the two traversal directions. This is the subject-neutral content of the S229 $2SIN + 1COS$ role quotient. Collapsing RETURN into OUTBOUND or CLOSURE destroys the reciprocal architecture.

213.2 K2 - Invariant holding

For the declared identity/equivalence invariant I ,

$$T_+(I) \sim I, \quad T_-(I) \sim I, \quad (T_- \circ T_+)(I) \sim I.$$

The state may change. TENET requires holding through admissible transformation, not immobility.

213.3 K3 - Trace and provenance

Realized events must not be silently erased. The RETURN must bind the same declared root/lineage or an explicitly documented migration:

$$\text{Trace}(P) \text{ append-only, } \quad \text{Root}(\text{RETURN}) = \text{Root}(\text{OUTBOUND})$$

within the declared contract. A numerically successful return to the wrong root is not TENET closure.

213.4 K4 - Typed transport

Carrier changes must preserve type and native provenance through exact round trip or documented migration. Surface-value recovery without type recovery is insufficient:

$$\text{Dec}(\text{Enc}(x)) = x$$

only inside the declared domain, with the original type/provenance still attached.

213.5 K5 - Protected non-sacrifice

A closure cannot be purchased by hiding or worsening a protected residual. For a declared protected residual order,

$$\mathbf{R}_{\text{prot}}^{\text{after}} \preceq \mathbf{R}_{\text{prot}}^{\text{before}},$$

unless a separately typed exchange law explicitly authorizes the change. This is the gate that separates a raw Sator-like successful round trip from a protected TENET closure.

213.6 K6 - Exact closure and RETURN

The declared complete closure family must close exactly, not approximately or on an incomplete projection:

$$\mathcal{R} = 0$$

for the complete declared residual object/family, together with valid RETURN_PROOF and fixed-point/idempotence in the registered scope:

$$F(F(S)) = F(S).$$

COS alignment alone is not enough.

214 PASS019 executable ablation

The executable bank instantiates one positive packet and six negative packets. Each negative packet violates exactly one gate while satisfying the other five.

Countermodel	Missing gate	Why it must fail TENET
Role collapse	K1	traversal/closure roles have been collapsed
Invariant drift	K2	round trip returns but the declared invariant changes
Erased / wrong root	K3	trace is erased or RETURN binds another root
Type-loss round trip	K4	surface value returns after loss of carrier semantics
Compensating Sator	K5	raw success is paid by protected sacrifice
Incomplete fake zero	K6	apparent zero is obtained from an incomplete closure family

PASS019 exhaustively evaluates all $2^6 = 64$ subsets of the six candidate gates. Exactly one subset separates the positive witness from all six negative witnesses:

$$\{K_1, K_2, K_3, K_4, K_5, K_6\}.$$

Therefore the minimal separating size is six and the minimal kernel is unique in this declared finite bank.

The ablation result is stronger than a simple set-cover summary: removing any one gate causes exactly its paired countermodel to escape.

TENET-H20 - Irredundant Reciprocal Closure Kernel. In the declared PASS019 countermodel bank, a TENET packet is separated from all registered single-defect false closures exactly by the conjunction $K_1 \wedge K_2 \wedge K_3 \wedge K_4 \wedge K_5 \wedge K_6$. Every one of the six gates is individually necessary in that bank.

215 Compressed TENET definition

PASS019 therefore adopts the following domain-neutral core candidate:

$$TENET_{core}(P) \iff K_1(P) \wedge K_2(P) \wedge K_3(P) \wedge K_4(P) \wedge K_5(P) \wedge K_6(P).$$

In prose:

Causal TENET core. A TENET is a reciprocal transformational closure in which a declared identity invariant survives an oriented OUTBOUND and RETURN through typed transport, realized trace/provenance is not erased, protected residuals are not sacrificed, and a distinct closure role certifies exact RETURN/fixed-point closure on the same declared object.

This definition is deliberately narrower than the whole philosophy developed around TENET. It says nothing by itself about humans, consciousness, love, destiny, time travel or the soul.

216 Why the Ship-of-Theseus branch is outside the core

The executable domain-neutrality audit contains two positive TENET objects with identical six-gate structure: one is a non-personal reciprocal machine/transaction, the other is a human application with an additional same-person premise. The core accepts both. Adding SAME_PERSON as a seventh universal gate rejects the non-personal case even though none of the reciprocal-closure defects is present.

TENET-NG15 - Personal-Identity Scope-Overreach No-Go. Numerical same-person identity is not a constitutive requirement of a subject-neutral TENET. Making it a universal core gate overconstrains the concept by rejecting valid non-personal reciprocal closures. Personal numerical identity can only be an application-specific bridge.

This does not solve the Ship-of-Theseus problem. It locates it correctly.

For a human application one may still write

$$TENET_{human} = TENET_{core} \wedge H_{human},$$

where H_{human} can contain guidance, will, relational protection, personal lineage or a same-person premise. Those predicates inherit their own status and may remain OPEN without reopening the six-gate core.

217 Reconciliation with earlier TENET hypotheses

PASS019 does not create six unrelated new axioms. It compresses the older lineage:

- H2/H4 supply transformational and reciprocal holding $\rightarrow K_2$;
- H5 and S229 supply the typed OUTBOUND/RETURN/CLOSURE carrier $\rightarrow K_1, K_4$;
- non-erasure, append-only lineage and RETURN binding $\rightarrow K_3$;
- PASS004/S263 protected closure $\rightarrow K_5$;
- S222/S244 exact closure, RETURN_PROOF and fixed point $\rightarrow K_6$;
- H6-H19 remain application, certification or boundary results unless independently required by one of the six gates.

The important compression is therefore

variation $\rightarrow$ transformation $\rightarrow$ reciprocal return $\rightarrow$ protected exact closure,

with identity and trace held through the entire route.

218 Current status after PASS019

Object	Status	Meaning
Six-gate kernel	CLOSED CONDITIONAL / finite bank	unique minimal separator in registered PASS019 countermodels
Domain-neutral TENET definition	WORKING CORE CANDIDATE	compressed subject-neutral definition
Personal numerical identity	OPEN / OUTSIDE CORE	retained Ship-of-Theseus boundary
Retrograde physical transport	OPEN EMPIRICAL	not required to define the core
Higher self = same person	OPEN APPLICATION BRIDGE	not a constitutive gate
2SIN/1COS physical realization	OPEN EMPIRICAL	role grammar is stronger than physical realization

219 Direction for PASS020

PASS020 should no longer reopen personal identity. It must stress the six-gate kernel itself under representation change, reversal of declared orientation, equivalent carrier encodings, partial projection, compensated closure and permutation of route descriptions. The target is to discover whether a hidden seventh obligation exists or whether the six-gate kernel survives as the final TENET core.

PASS020 - Final Core Stress Test: attack the six-gate kernel under representation, reversal and route-equivalence changes before final closure.

220 PASS020 - Representation Covariance and Final Kernel Stress Test

Status: CORE STRESS / REPRESENTATION COVARIANCE / NO-SEVENTH-GATE RESULT IN REGISTERED BANK.

PASS019 isolated an irredundant six-gate TENET kernel. PASS020 attacks that kernel without reopening the Ship-of-Theseus branch. The question is now structural: *does the core keep the same verdict when the same reciprocal object is represented through another exact carrier, another address, another certified route, or the opposite declared transaction origin?*

The test also attacks the two most dangerous false closures: a zero visible only after projection, and a scalar total that hides a protected component regression.

221 Admissible representation map

Let P be a TENET packet satisfying the PASS019 predicates $K_1, \dots, K_6$. Let

$$\Phi : P \longrightarrow P'$$

be a representation change. PASS020 calls Φ *admissible* only if it transports the entire declared contract:

1. the role partition is preserved, allowing a simultaneous reversal of declared source and target so that OUTBOUND and RETURN exchange labels while CLOSURE remains distinct;
2. the invariant/equivalence relation is preserved;
3. root, trace and append-only lineage remain bound;
4. carrier type and native provenance have an exact inverse or documented migration;
5. the protected residual preorder is preserved componentwise;
6. completeness, exact-zero and RETURN/fixed-point semantics are preserved.

This is stronger than matching surface values. It is an isomorphism of the declared TENET contract, not merely a relabeling of outputs.

222 TENET-H21 - Representation-Covariant Reciprocal Closure

Because the core is the conjunction

$$TENET_{core}(P) = \bigwedge_{i=1}^6 K_i(P),$$

and an admissible Φ preserves every K_i predicate,

$$K_i(P) \iff K_i(\Phi(P)) \quad (i = 1, \dots, 6),$$

we obtain

$$\boxed{TENET_{core}(P) \iff TENET_{core}(\Phi(P)).}$$

TENET-H21 - Representation-Covariant Reciprocal Closure Theorem. Within a declared class of representation isomorphisms that preserve K1-K6 semantics, TENET status is representation-covariant. Representation covariance is therefore a theorem derived from the six-gate kernel, not an independent seventh constitutive gate.

This is CLOSED CONDITIONAL for the registered transformation class. It does not prove that every physically possible representation change is in that class.

223 Orientation reversal is not a new ontology

S229 already supplies a bilateral reciprocal pair only up to exchange. PASS020 makes the consequence explicit. If the declared transaction origin is reversed,

$$A \rightarrow B \rightsquigarrow B \rightarrow A,$$

then the traversal labels exchange:

$$OUTBOUND \leftrightarrow RETURN,$$

while the closure/reference role remains distinct. The reciprocal structure is unchanged provided K1-K6 are transported covariantly.

Thus the words “outbound” and “return” are origin-relative role labels, not absolute ontological substances.

224 Registered named stress bank

The executable named bank contains fourteen cases. Six are lawful descriptions of the same core object and eight are attacks.

Case	Verdict	Existing owner
Base good packet	PASS	K1-K6
Bijective re-encoding	PASS	exact K4 transport; all gates preserved
Address isomorphism	PASS	K4 plus K3 provenance
Reversed declared origin	PASS	K1 reciprocal exchange
Equivalent certified route	PASS	K2-K6 preserved
Vocabulary rename	PASS	names do not alter typed roles
Role collapse by relabeling	FAIL	K1
Non-covariant invariant map	FAIL	K2
Root alias without binding	FAIL	K3
Lossy encoding	FAIL	K4
Scalar compensation	FAIL	K5
Partial-projection zero	FAIL	K6
COS-only alignment	FAIL	K6
Approximate round trip	FAIL	K4 and K6

Every named target is classified correctly by the existing kernel.

225 Exhaustive finite representation family

PASS020 also expands the stress bank over an explicit finite representation family. Payloads are encoded by every invertible affine map on $\mathbb{Z}_{11}$,

$$x \mapsto ax + b, \quad a \in \mathbb{Z}_{11}^\times, \quad b \in \mathbb{Z}_{11},$$

which gives $10 \times 11 = 110$ exact encodings. Each is crossed with two declared orientations and three certified route descriptions:

$$110 \times 2 \times 3 = 660$$

lawful representation variants.

For each lawful variant PASS020 also injects each of the six single-gate failures, producing

$$660 \times 6 = 3960$$

adversarial cases. Total:

4620 finite stress cases.

The executable result is

4620/4620 expected verdicts, 0 errors.

This does not establish universal exhaustivity. It closes the stated finite family and shows that no additional gate is needed to separate these representation attacks.

226 TENET-NG16 - Partial-Projection False Closure

Let the parent residual be

$$\mathcal{R} \mapsto (\mu, R_{info}, \rho).$$

A visible closure coordinate may vanish while the parent object remains nonzero. For example,

$$(\mu, R_{info}, \rho) = (3, 2, 0)$$

has

$$\rho = 0$$

but

$$\mathcal{R} \neq 0.$$

Therefore

$$\Pi_{close}(\mathcal{R}) = 0 \not\Rightarrow \mathcal{R} = 0$$

unless the declared projection family is complete and the completeness theorem applies.

TENET-NG16 - Partial-Projection False Closure No-Go. Zero in a non-injective or incomplete closure projection cannot certify zero of the parent TENET residual. A quotient can hide unpaid drive, information or other typed debt.

This is already owned by K6 rather than requiring a new constitutive gate.

227 TENET-NG17 - Scalar-Compensation False Closure

A scalar aggregate can remain unchanged while a protected component worsens. In the executable witness,

$$(5, 5) \longrightarrow (9, 1),$$

so the scalar total stays 10, yet the first protected residual increases from 5 to 9.

Hence

$$\sum_i R_i^{after} = \sum_i R_i^{before} \not\Rightarrow \mathbf{R}_{prot}^{after} \preceq \mathbf{R}_{prot}^{before}.$$

TENET-NG17 - Scalar-Compensation False Closure No-Go. Conservation or cancellation of an aggregate score cannot certify TENET closure when protected residuals are typed componentwise. A local regression cannot be paid by an unrelated improvement without an explicit exchange law.

This is exactly K5.

228 Why K7 is redundant in the registered bank

The natural K7 candidate was:

“TENET status must survive admissible changes of representation.”

But admissibility itself means that the representation map preserves the semantics already owned by K1-K6. Therefore, for this class,

$$K_7(\Phi) = \bigwedge_{i=1}^6 [K_i(P) \Leftrightarrow K_i(\Phi(P))]$$

is a covariance theorem about the existing predicates, not a new predicate on the object.

Conversely every tested non-covariant transformation fails at least one existing gate before K7 would add information.

TENET-C20-1 - No-Seventh-Gate result. In the registered PASS020 transformation/countermodel bank, K1-K6 are sufficient under representation, orientation, route, projection and compensation stress. No independent seventh constitutive gate is required. Claim ceiling: finite/declared bank only.

229 Core TENET after PASS020

The domain-neutral core therefore remains

$$TENET_{core} = K_1 \wedge K_2 \wedge K_3 \wedge K_4 \wedge K_5 \wedge K_6.$$

Its new strengthened reading is:

A TENET is a reciprocal transformational closure whose oriented traversals, invariant, historical trace, typed transport, protected residual order and exact RETURN remain valid under every declared admissible re-expression of the same object.

This statement does not require the object to be a person, does not require physical backward time travel, and does not solve numerical personal identity.

230 PASS020 ledger update

ID	Object	Status
TENET-H21	Representation-covariant reciprocal closure	CLOSED CONDITIONAL / WORKING CORE THEOREM
TENET-NG16	Partial-projection zero does not certify parent closure	CLOSED CONDITIONAL
TENET-NG17	Scalar compensation does not certify protected non-sacrifice	CLOSED CONDITIONAL
TENET-C20-1	No independent K7 in registered stress bank	CLOSED CONDITIONAL / FINITE BANK
TENET-K1–K6	Irredundant reciprocal-closure kernel	RETAINED
Personal numerical identity	Ship-of-Theseus branch	OPEN / OUTSIDE CORE

Physical retrograde
realization

External bridge

OPEN EMPIRICAL

231 Direction for PASS021

PASS020 found no seventh gate inside the registered representation family. PASS021 should therefore stop inventing internal structure and perform an adversarial *externalization audit*: attack the assumptions behind the six gates themselves, search for counterexamples not generated by the current toy grammars, test necessity versus convention, and determine the exact final claim ceiling before the FINAL TENET CLOSURE.

PASS021 - External adversarial audit: attack the completeness and necessity assumptions of K1-K6 before final closure.

232 PASS022 - FINAL TENET CLOSURE

Status: EXTERNAL ADVERSARIAL AUDIT / LAYERED KERNEL / FINAL FORMAL CLAIM CEILING.

PASS020 found no hidden seventh gate inside the registered representation bank. PASS021 asks a harder question: *what authority does the six-gate result actually have outside that bank?* A reversible map, an involution, an isomorphism, a palindrome, a feedback loop and a causal TENET are not automatically the same object. The audit therefore stops treating “reversible” as a synonym for “TENET”.

The result is a refinement rather than a rejection of PASS019-PASS020: K1-K6 remain irredundant for the *full registered TENET certificate*, but several gates are enrichments that are not logical consequences of bare reciprocal transport.

233 Split of K1: reciprocal pair versus distinct closure lane

PASS019 bundled two ideas inside K1:

1. **K1a - reciprocal oriented pair:** there are two typed traversal directions whose composition can be audited as a return;
2. **K1b - distinct CLOSURE/reference role:** the CT/ITC realization keeps CLOSURE distinct from both traversal roles, as in $2SIN + 1COS$.

The external audit shows that K1a belongs to the abstract reciprocal skeleton, whereas K1b is not forced by generic invertibility. A group involution or typed isomorphism can have a valid forward/backward structure without materializing a third reference lane as a separate object.

TENET-NG20 - Reciprocal-Closure-to-Distinct-COS No-Go. A reciprocal exact round trip does not by itself imply a distinct COS/CLOSURE carrier. The separate closure/reference role is a registered ITC realization requirement, not a universal theorem of reversibility.

This does not weaken S229 inside its declared protocol: there the third role is part of the protocol contract. It only prevents exporting that role count as a theorem about every reciprocal system.

234 Layered decomposition of the TENET kernel

PASS021 replaces the flat reading of K1-K6 by a typed ladder.

R - Reciprocal transport skeleton.

$$R = K_{1a} \wedge K_4 \wedge K_6$$

There must be a reciprocal pair, typed transport, and exact round-trip/closure semantics.

T - Transformational TENET.

$$T = R \wedge K_2$$

The declared invariant must hold through the admissible transformation, not merely reappear at the end.

C - Causal TENET.

$$C = T \wedge K_3$$

The realized history is not erased: root, trace and provenance remain causally bound.

P - Protected causal TENET.

$$P = C \wedge K_5$$

Closure cannot be certified by silently moving debt into another protected component.

ITC full TENET realization.

$$TENET_{ITC} = P \wedge K_{1b}$$

The reciprocal/protected object is realized with the distinct OUTBOUND, RETURN and CLOSURE/reference roles of the registered *2SIN + 1COS* protocol.

TENET-H22 - Layered TENET Kernel Decomposition. The registered six-gate contract factors into reciprocal transport, transformational holding, causal non-erasure, protected closure and an ITC-specific distinct closure/reference realization. This is a formal classification of the dossier's contract, not a claim that Nature must instantiate every layer.

235 External countermodels

The executable PASS021 bank contains twelve adjacent archetypes. They are deliberately not claimed as an exhaustive ontology; their role is to test implication.

Archetype	Highest layer reached	Audit signal
Full ITC TENET Group involution	ITC full transformational	all registered obligations paid reciprocal/invariant/exact, no intrinsic trace, protection or COS lane
Category isomorphism Exact codec round trip	transformational transformational	typed inverse does not force causal history payload returns exactly; provenance/protection remain optional
Reversible swap that breaks I midway	reciprocal only	exact return does not imply invariant holding throughout
Traceless reversible machine Destructive but reversible process	transformational causal	reversibility does not imply non-erasure ledger trace may survive while a protected component regresses
ITC-like reciprocal object without separate COS lane	protected causal	K1b is realization-specific
Palindrome surface	symmetry only	visible reversal is not typed causal return
Detailed-balance chain	statistical reciprocity	probabilistic reversal is not exact RETURN
Feedback fixed point	one-way closure	convergence is not a reciprocal pair
Quotient false closure	false closure	visible zero can hide parent residual

The implication audit finds explicit witnesses against each attempted inference

$$R \Rightarrow K_2, \quad R \Rightarrow K_3, \quad R \Rightarrow K_5, \quad R \Rightarrow K_{1b}.$$

All four implications are false in the registered external bank.

236 TENET-NG18 - Reversibility does not imply causal trace

A stateless involution can satisfy exact transport and exact return without an append-only historical witness. Therefore

$$R \not\Rightarrow K_3.$$

TENET-NG18 - Reversibility-to-Causal-Trace No-Go. Exact reciprocal mathematics does not, by itself, establish causal non-erasure, historical provenance or a persistent trace. K3 is the enrichment that turns transformational holding into a causal TENET.

This aligns with the CT non-erasure family: K3 is not cosmetic metadata. It is the additional causal assertion that a realized traversal remains historically relevant.

237 TENET-NG19 - Reversibility does not imply protected closure

An invertible process can return its selected state while transferring cost or damage into a component outside the chosen state description. Therefore

$$C \not\Rightarrow K_5.$$

TENET-NG19 - Reversibility-to-Protected-Closure No-Go. Invertibility and causal trace do not imply componentwise non-sacrifice. K5 is an explicit protected-closure axiom relative to the declared protected residual set.

K5 is therefore not a theorem of algebraic reversibility. Within the full TENET contract it remains constitutive because PASS004/PASS019 deliberately reject Sator-like or externalizing closures that succeed only by sacrificing protected coherence.

238 What remains irredundant

Splitting K1 does not make the full registered contract redundant. PASS021 repeats single-gate ablation over the seven explicit clauses

$$K_{1a}, K_{1b}, K_2, K_3, K_4, K_5, K_6.$$

Removing any one clause prevents qualification as the *full ITC/protected causal TENET* by definition and by the registered countermodel bank.

Thus two statements must be kept separate:

Full registered TENET contract is irredundant	CLOSED CONDITIONAL,
Every reversible object in mathematics/physics must satisfy it	NOT ESTABLISHED.

239 Necessity, sufficiency and empirical status

PASS021 fixes three different meanings of “necessary”.

1. **Definitional necessity.** Relative to the full registered TENET certificate, all clauses are required. This is closed by the stated contract and ablation bank.
2. **Model-class necessity.** Relative to a declared class of false closures/countermodels, the gates separate accepted from rejected objects. This is closed only for that declared class.
3. **Natural-law necessity.** The claim that every real causal system must instantiate the same gates is not proved. It requires independent physical theory and experiment.

Likewise, formal sufficiency means only

$$K_1 \wedge \dots \wedge K_6 \implies \text{certificate in the declared TENET contract.}$$

It does not imply physical time travel, universal metaphysics, higher-self identity, or empirical realization of $2SIN + 1COS$.

TENET-C21-1 - Final claim ceiling. K1-K6 are an irredundant, representation-covariant certificate for the registered full ITC/protected causal TENET contract. PASS021 does not promote that certificate into a universal law of reversibility or a demonstrated law of Nature.

240 Final abstract definition before closure

The external audit suggests that the most portable definition should not begin with the material name COS. It should begin with the functional structure:

Causal TENET - pre-final abstract form. A causal TENET is a typed reciprocal transformation in which a declared invariant survives the admissible traversals, the realized history remains non-erased and provenance-bound, protected obligations are not silently sacrificed, and exact return/closure is achieved in the declared complete carrier.

The ITC realization then specializes this abstract form as

$$SIN1 = OUTBOUND, \quad SIN2 = RETURN, \quad COS = CLOSURE/reference.$$

This ordering is important: the abstract law owns the semantics; $2SIN + 1COS$ is one registered realization of them.

241 PASS021 ledger update

ID	Object	Status
TENET-H22	Layered TENET kernel decomposition $R \rightarrow T \rightarrow C \rightarrow P \rightarrow ITC$	CLOSED CONDITIONAL / FORMAL CLASSIFICATION
TENET-NG18	Reversibility does not imply causal trace/non-erasure	CLOSED CONDITIONAL / COUNTERMODEL
TENET-NG19	Reversibility does not imply protected non-sacrifice	CLOSED CONDITIONAL / COUNTERMODEL
TENET-NG20	Reciprocal closure does not imply a distinct COS lane	CLOSED CONDITIONAL / COUNTERMODEL
TENET-C21-1	Full contract is registered/conditional, not universal natural law	FINAL CLAIM CEILING
K1-K6 full contract	Irredundant full ITC/protected causal certificate	RETAINED
Personal numerical identity	Ship-of-Theseus branch	OPEN / OUTSIDE CORE
Physical retrograde realization	External empirical bridge	OPEN EMPIRICAL
Universal natural-law status	External theory/experiment question	OPEN

242 Direction for PASS022

PASS021 has completed the external authority audit. PASS022 should be the **FINAL TENET CLOSURE**: freeze the definition, theorem stack, no-go stack, realization map and status table; remove exploratory wording that has been superseded; distinguish formal closure from empirical OPEN branches; and state precisely what the TENET research has and has not established.

PASS022 - FINAL TENET CLOSURE: freeze the causal TENET law, its registered ITC realization, its no-go boundaries and its empirical claim ceiling.

243 PASS022 - FINAL TENET CLOSURE

Status: FORMAL DOSSIER FROZEN / CONDITIONAL CLOSURE / EMPIRICAL BRIDGES REMAIN OPEN.

PASS022 is a freeze pass. It adds no new constitutive hypothesis, no new physical claim and no new metaphysical identity primitive. Its task is to state the strongest result that survives PASS001–PASS021 without silently promoting any OPEN bridge.

244 Final layered object

The authority-preserving chain is frozen as

$$R = K_{1a} \wedge K_4 \wedge K_6$$

for reciprocal typed transport,

$$T = R \wedge K_2$$

for transformational holding,

$$C = T \wedge K_3$$

for causal non-erasure/provenance,

$$P = C \wedge K_5$$

for protected causal closure, and

$$TENET_{ITC} = P \wedge K_{1b}$$

for the registered ITC realization with a distinct closure/reference role.

PASS022 takes P as the final portable abstract target of the TENET research. K_{1b} is not promoted into a universal law: it belongs to the registered ITC implementation layer.

TENET-F22-1 - Final protected causal TENET definition. A TENET is a typed reciprocal transformation for which the declared invariant survives admissible traversal, the realized history remains non-erased and provenance-bound, protected residual obligations are not silently sacrificed, and exact return/closure is obtained in the declared complete carrier.

In compact form,

$$TENET_{final} = K_{1a} \wedge K_2 \wedge K_3 \wedge K_4 \wedge K_5 \wedge K_6.$$

245 Registered ITC realization

The final ITC specialization remains

$$SIN1 = OUTBOUND, \quad SIN2 = RETURN, \quad COS = CLOSURE/reference.$$

This is a typed role realization, not a theorem that physical spacetime literally consists of three scalar lanes. The parent causal residual remains one object with typed drive, information and closure projections; a partial zero never substitutes for whole-object closure.

TENET-F22-2 - Registered ITC realization. The $2SIN + 1COS$ architecture is the dossier's registered realization of the protected causal TENET contract. Its internal role semantics are conditionally closed; literal external physical realization remains OPEN EMPIRICAL.

246 One-sentence closure

The entire search can now be compressed without the film vocabulary:

A TENET is what can go out, transform, return, and close without ceasing to satisfy its invariant, without erasing what happened, and without hiding the cost of closure in a protected component.

This formulation preserves the structural insight from the SATOR/TENET palindrome tradition and the temporal-pincer exploration while no longer depending on either one for formal authority.

247 What is formally closed

The following results are frozen at their stated scope:

1. **Irredundancy.** PASS019 gives an irredundant six-gate full contract in its registered ablation bank (TENET-H20).
2. **Representation covariance.** PASS020 shows that admissible representation, orientation and route changes preserve verdict when they transport the full contract (TENET-H21); no independent K7 appeared in the registered stress bank.
3. **Layered authority.** PASS021 separates reciprocal, transformational, causal, protected and ITC-specific layers (TENET-H22).
4. **No-go boundaries.** NG2–NG20 forbid the main false promotions accumulated through the search. The historical dossier contains no NG1; PASS022 preserves that numbering rather than inventing one.
5. **Scope cut.** Numerical personal identity/Ship-of-Theseus is not constitutive of the TENET core.

TENET-F22-3 - Final formal closure certificate. Relative to the declared objects, gates, typed carriers and registered countermodel banks, the constitutive TENET research is CLOSED CONDITIONAL and documentarily complete. No additional internal gate is currently owed.

248 What is not established

PASS022 explicitly refuses the following promotions:

- no physical retrograde-transport realization has been established;
- no literal physical $2SIN + 1COS$ carrier has been established;
- no theorem shows that every reversible mathematical or physical process must satisfy the protected TENET contract;
- no exhaustive open-world theorem eliminates every unknown forward mechanism;
- no result establishes that a terminal/future source is a higher self;
- no result solves numerical personal identity or proves a soul as a physical primitive;
- no internal closure substitutes for independent empirical calibration.

TENET-F22-4 - Final empirical claim ceiling. The formal TENET certificate and its registered ITC realization do not, by themselves, establish time travel, retrocausality, higher-self agency, universal natural-law status or a physical theory of personal numerical identity. Those claims remain OPEN at their recorded bridges.

249 Ship-of-Theseus branch closure

PASS013–PASS018 were useful because they showed exactly where the human interpretation becomes a different problem. Copyable pattern, memory, lineage, a history fiber and quantum anti-copy constraints can constrain continuity without solving numerical same-person identity.

PASS022 therefore freezes:

personal numerical identity = OPEN / OUTSIDE CORE TENET.

TENET-F22-5 - Scope cut. A future state may be a valid causal/lineage continuation in a TENET analysis without the dossier claiming a solution to the metaphysical Ship-of-Theseus question.

250 Final theorem registry

The H-series is preserved as the historical development path. Early H-items include exploratory or branch-specific candidates; H20–H22 contain the final formal kernel/covariance/layering results. Their recorded latest statuses are frozen below.

ID	Object	Latest recorded status
TENET-H1	Temporal Self-Closure	CANDIDATE
TENET-H2	Law of Transformational Holding	CANDIDATE CORE
TENET-H3	2SIN/1COS + WILL + LOVE Gate	REFINED ARCHITECTURE LOVE DERIVED
TENET-H4	Reciprocal Transformational Holding	STRENGTHENED BY TYPED CARRIER
TENET-H5	Carrier-Bound Reciprocal Holding	CANDIDATE OPERATIONAL CORE
TENET-H6	Boundary-Guided Free Closure	CANDIDATE OPERATIONAL TOY
TENET-H7	Guidance Provenance Principle	CANDIDATE CORE
TENET-H8	Terminal-Source Witness Principle	CANDIDATE DISCRIMINATOR CORE
TENET-H9	Deferred-Verification Pincer Receipt Principle	CANDIDATE CORE CLOSED TOY ARCHITECTURE
TENET-H10	Independent Chronology Quorum Principle	CANDIDATE CORE CLOSED CONDITIONAL TOY
TENET-H11	Causal-Cone Exclusion Witness Principle	CANDIDATE CORE CLOSED CONDITIONAL TOY
TENET-H12	Adversarial Explanation Saturation Principle	CANDIDATE CORE CLOSED FINITE TOY
TENET-H13	Terminal Lineage Witness Principle	CANDIDATE CORE / CLOSED TOY
TENET-H14	Linear Personal-Continuity Candidate	CANDIDATE PHYSICAL BRIDGE OPEN
TENET-H15	Unique Committed Lineage Reduction	CLOSED CONDITIONAL FORMAL
TENET-H16	Observable-Difference Requirement for B self	WORKING CANDIDATE
TENET-H17	Independent History-Fiber Requirement HF1-HF6	CANDIDATE
TENET-H18	History/Identity Separation Principle	WORKING CANON OF DOSSIER
TENET-H19	Quantum Anti-Duplication Inheritance Principle	CANDIDATE
TENET-H20	Irredundant Reciprocal Closure Kernel	CLOSED CONDITIONAL FINITE BANK / WORKING CORE CANDIDATE
TENET-H21	Representation-Covariant Reciprocal Closure Theorem	CLOSED CONDITIONAL REGISTERED CLASS / WORKING CORE THEOREM
TENET-H22	Layered TENET Kernel Decomposition	CLOSED CONDITIONAL / FORMAL CLASSIFICATION

251 Final no-go registry

The no-go sequence begins historically at NG2. No NG1 is created by PASS022.

ID	No-go	Latest recorded status
TENET-NG2	Self-Certifying Chronology No-Go	STRENGTHENED EPISTEMIC GUARD
TENET-NG3	Declared-Cone Completeness No-Go	STRENGTHENED EPISTEMIC GUARD
TENET-NG4	Open-World Last-Survivor No-Go	STRENGTHENED EPISTEMIC GUARD
TENET-NG5	Transcript Identity No-Go	CLOSED IDENTIFIABILITY GUARD
TENET-NG6	Adaptive-Agency No-Go	CLOSED IDENTIFIABILITY GUARD
TENET-NG7	Copyable-Invariant Identity No-Go	CLOSED LOGICAL NO GO
TENET-NG8	Fission Numerical-Identity No-Go	CLOSED LOGICAL NO GO
TENET-NG9	Formal-Record Personal-Identity Underdetermination	CLOSED LOGICAL NO GO
TENET-NG10	Operational Personal-Identity Indistinguishability	CLOSED CONDITIONAL

TENET-NG11	Reconstructible-History Collapse No-Go	CLOSED CONDITIONAL
TENET-NG12	Cloneable-History Identity No-Go	CLOSED CONDITIONAL
TENET-NG13	Quantum Anti-Copy-to-Person No-Go	CLOSED CONDITIONAL
TENET-NG14	Destructive-Transfer Identity Gap	CLOSED CONDITIONAL
TENET-NG15	Personal-Identity Scope-Overreach No-Go	CLOSED CONDITIONAL SCOPE
TENET-NG16	Partial-Projection False Closure No-Go	CLOSED CONDITIONAL
TENET-NG17	Scalar-Compensation False Closure No-Go	CLOSED CONDITIONAL
TENET-NG18	Reversibility-to-Causal-Trace No-Go	CLOSED CONDITIONAL / COUNTERMODEL
TENET-NG19	Reversibility-to-Protected-Closure No-Go	CLOSED CONDITIONAL / COUNTERMODEL
TENET-NG20	Reciprocal-Closure-to-Distinct-COS No-Go	CLOSED CONDITIONAL / COUNTERMODEL

252 Final status table

Object	Final status	Meaning
Protected causal TENET <i>P</i>	CLOSED CONDITIONAL	final portable formal target of this dossier
Registered ITC realization	CLOSED CONDITIONAL / protocol	role grammar closed internally; physical realization open
Representation covariance	CLOSED CONDITIONAL / registered class	no K7 found in registered stress family
Universal natural-law status	OPEN	requires external theory and experiment
Physical retrograde transport	OPEN EMPIRICAL	no realization established
Higher-self attribution	OPEN / INTERPRETIVE	source and same-person bridges not established
Personal numerical identity	OPEN / OUTSIDE CORE	Ship-of-Theseus branch deliberately cut
External model exhaustivity	OPEN	unknown-forward/hidden-channel possibility not globally eliminated

253 Closure declaration

The constitutive TENET research begun in PASS001 is now closed at the formal/documentary level. Future work should not continue by adding internal TENET gates unless a new source, theorem, counterexample or experiment reopens an obligation. Physical tests of retrograde transport, history fibers or ITC carriers belong to a separate empirical program.

FINAL TENET CLOSURE - PASS022. Formal core frozen. Registered ITC realization frozen. No-go boundaries frozen. Personal numerical identity scoped out. Physical and universal-law bridges remain explicitly OPEN.